7355

Digital Multimeter

Operation Manual

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ADC CORPORATION

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Safety Summary

To ensure thorough understanding of all functions and to ensure efficient use of this instrument, please read the manual carefully before using. Note that ADC Corporation (hereafter referred to as ADC) bears absolutely no responsibility for the result of operations caused due to incorrect or inappropriate use of this instrument.

If the equipment is used in a manner not specified by ADC, the protection provided by the equipment may be impaired.

- **Warning Labels**

  Warning labels are applied to ADC products in locations where specific dangers exist. Pay careful attention to these labels during handling. Do not remove or tear these labels. If you have any questions regarding warning labels, please ask your nearest ADC dealer. Our address and phone number are listed at the end of this manual.

  Symbols of those warning labels are shown below together with their meaning.

  DANGER: Indicates an imminently hazardous situation which will result in death or serious personal injury.

  WARNING: Indicates a potentially hazardous situation which will result in death or serious personal injury.

  CAUTION: Indicates a potentially hazardous situation which will result in personal injury or a damage to property including the product.

- **Basic Precautions**

  Please observe the following precautions to prevent fire, burn, electric shock, and personal injury.

  - Use a power cable rated for the voltage in question. Be sure however to use a power cable conforming to safety standards of your nation when using a product overseas.
  - When inserting the plug into the electrical outlet, first turn the power switch OFF and then insert the plug as far as it will go.
  - When removing the plug from the electrical outlet, first turn the power switch OFF and then pull it out by gripping the plug. Do not pull on the power cable itself. Make sure your hands are dry at this time.
  - Before turning on the power, be sure to check that the supply voltage matches the voltage requirements of the instrument.
  - Connect the power cable to a power outlet that is connected to a protected ground terminal. Grounding will be defeated if you use an extension cord which does not include a protective conductor terminal.
  - Be sure to use fuses rated for the voltage in question.
  - Do not use this instrument with the case open.
  - If there is any abnormality such as abnormal heart, smoke, smell or noise, immediately stop using the instrument, turn off the power and disconnect the plug from the outlet.

- Do not place anything on the product and do not apply excessive pressure to the product. Also, do not place flower pots or other containers containing liquid such as chemicals near this product.
- When the product has ventilation outlets, do not stick or drop metal or easily flammable objects into the ventilation outlets.
- When using the product on a cart, fix it with belts to avoid its drop.
- When connecting the product to peripheral equipment, turn the power off.

- **Caution Symbols Used Within this Manual**

Symbols indicating items requiring caution which are used in this manual are shown below together with their meaning.

DANGER: Indicates an item where there is a danger of serious personal injury (death or serious injury).

WARNING: Indicates an item relating to personal safety or health.

CAUTION: Indicates an item relating to possible damage to the product or instrument or relating to a restriction on operation.

- **Safety Marks on the Product**

The following safety marks can be found on ADC products.

 : ATTENTION - Refer to manual.

 : Protective ground (earth) terminal.

 : DANGER - High voltage.

 : CAUTION - Risk of electric shock.

- **Replacing Parts with Limited Life**

The following parts used in the instrument are main parts with limited life.
Replace the parts listed below before their expected lifespan has expired to maintain the performance and function of the instrument.
Note that the estimated lifespan for the parts listed below may be shortened by factors such as the environment where the instrument is stored or used, and how often the instrument is used.
The parts inside are not user-replaceable. For a part replacement, please contact the ADC sales office for servicing.

Each product may use parts with limited life.
For more information, refer to the section in this document where the parts with limited life are described.

Main Parts with Limited Life

Part name	Life
Unit power supply	5 years
Fan motor	5 years
Electrolytic capacitor	5 years
LCD display	6 years
LCD backlight	2.5 years
Memory backup battery	5 years

- **Hard Disk Mounted Products**

The operational warnings are listed below.

- Do not move, shock and vibrate the product while the power is turned on.
  Reading or writing data in the hard disk unit is performed with the memory disk turning at a high speed. It is a very delicate process.

- Store and operate the products under the following environmental conditions.
  An area with no sudden temperature changes.
  An area away from shock or vibrations. An area free from moisture, dirt, or dust.
  An area away from magnets or an instrument which generates a magnetic field.

- Make back-ups of important data.
  The data stored in the disk may become damaged if the product is mishandled. The hard disc has a limited life span which depends on the operational conditions. Note that there is no guarantee for any loss of data.

- **Precautions when Disposing of this Instrument**

When disposing of harmful substances, be sure dispose of them properly with abiding by the state-provided law.

Harmful substances: (1) PCB (polycarbon biphenyl)
(2) Mercury
(3) Ni-Cd (nickel cadmium)
(4) Other
Items possessing cyan, organic phosphorous and hexadic chromium and items which may leak cadmium or arsenic (excluding lead in solder).

Example: fluorescent tubes, batterie

Environmental Conditions

This instrument should be only be used in an indoor area which satisfies the following conditions:

- An area free from corrosive gas
- An area away from direct sunlight
- A dust-free area
- An area free from vibrations
- Altitude of up to 2000 m

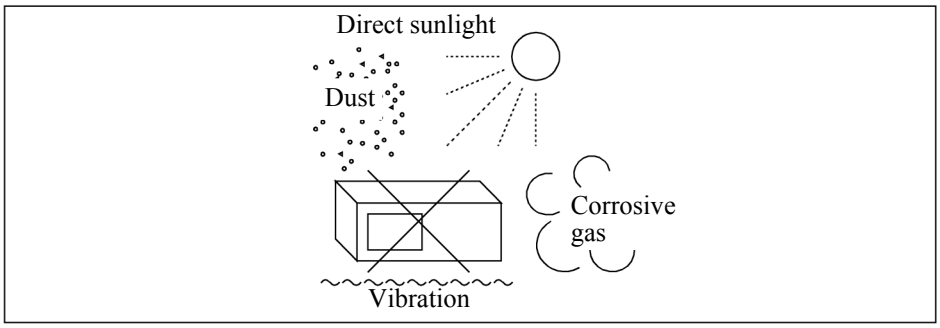


Figure-1 Environmental Conditions

- Operating position

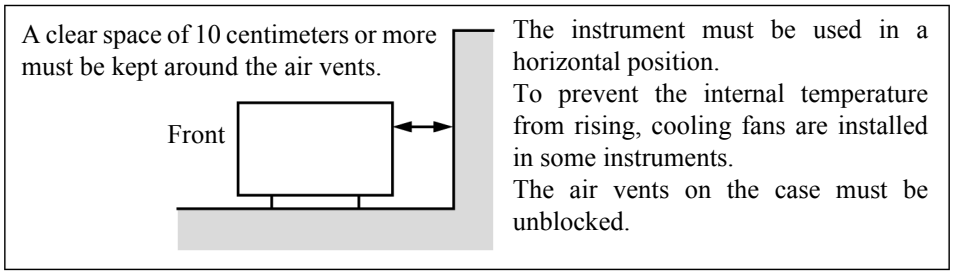


Figure-2 Operating Position

- Storage position

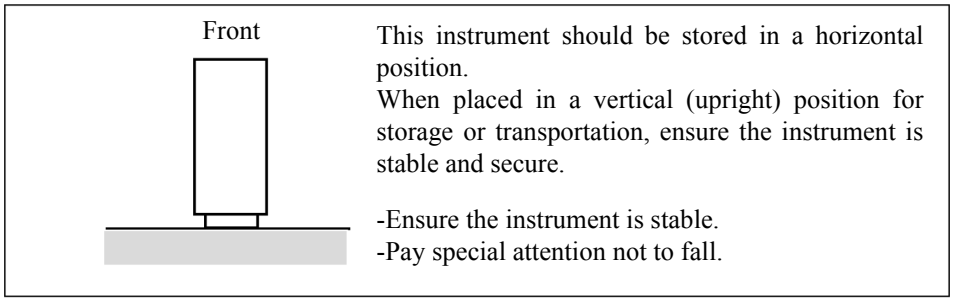


Figure-3 Storage Position

- The classification of the transient over-voltage, which exists typically in the main power supply, and the pollution degree is defined by IEC61010-1 and described below.

  Impulse withstand voltage (over-voltage) category II defined by IEC60364-4-443

  Pollution Degree 2

Types of Power Cable

Replace any references to the power cable type, according to the following table, with the appropriate power cable type for your country.

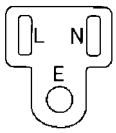
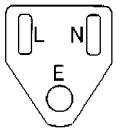
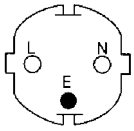
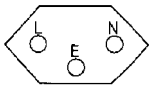
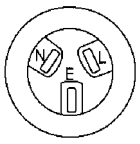
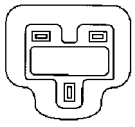
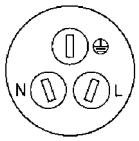
Plug configuration	Standards	Rating, color and length	Model number (Option number)
	PSE: Japan Electrical Appliance and Material Safety Law	125 V at 7 A Black 2 m (6 ft)	Straight: A01402 Angled: A01412
	UL: United States of America CSA: Canada	125 V at 7 A Black 2 m (6 ft)	Straight: A01403 (Option 95) Angled: A01413
	CEE: Europe DEMKO: Denmark NEMKO: Norway VDE: Germany KEMA: The Netherlands CEBEC: Belgium OVE: Austria FIMKO: Finland SEMKO: Sweden	250 V at 6 A Gray 2 m (6 ft)	Straight: A01404 (Option 96) Angled: A01414
	SEV: Switzerland	250 V at 6 A Gray 2 m (6 ft)	Straight: A01405 (Option 97) Angled: A01415
	SAA: Australia, New Zealand	250 V at 6 A Gray 2 m (6 ft)	Straight: A01406 (Option 98) Angled: ---------
	BS: United Kingdom	250 V at 6 A Black 2 m (6 ft)	Straight: A01407 (Option 99) Angled: A01417
	CCC: China	250 V at 10 A Black 2 m (6 ft)	Straight: A114009 (Option 94) Angled: A114109

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1. INTRODUCTION

This chapter describes the contents of this manual and the product overview of the 7355 Digital Multimeter.

1.1 Contents of This Manual

This manual can be used by novices or experienced users of the 7355. This manual can be read from Chapter 1 to learn more about this instrument or it can be used as a reference. Refer to the table of contents, which is found at the beginning of each chapter, and directly jump to the required section.

The contents of each chapter are as follows.

Chapter 1. INTRODUCTION	This chapter describes the contents of this manual and the product overview.
Chapter 2. PRECAUTIONS	This chapter describes precautions when using the 7355. Read this chapter before use.
Chapter 3. SETUP	This chapter describes how to setup the 7355. After setting it up in an appropriate location, turn on the power and check that it starts correctly.
Chapter 4. QUICK START	This chapter describes the function of each section on the panels of the 7355 and how to operate it.
Chapter 5. MENU OPERATION AND FUNCTION DESCRIPTION	This chapter describes the menu structure and functions of the 7355.
Chapter 6. HOW TO USE INTERFACE	This chapter describes how to use the interface of the 7355.
Chapter 7. SPECIFICATIONS	This chapter describes the specifications of the 7355.
Chapter 8. MAINTENANCE	This chapter describes how to care for the 7355 such as cleaning, calibration, and storage to maintain the high performance and smooth functioning of it. This chapter also describes how to identify problems and the relevant procedures to follow.

1.2 Product Overview

The 7355 is a digital multimeter adopting an integrating type A/D converter.

It is convenient and versatile for both bench-top and system use.

The factory default setting is 6 1/2-digit display. To use with the 5 1/2-digit display, refer to the following chapters.

- 6.8.1.1 7351-Compatible Mode
- 5.6 Number of Measurement Display Digits Setting

The main features of the 7355 are as follows:

- Simple and easy panel operations, with flexible and various system functions by using the menu
- Eye-friendly VA LCD with large characters
- Wide dynamic range: Maximum display of 1999999
- DC voltage measurement with high sensitivity of 100 nV
- DC current measurement in high dynamic range between 100 nA to 10 A
- Low power resistance measurement function that reduces the self-heating effect of the device
- USB, GPIB, RS232, LAN, trigger input/complete signal and comparator calculation output for easy connection to personal computers
- A variety of combinable calculation functions
- Four files of measurement parameters can be saved.
- Conventional and SCPI command systems are included as standard.

CAUTION: GPIB, RS232, comparator output, external trigger input, and complete output are factory options. No options can be added after purchase.

Table 1-1 Interface Configuration

Interface	Standard	Option +01	Option +03
USB	✓	✓	✓
GPIB		✓	
RS232			✓
LAN	✓	✓	✓
Comparator output			✓
External trigger input		✓	✓
Complete output		✓	✓

2. PRECAUTIONS

This chapter describes precautions when using the 7355. Read this chapter before use.

CAUTION: Using the 7355 while it is stood on end or stacked with other products is unstable and dangerous. Do not push or place heavy loads on the 7355.

2.1 If Fault Occurs

If any smoke, smell, or noise emanates from the 7355, disconnect it from the power supply by removing the power cable. Contact an ADC CORPORATION sales representative immediately.

2.2 Removing Mainframe Cover

The mainframe cover of the 7355 should only be opened by ADC CORPORATION service engineers.

WARNING: The 7355 contains high-voltage and high temperature parts which may cause electrical shocks or burns.

2.3 Overcurrent Protection

The 7355 is protected from overcurrent by a power fuse.
The power fuse, which is located on the rear panel, forcibly cuts the power supply if an overcurrent flows. If the power fuse is blown, remove the power cable and separate between the power supply and the 7355. There may be some problems in the 7355 and request an ADC CORPORATION sales representative to repair it.

2.4 Electromagnetic Interference

The 7355 may cause electromagnetic interference and affect television and radio. If its power is turned off and any electromagnetic interference that may be present is reduced, then the 7355 has caused the interference.
Electromagnetic interference from the 7355 may be prevented by the following precautions.

- Changing the direction of the antenna of the television or radio.
- Placing the 7355 on the opposite side of the television or radio.
- Placing the 7355 away from the television or radio.
- Using a different power source for the television or radio, and the 7355.

In an area where the electric field strength is high, high-frequency noise will occur on the DUT or input cable, and that noise may affect the measured value. In this case, shield the DUT and input cable from high-frequency noise.

2.5 Note When Turning on Power

When turning on the power, do not connect any DUTs.

3. SETUP

This chapter describes how to set up the 7355 on delivery. The following topics are covered in this chapter.

3.1 Inspection on Delivery
3.2 Installation Environment
3.3 Accessories
3.4 Power Requirements

3.1 Inspection on Delivery

After receiving the product, inspect the outside and the accessories according to the following procedure.

1. Check that the shipping container and the cushioning material are not damaged.

IMPORTANT: If the shipping container or the cushioning material is damaged, keep them until the following inspections are complete.

---

2. Check that the outside of the product is not damaged.

WARNING: If any outside components of the product such as the cover, panel (front or rear), power switch, or connector are damaged, do not turn on the power. Doing so may cause an electrical shock.

---

3. Check that the supplied accessories listed in Table 3-1 are complete and not damaged.

If any of the following occur, contact an ADC CORPORATION sales representative.

- The shipping container or the cushioning material is damaged, or signs of stress are found.
- The outside of the product is damaged.
- The supplied accessories are incomplete or are damaged.
- Defects are found in the operation check.

Table 3-1 Supplied Accessory List

Name	Part number	Quantity	Remarks
Power cable	A01402	1	*1
Input cable (red, black)	CC010001	1 each	
Power fuse (for 100 V/120 V)	DFT-AAR25A-1	1	*2
Power fuse (for 220 V/240 V)	DFT-AAR2A-1		
Overcurrent protection fuse	DFS-AN2A-1	1	
Quick Manual	E7355 (Q)	1	

*1: The power cable included in this instrument depends on the option that was specified when this instrument was purchased (see “Safety Summary”).
Quote the model name or option number when ordering the power cable.
*2: Either one is included according to the specified option

NOTE: Quote the model name of the accessory when ordering.

3.2 Installation Environment

This section describes the environment into which the 7355 should be installed.

3.2.1 Operating Environment

The 7355 must be installed in an environment meeting the following conditions:

- Ambient temperature 0 °C to +50 °C (operating temperature)
  -25 °C to +70 °C (storage temperature)
- Relative humidity: 85% or lower without condensation
- Location not subject to corrosive gasses
- Location away from direct sunlight
- Dust free
- Vibration free
- Noise free
  The 7355 is designed with full consideration given to the noise contained in the AC power line. Nevertheless, it is recommended that the 7355 be used in an environment with as little line noise as possible. If line noise is unavoidable, use a noise filter.

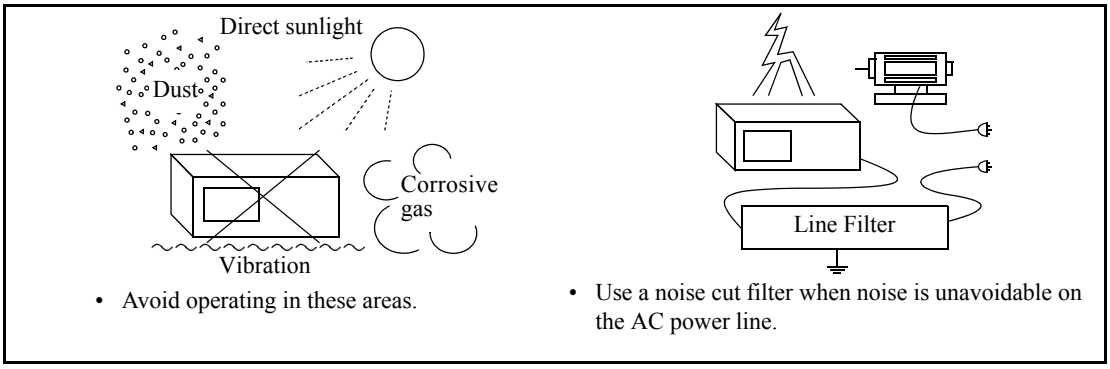


Figure 3-1 Operating Environment

NOTE: Warm-up
After the 7355 reaches room temperature, turn on the power and warm up for 60 minutes or more to perform the high accuracy measurement.

3.2.2 Protecting Against Electrostatic Discharge

To prevent semiconductors from being damaged by electrostatic discharge (ESD), the precautions shown below should be taken. We recommend combining two or more countermeasures to prevent damage from ESD. (Static electricity can be generated easily by the movement of a person or the friction against insulation.)

Table 3-2 ESD Countermeasures

Human body	Use a wrist strap (See Figure 3-2).
Work floor	Install a conductive mat, use conductive shoes, and connect to earth. (See Figure 3-2)
Workbench	Install a conductive mat and connect to earth. (See Figure 3-3)

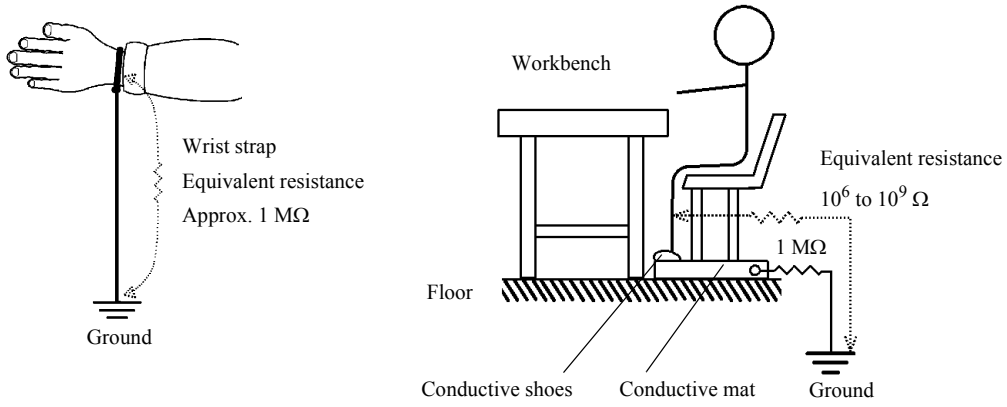


Figure 3-2 Electrostatic Countermeasures for Work Floor and Human Body

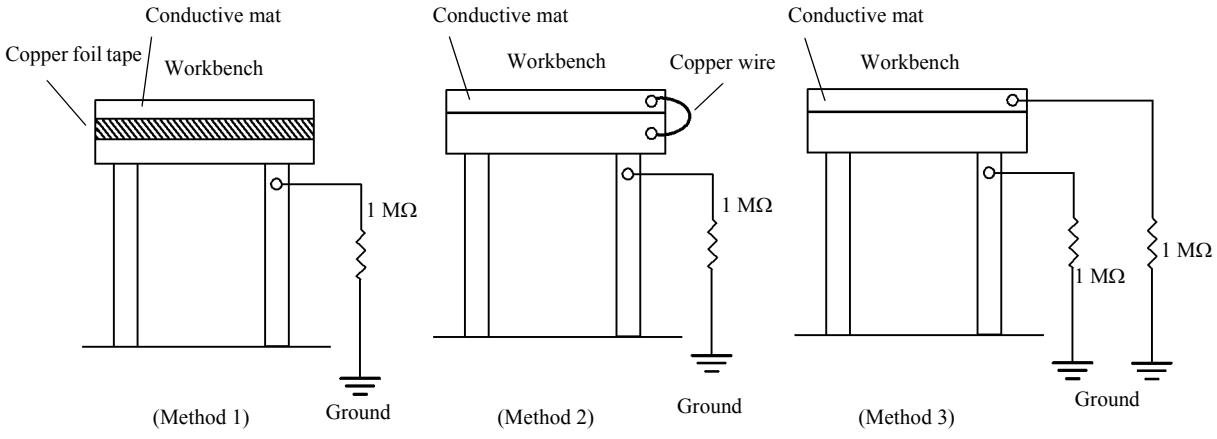


Figure 3-3 Electrostatic Countermeasures for Workbench

3.3 Accessories

The 7355 optional accessories are listed below.

Specify the part number when ordering.

Table 3-3 Optional Accessory List

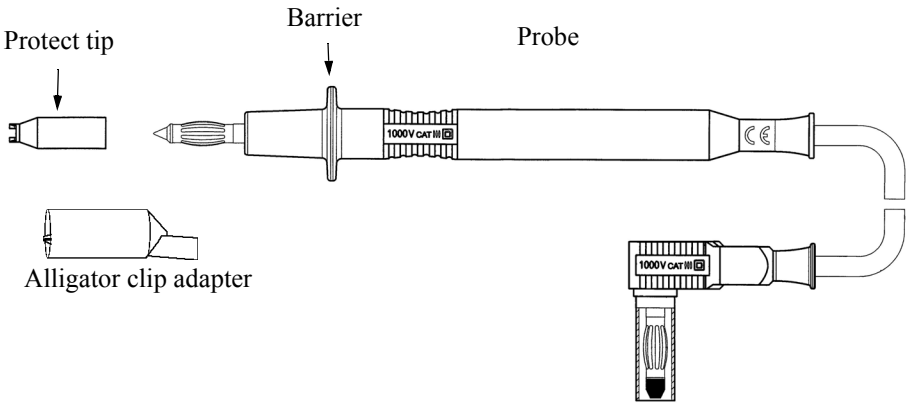
Name	Model	Remarks
Input cable	CC010001	Supplied accessory
	A01001	Shielded cable
	A01006	For 4-wire resistance measurement
Alligator clip adapter	CC015001	
Internal temperature detecting sensor	1104-010	Jpt100, probe type
Sheath-type platinum resistance bulb	1104-001	JPt100, sheath outer diameter: φ3.2 mm
	1104-002	JPt100, sheath outer diameter: φ1.0 mm
JIS rack mount set	A02263	
	A02264	Twin
EIA rack mount set	A02463	
	A02464	Twin
Panel mount set	A02039	
	A02040	Twin

3.3.1 Input Cable

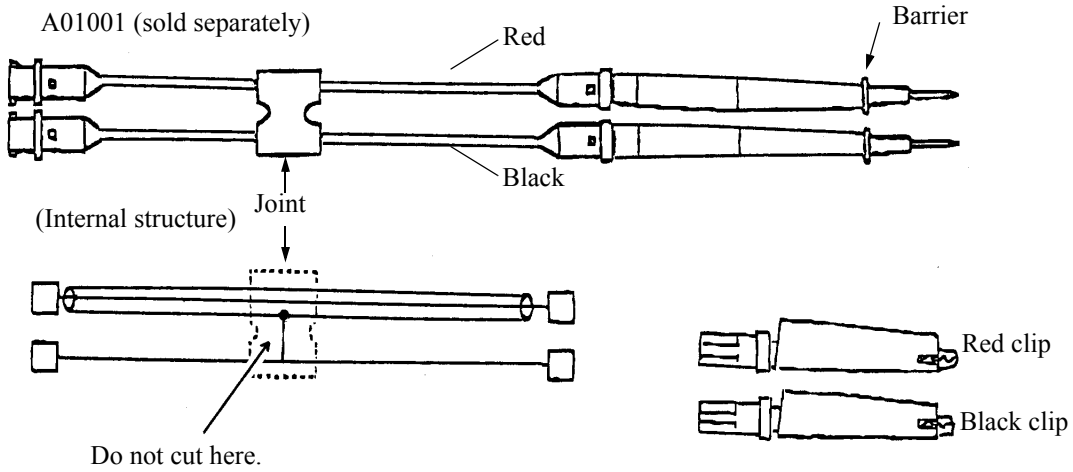
1. The standard input cable (CC010001) consists of red and black and it can be used easily.

CAUTION: Handle the input cable with caution because it has a sharp tip. The rating of the alligator clip adapter is 33 VAC and 70 VDC or less.

WARNING: Do not touch the metal part of the input cable to prevent any damage of electrical shock.



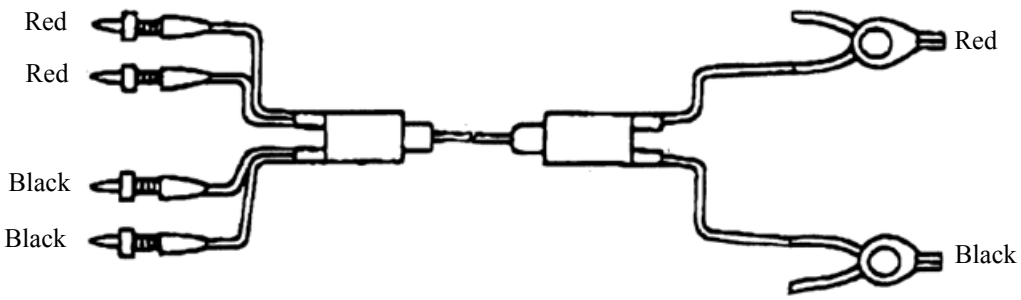
2. To reduce the effects of short-term stability in high resistance (in the order of MΩ) or high sensitivity (in the order of μV) measurements, use the optional input cable A01001 in which the HI side is shielded.



The red wire of A01001 is shielded. Do not cut the joint.
The rating of the probe decreases to 33 VAC and 70 VDC when the alligator clip adapter is attached.

3. Use the dedicated input cable A01006 in the 4WΩ measurement.

A01006 (sold separately)



3.3.2 Caution when Connecting Peripherals

When making up a system, which consists of two or more devices, be careful of the following:

1. Before connecting the 7355 to a controller and peripherals, check that each device operates correctly according to its manual.
2. Do not use too long a cable to connect the instrument or controller.
3. Use shielded cables which are connected between peripherals.

Setting up the USB port

To use the USB interface of this instrument, the USB driver of our ADC instrument must be installed in a personal computer.

The ADC instruments USB driver can be downloaded from the following website.

URL https://www.adcmt.com/en/download/usbdriver

For more information on how to install the driver, refer to the operation manual that is included in the driver file.

3.4 Power Requirements

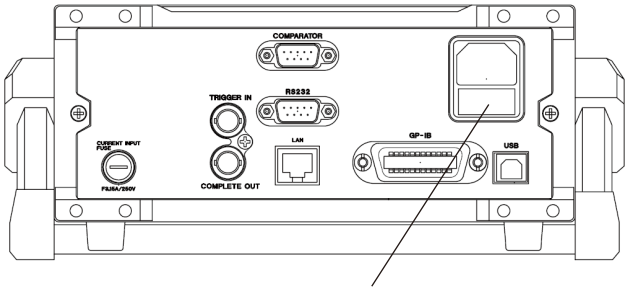
This section describes power requirements and how to connect the power cable.

3.4.1 Checking Power Supply Voltage

Check that the power supply voltage setting on the rear panel of the 7355 accords with the AC power supply voltage to be used.

Table 3-4 Relation Table between AC Power Supply Voltage and Setting Power Supply Voltage

AC power supply voltage	Setting power supply voltage	Applicable fuse	Frequency	Power consumption
90 V - 110 V	100 V	T250 mA/250 V	48 Hz-66 Hz	15 VA or less
103 V - 132 V	120 V			
198 V - 242 V	220 V	T200 mA/250 V		
207 V - 250 V	240 V			



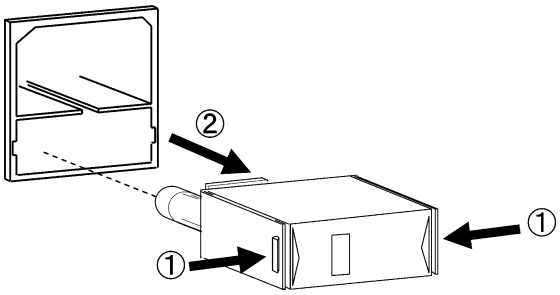
Indication of setting power supply voltage (on rear panel)

Figure 3-4 Indication of Setting Power Supply Voltage

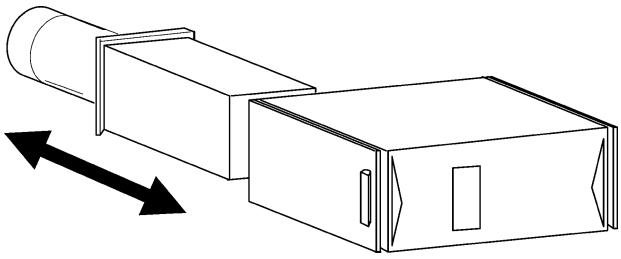
3.4.2 Changing Power Supply Voltage

1. Remove the fuse holder assembly from the rear panel.

   Push both side of the fuse holder assembly by using flathead screwdrivers (①) and then pull it out (②).

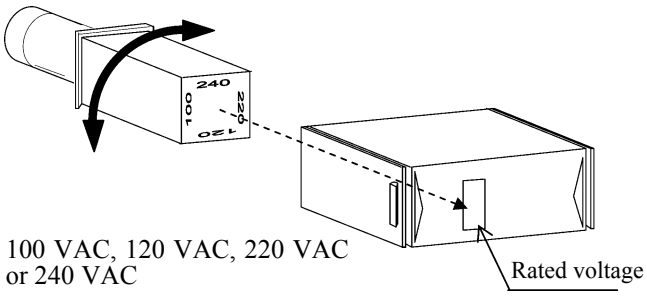


2. Remove the power supply voltage selector from the assembly.



   100 VAC or 120 VAC: T250 mA fuse
   220 VAC or 240 VAC: T200 mA fuse

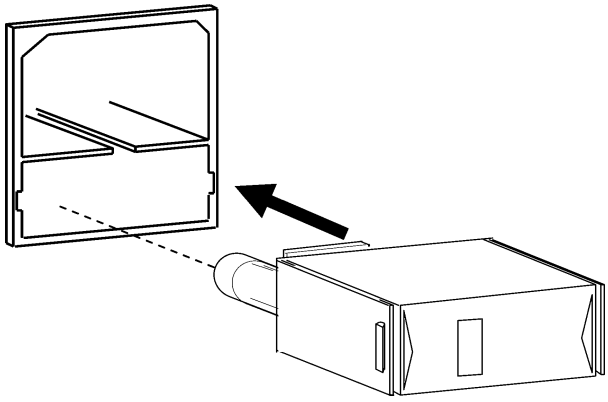
3. Rotate the voltage selector until the correct voltage appears in the window.



   100 VAC, 120 VAC, 220 VAC or 240 VAC

4. Insert a rated fuse (Refer to Table 3-1).

5. Return the fuse holder assembly into the rear panel.



6. Check that the correct power supply voltage is indicated in the window.

3.4.3 Connecting Power Cable

The 7355 is provided with a 3-pin power cable including a ground pin. Always use this power cable to ground the 7355 to a 3-pin power outlet for prevention of electrical shock.

1. Check for any damage to the power cable.

WARNING: Never use a damaged cable. It may result in electric shock.

2. Connect the AC power connector on the rear panel to a 3-pin power outlet including a ground terminal by using the power cable (See Figure 3-5)

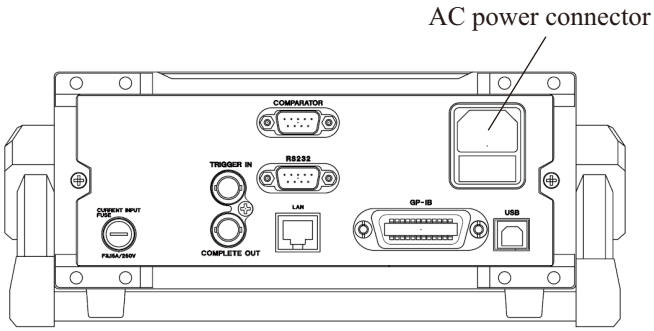


Figure 3-5 Connecting Power Cable

WARNING:

1. *Use a power cable that conforms to the power outlet voltage and type.*
2. *To prevent electric shock, connect the power cable to a 3-pin power outlet with a ground terminal. If an extension cable without a ground terminal is used, the protective ground feature will be rendered ineffective.*

3.4.4 Selecting Power Frequency

The power frequency is set automatically when the power is turned on, depending on your location. To check the power frequency or select it manually, use the following procedure.

Select the following from 'LINE' in the category **9 SYS** in **MENU**.

50 The commercial power frequency is 50 Hz.

60 The commercial power frequency is 60 Hz.

For more information on navigating the menu, refer to Chapter 5, “MENU OPERATION AND FUNCTION DESCRIPTION”.

4. QUICK START

The high-speed measurement and high-accuracy measurement can be performed in this digital multimeter. To make the performing operations easier, the 7355 includes the following functions on the front panel.

Setting the measurement function

Setting the measurement range

Sampling conditions

Setting the NULL and smoothing calculations to ON or OFF

Chapter 4 describes the front and rear panels and panel operations which do not use the menu. Other settings are made in the layered menu. For more information on function descriptions and how to navigate the menu, refer to Chapter 5.

4.1 Panel Descriptions

4.1.1 Front Panel

The front panel consists of the display section, operation keys, and measurement input terminal section.

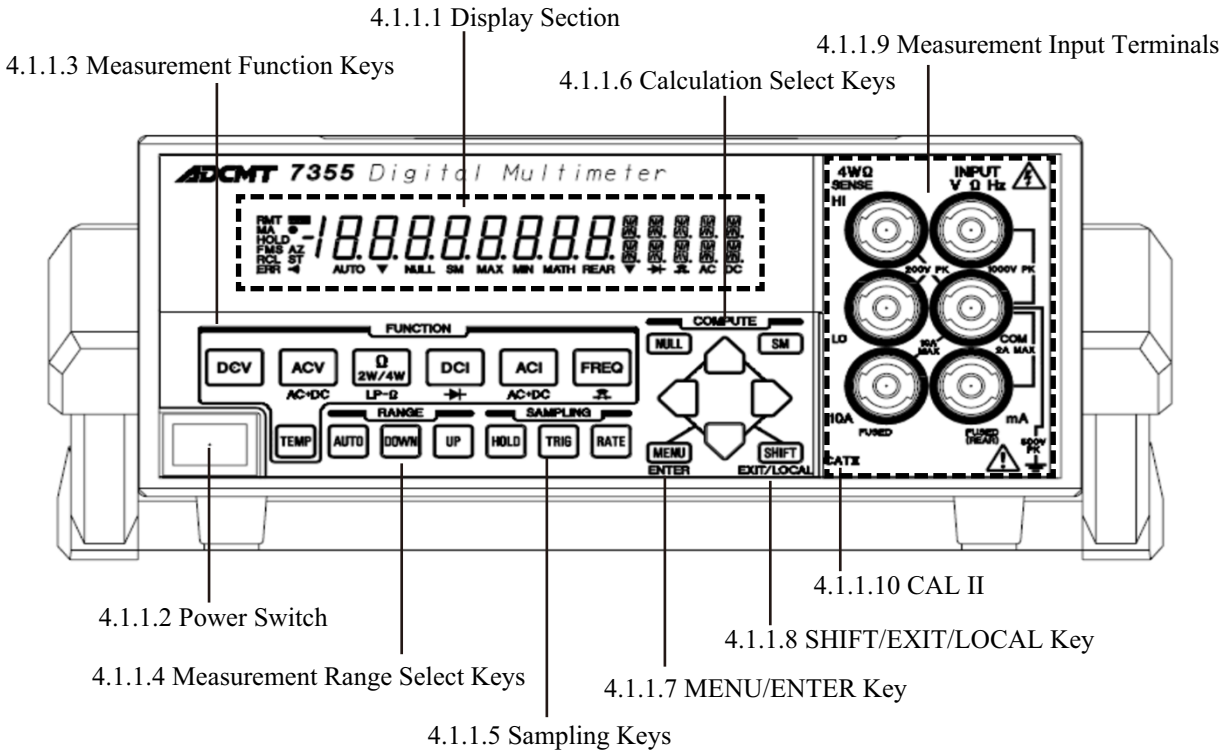


Figure 4-1 Front Panel

4.1.1.1 Display Section

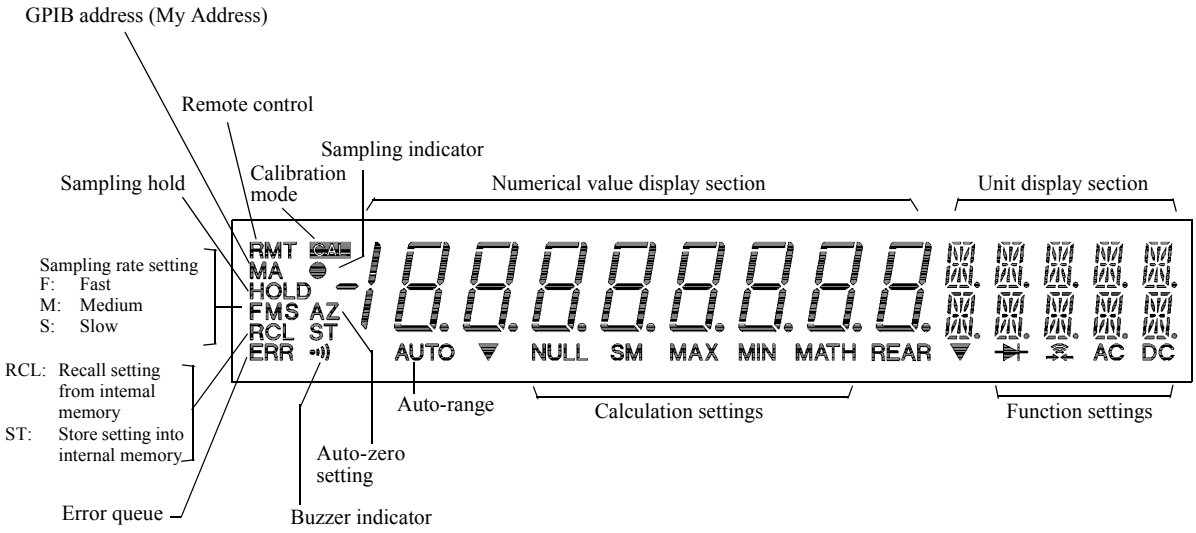


Figure 4-2 Display Section

4.1.1.2 Power Switch

1.	POWER switch	Press the power switch to the “I” position to supply power, and press it to the “O” position to cut off power.

4.1.1.3 Measurement Function Keys

1.	DCV	Selects DC voltage measurement.
2.	ACV	Selects AC voltage measurement.
3.	2WΩ/4WΩ	Selects the resistance measurement. Either 2WΩ or 4WΩ can be selected alternately.
4.	DCI	Selects DC current measurement.
5.	ACI	Selects AC current measurement.
6.	FREQ	Selects frequency measurement.
7.	AC + DC (SHIFT + ACV)	Selects the AC+DC coupling in the AC voltage measurement.
8.	LP-Ω (SHIFT + Ω)	Selects the low power resistance measurement. Either 2WΩ or 4WΩ can be selected alternately.
9.	⊣▶⊢ (SHIFT + DCI)	Selects the diode measurement (⊣▶⊢).

10.	AC + DC (SHIFT + ACI)	Selects the AC+DC coupling in the AC current measurement.
11.	(SHIFT+FREQ)	Selects the continuity measurement ( ).
12.	TEMP	Selects the temperature measurement (TEMP).

4.1.1.4 Measurement Range Select Keys

1.	AUTO	Switches the measurement range between auto-range and fixed range.
2.	DOWN	Decreases the measurement range by one and fixes the range.
3.	UP	Increases the measurement range by one and fixes the range.

4.1.1.5 Sampling Keys

1.	HOLD	Sets the sampling mode to ON (sampling HOLD) or cancel it.
2.	TRIG	Starts the measurement in the sampling HOLD status.
3.	RATE	Set the sampling rate to the high (FAST), medium (MED), or low speed (SLOW1 or SLOW2).

4.1.1.6 Calculation Select Keys

1.	NULL	Sets or cancels the NULL calculation.
2.	SM	Sets or cancels the smoothing calculation.

4.1.1.7 MENU/ENTER Keys

1.	MENU	This key is used to enter the menu.
2.	ENTER	Confirms settings.
3.	△ ▽ ◁ ▷	Selects the menu item and its layer.

4.1.1.8 SHIFT/EXIT/LOCA Keys

1. SHIFT — Sets or cancels the SHIFT mode.

• In the setting mode

2. EXIT — Exits from the menu and returns to the measurement state display. (Data is not saved while changing settings.)

• In the remote operation

3. LOCAL — Used to switch to the local operation.

NOTE: The local operation cannot be set if the LLO (LOCAL LOCKOUT) command is set from the GPIB interface.

4.1.1.9 Measurement Input Terminals

1. INPUT V Ω Hz HI terminal — HI terminal used for the DC voltage, AC voltage, resistance, diode, and frequency measurements (Source HI terminal used for the 4WΩ or temperature measurement)
2. COM terminal — LO terminal used for the DC voltage, AC voltage, DC current, AC current, resistance, diode, and frequency measurements (Source LO terminal used for the 4WΩ or temperature measurement)
3. mA terminal — HI terminal used for the 200 mA and 2000 mA ranges of the DC current and AC current measurements
4. 10 A terminal — HI terminal used for the 10 A range of the DC current and AC current measurements
5. 4WΩ SENSE HI terminal — SENSE HI terminal used for 4WΩ or temperature measurement
6. 4WΩ SENSE LO terminal — SENSE LO terminal used for 4WΩ or temperature measurement

4.1.1.10 CAT II (Marking)

This display shows that the safety of this instrument satisfies the measurement category II which is compliant with IEC 61010-1 Ed.3. Measurement category II is for measurements performed on circuits directory connected to the low voltage installation.

4.1.2 Rear Panel

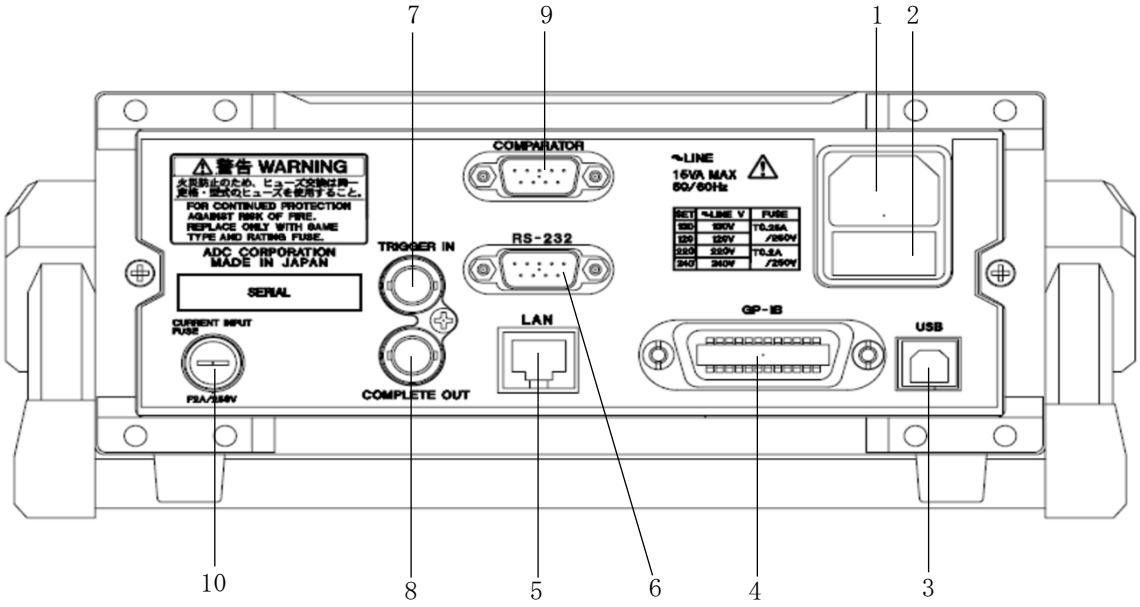


Figure 4-3 Rear Panel

1.	Power connector	Connector for the AC power supply The included power cable is connected here.
2.	Fuse holder assembly	The power supply voltage can be selected from 100 V, 120 V, 220 V, and 240 V. A slow-blow fuse is stored here.
3.	USB connector	Connector for USB Used to output data and control remotely.
4.	GPIB connector (Option +01)	Connector for GPIB Used to output data and control remotely.
5.	LAN connector	Connector for LAN Used to output data and control remotely.
6.	RS232 connector (Option +03)	Connector for RS232 Used to output data and control remotely.
7.	TRIGGER IN (Option +01, +03)	Used to input the external trigger.
8.	COMPLETE OUT (Option +01, +03)	Outputs the measurement complete signal.
9.	COMPARATOR (Option +03)	Outputs the comparator calculation result.
10.	Overcurrent protection fuse	Protects the current input terminal. (200 mA and 2000 mA ranges)

4.2 Basic Operation

When the power is turned on, the 7355 automatically performs the self-test and then proceeds to the measurement sequence.
For more information on the detail of the self-test, refer to Chapter 8, “MAINTENANCE”.
When the self-test is complete correctly, the 7355 enters the measurement state. If any error occurs, it enters the error display state. Check the error number by operating the menu.

For more information on navigating the menu, refer to Section 5.1, “Menu”.

The 7355 saves the set conditions when the power is turned off. Chapter 4 is described assuming factory defaults. For more information on how to initialize the 7355, refer to Chapter 8.

4.2.1 Measurement Functions

4.2.1.1 DC Voltage Measurement (DCV)

1. Connect a DUT between the COM and V terminals on the front panel.
2. Press **DCV**.

WARNING: Do not apply voltage that exceeds the maximum allowable input voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-1 Maximum Allowable Input Voltage (DCV)

V terminal – COM terminal	1000 V peak
COM terminal – chassis (ground)	500 V peak

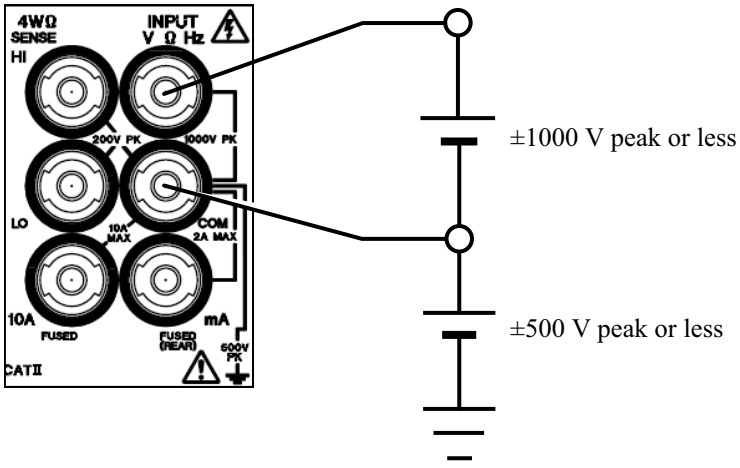


Figure 4-4 DC Voltage Measurement (DCV)

Table 4-2 Measurement Range and Input Impedance (DCV)

Measurement range	Input impedance
200 mV, 2000 mV	1 GΩ or higher
20 V, 200 V, 1000 V	10 MΩ ± 1 %

4.2.1.2 AC Voltage Measurement (ACV, ACV (AC+DC))

For AC voltage superimposed on DC voltage, select either the ACV function, which measures the RMS value of AC voltage only, or the ACV (AC+DC) function, which measures the root mean square value of AC voltage and DC voltage.

1. Connect a DUT between the COM and V terminals on the front panel.
2. <ACV measurement>

   Press **ACV**.

   The RMS value of the AC-coupled AC voltage is measured.

   <ACV (AC+DC) measurement>

   Press **SHIFT** **ACV**.

   ACV and DCV are measured and the root-mean-square value $\sqrt{ACV^2+DCV^2}$ is displayed.

   The AC and DC indicators are activated to show the AC+DC coupling.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-3 Maximum Allowable Input Voltage (ACV, ACV (AC+DC))

V terminal – COM terminal	1000 V peak, 700 V rms, 2.2×10^7 VHz
COM terminal – chassis (ground)	500 V peak

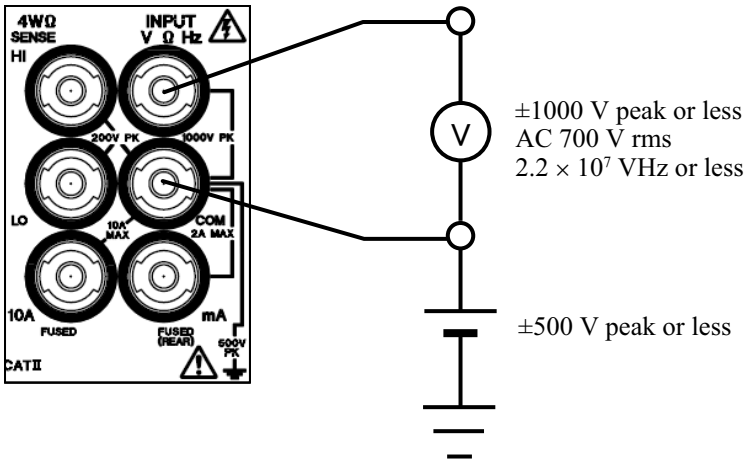


Figure 4-5 AC Voltage Measurement (ACV, ACV (AC+DC))

Measurement method:	True RMS measurement
Input range:	5% or more of a full scale
Crest factor:	3 (Restricted to the maximum allowable input)
Filter response time:	Approx. 1 s (Time until the measured value reaches within 0.1% of the final value in the same range)

4.2.1.3 Resistance Measurement (2WΩ, 4WΩ, LP-2WΩ, and LP-4WΩ)

1. Connect a DUT between the Ω and COM terminals on the front panel.
2. Press **Ω 2W/4W**.
   The resistance is measured by using the 2WΩ or 4WΩ which is set before this measurement. Pressing it again will switch between 2WΩ and 4WΩ.
3. Press **SHIFT** and **Ω 2W/4W**.
   The resistance is measured in low-power in which the measurement current is reduced.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-4 Maximum Allowable Input Voltage (2WΩ , 4WΩ , LP-2WΩ , LP-4WΩ)

Ω terminal – COM terminal	1000 V peak
4WΩHI terminal – 4WΩLO terminal	200 V peak
4WΩHI/LO terminal – COM terminal	200 V peak
COM terminal – chassis (ground)	500 V peak

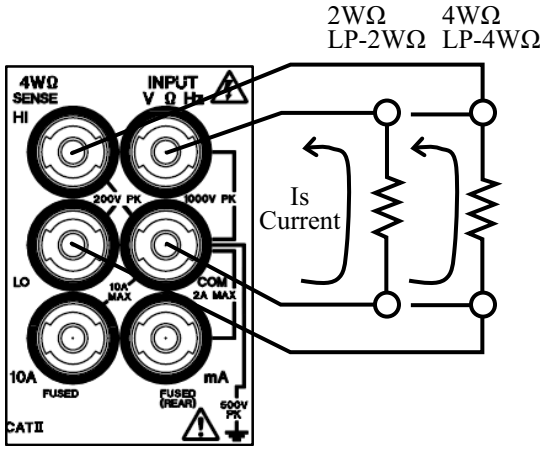


Figure 4-6 Resistance Measurement (2WΩ , 4WΩ , LP-2WΩ , LP-4WΩ)

The result of 2WΩ measurement includes the input cable resistance and contact resistance. To reject these resistance effects, short the tip of the input cable and set the NULL calculation to ON before the measurement.

To reduce the effect of inductive noise in high-resistance measurements, the following precautions should be taken:

- Shield the DUT.
- Use the A01001 input cable (sold separately) whose HI side is shielded.
- Secure the input cable to prevent it from moving.

4.2.1.4 DC Current Measurement (DCI)

1. Input terminals differ depending on the measurement range.
   <200 mA and 2000 mA ranges>
   Connect a DUT between the mA and COM terminals on the front panel.
   <10 A range>
   Connect a DUT between the 10 A and COM terminals on the front panel.
2. Press **DCI**.

Table 4-5 Maximum Allowable Input Current and Protection Function (DCI)

Input terminal	Measurement range	Maximum allowable input current	Protection function
mA	200 mA, 2000 mA	2 A	F2A / 250 V Fuse
10 A	10 A	10 A	F15A / 250 V Fuse

WARNING: Do not measure current whose potential is more than 250 V.
The 7355 may be damaged and injury accidents may be caused.

The mA and 10 A terminals are protected by a fast-acting fuse against overcurrent input.
For more information on how to replace the overcurrent protection fuse, refer to Chapter 8, "MAINTENANCE".

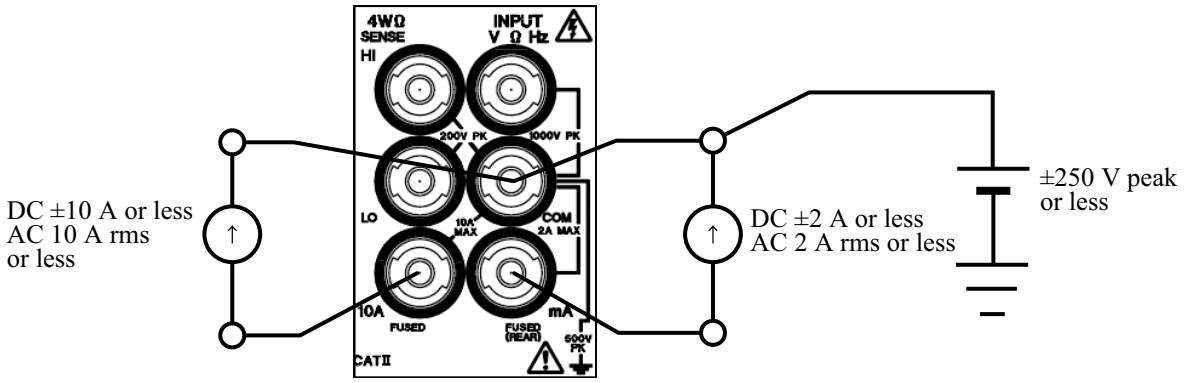


Figure 4-7 DC Current Measurement (DCI)

Auto-range restrictions

The 10 A range is a fixed range. The auto-range cannot function between the 10 A range and 200 mA or 2000 mA range.

NOTE: Applying current causes the temperature of the input terminals and internal circuits to rise, generating a thermoelectromotive force.
When making other function measurements, especially high-sensitivity measurements, after applying current, allow time for them to stabilize.

4.2.1.5 AC Current Measurement (ACI and ACI (AC+DC))

For AC current superimposed on DC current, select either the ACI function, which measures the RMS value of AC current only, or the ACI (AC+DC) function, which measures the root mean square value of AC current and DC current.

1. Input terminals differ depending on the measurement range.

   <200 mA and 2000 mA ranges>

   Connect a DUT between the mA and COM terminals on the front panel.

   <10 A range>

   Connect a DUT between the 10 A and COM terminals on the front panel.

2. <ACV measurement>

   Press **ACI**.

   The RMS value of the AC-coupled AC current is measured.

   <ACI (AC+DC) measurement>

   Press **SHIFT** **ACI**.

   ACI and DCI are measured and the root-mean-square value $\sqrt{ACI^2+DCI^2}$ is displayed.

   The AC and DC indicators are activated to show the AC+DC coupling.

Table 4-6 Maximum Allowable Input Current and Protection Function (ACI, ACI (AC+DC))

Input terminal	Measurement range	Maximum allowable input current	Protection function
mA	200 mA, 2000 mA	2 A	F2A / 250 V Fuse
10 A	10 A	10 A	F15A / 250 V Fuse

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage.
Do not measure current whose potential is more than 250 V.
Fire or electrical shock due to the failure of this instrument may occur.

The mA and 10 A terminals are protected by a fast-acting fuse against overcurrent input.
For more information on how to replace the overcurrent protection fuse, refer to Chapter 8, “MAINTENANCE”.

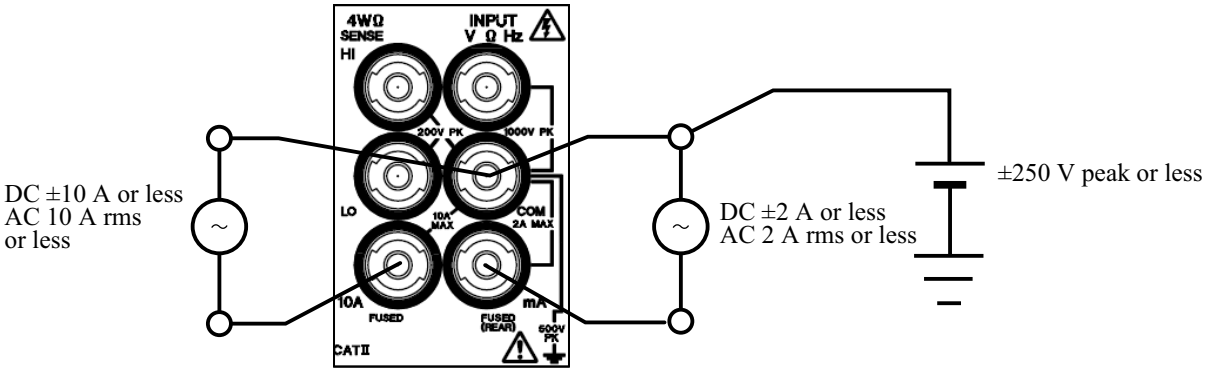


Figure 4-8 AC Current Measurement (ACI, ACI (AC+DC))

Measurement method:	True RMS measurement
Input range:	5% or more of a full scale
Crest factor:	3 (Restricted by the overcurrent protection fuse)
Filter response time:	Approx. 1 s (Time until the measured value reaches within 0.1% of the fin al value in the same range)

Auto-range restrictions

The 10 A range is a fixed range. The auto-range cannot function between the 10 A range and 200 mA or 2000 mA range.

4.2.1.6 Continuity Measurement

1. Connect a DUT between the Ω and COM terminals on the front panel.
2. Press **SHIFT** and **FREQ**.

   The continuity indicator ( ) is activated.

   The resistance is measured in 2WΩ. If the measured resistance is lower than the continuity threshold constant, the buzzer sounds.

Measurement range:	2000 Ω
Measurement current:	1 mA
Continuity threshold constant:	1 Ω to 1000 Ω

WARNING: Do not apply voltage that exceeds the maximum allowable iput voltage. Fire or electric shocks due to the failure of this instrument may occur.

Table 4-7 Maximum Allowable Input Voltage (Continuity)

Ω terminal – COM terminal	1000 V peak
COM terminal – chassis (ground)	500 V peak

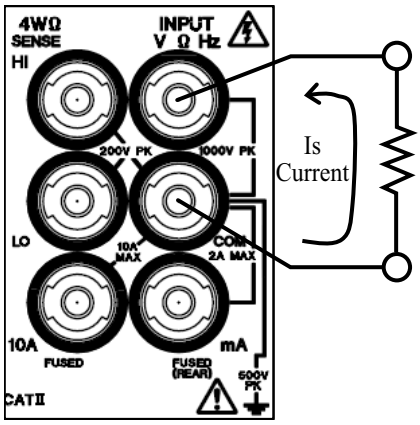


Figure 4-9 Continuity Measurement

4.2.1.7 Diode Measurement

1. Connect a DUT between the V and COM terminals on the front panel.
2. Press **SHIFT** and **DCI**.

   The diode indicator (⊣▶⊢) is activated.

   A current of approximately 1 mA is applied from the COM terminal to the V terminal, and the voltage generated between these terminals is measured.

   Measurement range: 2000 mV

   Measurement current: 1 mA

WARNING: Do not apply voltage that exceeds the maximum allowable input voltage. Fire or electric shocks due to the failure of this instrument may occur.

Table 4-8 Maximum Allowable Input Voltage (Diode)

Ω terminal – COM terminal	1000 V peak
COM terminal – chassis (ground)	500 V peak

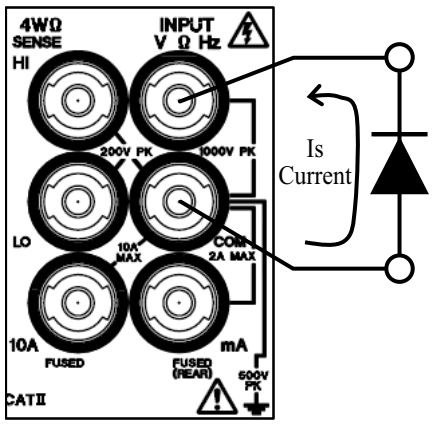


Figure 4-10 Diode Measurement

4.2.1.8 Frequency Measurement (FREQ)

1. Connect a DUT between the V and COM terminals on the front panel.
2. Press **FREQ**.

   The frequency in the ACV measurement function is displayed.

   The DC component of the input signal is cut by using the AC coupling and the frequency is measured at the zero-cross point.

   Measurement method: Reciprocal

   Input voltage range: 100 mV rms to 700 V rms and 10 % or more of the range
   The range does not mean the frequency range but the input voltage range.
   Measure the input voltage by using the ACV function and then measure the frequency by using an appropriate voltage range depending on the input voltage. Even if the input voltage exceeds the measurement range, the measurement is performed but the measured value may deviate from the accuracy specifications.

   Gate time

Sampling rate	Gate time	Measurement frequency range	Measurement period	Maximum display	
				5 1/2 digits	6 1/2 digits
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	999999	9999999
MED	100 ms	10 Hz to 300 kHz	220 ms	99999	999999
FAST	10 ms	100 Hz to 300 kHz	22 ms	9999	99999

If the input signal cannot be detected because the period of the input signal is longer than the gate time, 0 Hz is displayed.

NOTE: When the sampling rate is set to SLOW, the sampling rate indicator alternates between the dark "S" and bright "S".
When moving to the frequency function with the integration time selected without setting the sampling rate in any other function, the sampling rate is set to MED. When moving to another function again, set the integration time again.

WARNING: Do not apply voltage that exceeds the maximum allowable input voltage. Fire or electric shocks due to the failure of this instrument may occur.

Table 4-9 Maximum Allowable Input Voltage (FREQ)

V terminal – COM terminal	1000 V peak, 700 V rms, 2.2×10^7 VHz
COM terminal – chassis (ground)	500 V peak

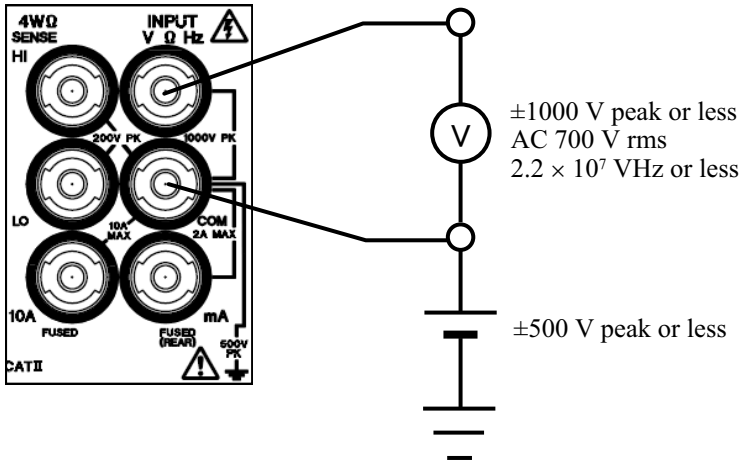


Figure 4-11 Frequency Measurement (FREQ)

4.2.1.9 Temperature Measurement (PT100, JPT100)

1. Connect a temperature sensor between the Ω and COM terminals on the front panel.
2. Press **TEMP**.
   The temperature is measured by using the 2W, 3W or 4W (formerly set method).
   The measurement conductor type for platinum resistance thermometer is set with "'WIRE" of the **10 TEMP** category in the **MENU**.

WIRE	Measurement conductor type
2W	2-wire measurement
3W	3-wire measurement
4W	4-wire measurement

   The JIS standard for platinum resistance thermometer is set with "P.TYPE" of the **10 TEMP** category in the **MENU**.

P.TYPE	JIS standard
PT	Pt100 (new JIS)
JPT	JPt100 (old JIS)

   The temperature unit is set with "UNIT" of the **10 TEMP** category in the **MENU**.

UNIT	Temperature unit
°C	Celsius temperature
°F	Fahrenheit temperature
K	Kelvin scale

   The measurement resolution is decided depending on the sampling rate.
   The sampling rate is switched in order by pressing the **RATE**.
   Choose the sampling rate according to the resolution required.

Sampling rate	Integration time	Measurement resolution
FAST	2 ms	0.1 °C
MED	1 PLC	0.001 °C
SLOW1	100 ms	0.001 °C
SLOW2	200 ms	0.001 °C

CAUTION: When leaving the temperature measurement function, the Auto-range of 2WΩ and 4WΩ is turned ON.

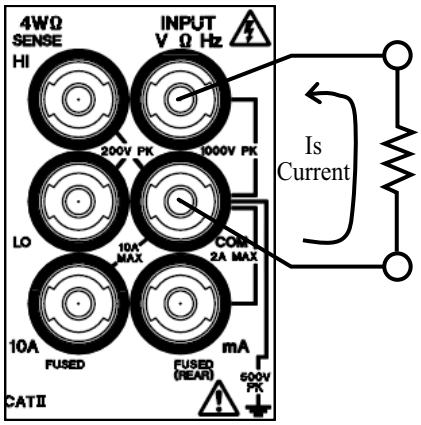


Figure 4-12 Temperature Measurement (PT 2W)

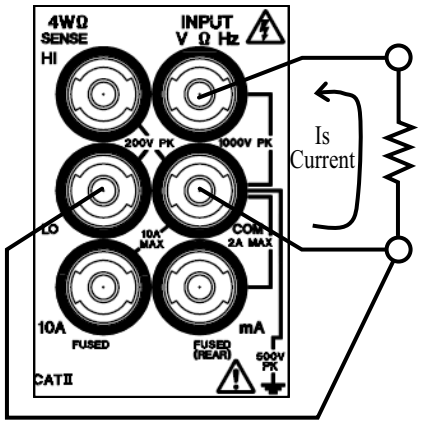


Figure 4-13 Temperature Measurement (PT 3W)

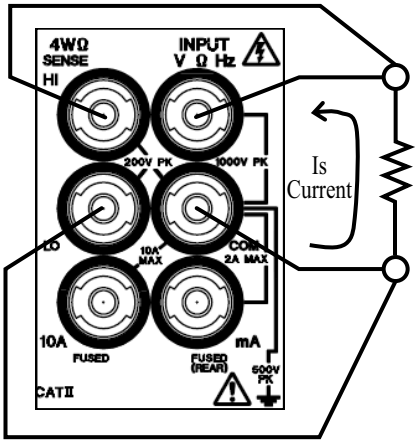


Figure 4-14 Temperature Measurement (PT 4W)

4.2.2 Setting Measurement Range

UP Increases the measurement range by one and fixes the range.

DOWN Decreases the measurement range by one and fixes the range.

AUTO Switches the measurement range between auto-range and fixed range.

4.2.3 Changing Measurement Speed and Display Digit

Press **RATE**.

The measurement speed and the number of display digits are switched as shown in Table 4-10 and Table 4-11.

In the frequency measurement, the sampling rate is used to set the gate time. For more information on the frequency measurement, refer to Section 4.2.1.8, “Frequency Measurement (FREQ)”.

Table 4-10 Sampling Rate and Measurement Speed

Sampling rate	FMS indicator	Measurement speed (Measurement time)				Integration time
		DCV DCI 2WΩ LP-2WΩ TEMP (PT-2W)	ACV ACI Continuity Diode	ACV (AC+DC) ACI (AC+DC)	4WΩ LP-4WΩ TEMP (PT-3W) TEMP (PT-4W)	
		Auto-zero ON	Auto-zero OFF			
FAST	F	69.4 times/s (14.4 ms)	149.2 times/s (6.7 ms)	30.3 times/s (33 ms)	56.1 times/s (17.8 ms)	2 ms
MED	M	19.8 times/s (50.4 ms)	40 times/s (25 ms)	20 times/s (50 ms)	18.5 times/s (53.8 ms)	1 PLC
SLOW1	S (bright)	4.7 times/s (210.4 ms)	9.5 times/s (105 ms)	4.7 times/s (210 ms)	4.6 times/s (214 ms)	100 ms
SLOW2	S (dark)	2.4 times/s (410.4 ms)	4.8 times/s (205 ms)	2.4 times/s (410 ms)	2.4 times/s (414 ms)	200 ms

Conditions: Auto-range: OFF, Calculation: OFF, and Trigger source: IMMED

Table 4-11 Sampling Rate and Number of Display Digits

Sampling rate	FMS indicator	DCV DCI 2WΩ LP-2WΩ 4WΩ LP-4WΩ TEMP Continuity Diode	ACV ACI	ACV (AC+DC) ACI (AC+DC)
FAST	F	4 1/2-digit display	4 1/2-digit display	3 1/2-digit display
MED	M	6 1/2-digit display	6 1/2-digit display	4 1/2-digit display
SLOW1	S (bright)	6 1/2-digit display	6 1/2-digit display	4 1/2-digit display
SLOW2	S (dark)	6 1/2-digit display	6 1/2-digit display	4 1/2-digit display

Also refer to Section 5.7, “Sampling Operation”.

Sampling HOLD setting

HOLD Sets the sampling mode to ON (sampling HOLD) or OFF.

TRIG Starts the measurement in the sampling HOLD status.

5. MENU OPERATION AND FUNCTION DESCRIPTION

5.1 Menu

The functions and conditions in the 7355 are set by using a layered menu.

5.1.1 How to Navigate Menu

The menu has a three-layer structure. (The menu partially has a four-layer structure.)

First layer	Category layer	Select categories in the menu.
Second layer	Select layer	Select parameters to set within the category.
Third layer	Input/Run layer	Input or select the set value.

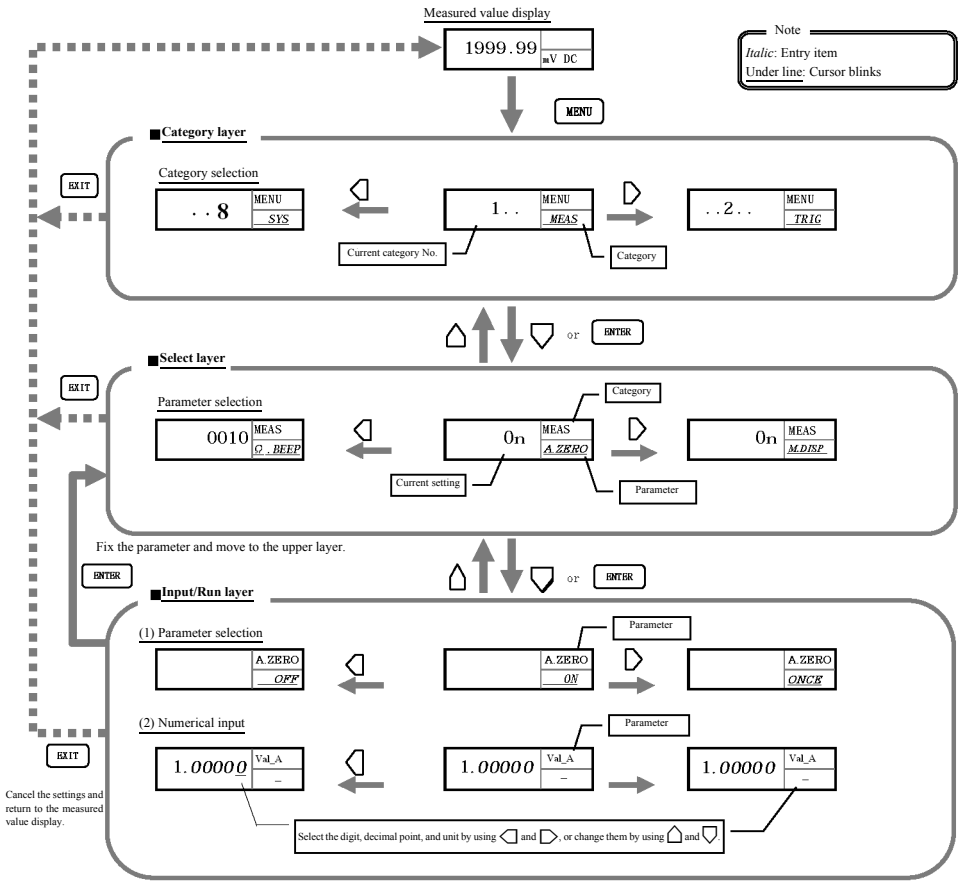


Figure 5-1 Menu Operation

1. Press **MENU**.
   Categories from **1 MEAS** to **10 TEMP** are displayed.
2. Select the category by using ◁ and ▷.
3. Press ▽.
   The Select layer is displayed.
4. Select the parameter by using ◁ and ▷.
5. Press ▽.
   The Input/Run layer is displayed.
6. Set the parameter by using △, ▽, ◁, ▷, and the **TRIG** key.
7. Press **ENTER** to fix the entered parameter and moves to the Selection layer.
   Pressing **EXIT/LOCAL** before conforming cancels the entered settings exits the menu.
   "DONE" after "BUSY" is displayed during the process.

Setting the selectable parameters

For the selectable parameter that is set by selecting the item, select the parameter by using ◁ and ▷, and determine it by pressing **ENTER**.

Setting the numerical parameters

For the numerical parameter that is set by entering the numerical value, select either numerical digit, decimal point position, or unit prefix by using ◁ and ▷.

The operation of △ and ▽ varies depending on the target item.

If there is no target item that blinks, move to the Selection layer by using △.

If a number blinks, the numerical value can be increased or decreased.

If a decimal point blinks, the decimal point can be moved.

If the unit display section blinks, the unit prefix can be changed in the following order: p, n, μ, m, none, k, M, G, and T

Pressing the **TRIG** key performs the measurement and can set the result as the value of the parameter.

Although any numerical value can be entered into the numerical parameter, it is rounded off to the resolution digit of each parameter.

Table 5-1 shows the key function in each layer.

Table 5-1 Menu Operations and Key Functions

	Category layer	Select layer	Input/Run layer		
			Numerical parameter		Selectable parameter/execution item
			Parameter: Dark display, No blinking	Target items: Numerical value/ decimal point/unit	
△	-	To the Category layer	To the Select layer	Increases the value	To the Select layer
▽	To the Select layer	To the Input/Run layer	-	Decreases the value	-
◁	To the left category	To the left parameter	To the target number	To the left target	Displays the previous parameter
▷	To the right category	To the right parameter	To the target unit	To the right target	Displays the next parameter
TRIG	-	-	Runs the measurement and reflects on the parameter value *1		-
Enter key	To the Select layer	To the Input/Run layer	Fixes the value and moves to the Select layer		Fixes the parameter and moves to the Select layer *2
HOME key	Exits the menu	Exits the menu	Exits the menu		

*1: Only the constant that is used to calculate the HIGH, LOW, and NULL values
*2: For the execution item, quits the menu.
Example: Parameter SAVE/LOAD

5.1.2 Menu List

MENU

Category layer	Select layer	Input/Run layer	Default	Type
1 MEAS (Measure)	A.ZERO	Auto-zero OFF/ON/ONCE	ON	Select
	IT SEC	Integration time (s) 200 µs to 2 s	200 ms	Input
	IT PLC	Integration time (PLC) 0.02 PLC to 100 PLC	10 PLC	Input
	M.DISP	Measured value display OFF/ON	ON	Select
	DIGIT	Measurement display digits 1999999 (6 1/2) to 1999___ (3 1/2)	6 1/2	Select
	Ω.BEEP	Continuity measurement function threshold 1 Ω to 1000 Ω	10	Input
2 TRIG (Trigger)	DELAY	Trigger delay 0 s to 3600 s	0 s	Input
	SRC	Trigger source IMMED/MAN/EXT/BUS	IMMED	Select
	N.Smpl	Number of sampling 1 to 20000	1	Input
	SI	Sampling interval 4 ms to 3600 s	205 ms	Input
	INITC *1	INITiate:CONTinus OFF/ON	ON	Select
3 CALC (Calculation)	COMP	Comparator calculation OFF/ON	OFF	Select
	MATH	Other calculations OFF/dB/dBm/SCL	OFF	Select
	MX/M	MAX/MIN and display OFF/ON/MAX/MIN	OFF	Select
	RSLT	MAX/MIN calculation result display SAMPL/MAX/MIN/AVG	SAMPL	Select
	PAS	Pass condition LGH/__H/_G_/L__/L_H/LG_/_GH/___	_G_	Select
	BEEP	Calculation result buzzer OFF/FAIL/PASS	OFF	Select
	SM.CNT	Smoothing count 2 to 100 (times)	10	Input
	D.OUT	Digital output OFF/ON	OFF	Select
4 CONST (Constant)	HIGH	Constant HIGH -9999.999E+9 to +9999.999E+9		Input
	LOW	Constant LOW -9999.999E+9 to +9999.999E+9	0	Input
	⊿NULL	Constant NULL (For only NULL ON) -9999.999E+9 to +9999.999E+9	0	Input
	Val_A	Constant A (scaling calculation) -9999.999E+9 to +9999.999E+9	1	Input
	Val_B	Constant B (scaling calculation) -9999.999E+9 to +9999.999E+9	0	Input
	Val_C	Constant C (scaling calculation) -9999.999E+9 to +9999.999E+9	1	Input
	ValDB	Constant D (dB/dBm calculation) -9999.999E+9 to +9999.999E+9	1	Input

(a)

(a)

5 PARAM (Parameter)

PON.LD	Parameters to be loaded at power ON P.OFF/USER0	P.OFF	Select
SAVE	File to save parameters USER0 to USER3 (ENTER)		Run
LOAD	File to load parameters USER0 to USER3/DFLT (ENTER)		Run

6 INIT (Initial)

PARAM	Parameter initialization including interface DFLT (ENTER)		Run

7 MEM (Memory)

STORE	Measurement data memory store OFF/ON	OFF	Select
RCL	Measurement data memory recall number 0 to 19999	0	Input
STAT	Statistical calculations and measurement result readings		Run

8 I/F (Interface)

BUS	Interface USB/GPIB/RS232/LAN/CDC	USB	Select
GP.Adr *2	GPIB address 0 to 30	1	Input
USBid *3	USB ID 1 to 127	1	Input
BAUD *4	Baud rate 300/600/1200/2400/4800/9600/19200	9600	Select
DATA *4	Data length 8 bits/7 bits	8 bit	Select
P-Bit *4	Parity NONE/EVEN/ODD	NONE	Select
STOP *4	Stop bit 1 bit/2 bits	1 bit	Select
ECHO *4	Echo OFF/ON	OFF	Select
IP.Adr *5	IP address 0.0.0.0 to 255.255.255.255	192.168.0.1	Input
G.WAY *5	Default gateway 0.0.0.0 to 255.255.255.255	192.168.0.0	Input
S.MASK *5	Subnet mask 0.0.0.0 to 255.255.255.255	255.255.255.0	Input
HEADR *6	Header OFF/ON	ON	Select
DFORM	Output data format ASCII/REAL	ASCII	Select
T.Only *2,*4	Talk Only OFF/ON	OFF	Select
LANG	Command language ADC/34401/SCPI	ADC	Select
CONT *7	Continue OFF/ON	OFF	Select
R.DIG	Number of measurement data output digits 6.5 (6 1/2) / 5.5 (5 1/2)	6 1/2	Select

(b)

(b)

9 SYS (System)

BEEP	Buzzer OFF/ON	OFF	Select
TEST	Self-test (ENTER)		Run
P/KEY	Panel/key test (ENTER)		Run
C.Widt *8	Complete signal width 5 µs/100 µs/1 ms	100 µs	Select
E.EDGE *8	External trigger polarity NEG/POS	NEG	Select
F.Inhi	Function enable/disable ENA (enabled)/DIS (disabled) for each function	ENA	Select
P.Lock	Panel lock OFF/ON	OFF	Select
PWD	Password 0000 to 9999	0000	Input
LINE	Power frequency 50 Hz/60 Hz	50 Hz	Select
MACH	Mold name and option	7355	Display
CAL	CAL mode OFF/ON/EXIT	OFF	Select
SER.No.	Serial No.		Display
QUEUE	Error log No.01 to No.20		Select
HIST	Event history No.01 to No.20		Select
735	7351-compatible mode OFF/ON	OFF	Select

10 TEMP (Temperature)

WIRE	Measurement conductor type of platinum resistance thermometer 2W/3W/4W	2W	Select
P.TYPE	Platinum resistance thermometer PT/JPT	PT	Select
UNIT	Temperature unit °C/°F/K	°C	Select

*1 Not displayed when the command language is set to ADC.
*2 Enabled when GPIB (Option +01) is installed and the interface is set to GPIB.
*3 Enabled when the interface is USB.
*4 Enabled when RS232 (Option +03) is installed and the interface is set to RS232.
*5 Enabled when the interface is LAN.
*6 Enabled when the command language is set to ADC.
*7 Enabled when the command language is set to SCPI.
*8 Enabled when the Option +01or Option +03 is installed.

5.2 Measurement Functions

For more information on the measurement function description, refer to Chapter 4.

This chapter describes the parameters of the measurement functions that are set in the menu.

5.2.1 Continuity Measurement

Setting the continuity threshold constant

Set the continuity threshold constant between 1 Ω and 1000 Ω by 'ΩBEEP' in category **1 MEAS** in **MENU**.

Press **ENTER** to apply the setting.

5.3 Measurement Range and Auto Range

5.3.1 Measurement Function and Range Structure

The following table shows the range structure corresponding to the measurement function.

Table 5-2 Measurement Function and Range Structure

DCV	ACV ACV (AC+DC)	2WΩ 4WΩ	LP-2WΩ LP-4WΩ	DCI	ACI ACI (AC+DC)	FREQ	Continuity	Diode
200 mV	200 mV	200 Ω	200 Ω			200 mV		
2000 mV	2000 mV	2000 Ω	2000 Ω			2000 mV	2000 Ω	2000 mV
20 V	20 V	20 kΩ	20 kΩ			20 V		
200 V	200 V	200 kΩ	200 kΩ	200 mA	200 mA	200 V		
1000 V	700 V	2000 kΩ	2000 kΩ	2000 mA	2000 mA	700 V		
		20 MΩ	20 MΩ	10 A	10 A			
		200 MΩ						

NOTE: The range settings are saved for each function.
For the frequency function, the range is not switched in the frequency range but switched in the voltage range.

5.3.2 Auto Range

When the Auto range is set to ON, the measurement range is set to the optimum range.

Judgment value for range change

Range down	Less than 9 % of the range, less than 180 V for the 1000 V and 700 V ranges
Range up	When the maximum display is exceeded

The Auto range is turned OFF under the following conditions:

- Frequency measurement
- Diode measurement
- Continuity measurement
- 10 A range for the DC and AC current measurements
- Temperature measurement

5.4 Auto Zero

The Auto zero function eliminates the offset error in the measurement system of the 7355.

When the Auto zero is ON, the internal offset is measured and its value is subtracted from the measured value of the input signal, thus eliminating the offset error. However, the measurement time is approximately twice as long.

Set the Auto-zero by 'A.ZERO' in **1 MEAS** in **MENU**.

The Auto zero is executed at the range under measurement for the following measurement functions and ranges.

Function	Auto zero	Indicator
DCV	Executed	ON
ACV ACV (AC+DC)	Not executed	OFF
2WΩ LP-2WΩ	Executed	ON
4WΩ LP-4WΩ	Not executed	OFF
DCI	Executed	ON
ACI ACI (AC+DC)	Not executed	OFF
FREQ	Not executed	OFF
Diode	Executed	ON
Continuity	Executed	ON
TEMP-2W	Executed	ON
TEMP-3W TEMP-4W	Not executed	OFF

The Auto zero can be performed only once according to the measurement condition or measurement timing.

When set to ONCE, the Auto zero is executed only once to eliminate the error, and then Auto zero is turned off.

5.5 Display OFF Setting

Setting the display OFF

Select either ON or OFF by 'M.DISP' in **1 MEAS** in **MENU**.

Press **ENTER** to apply the setting.

A time of approximately 1 ms is required for the output to be displayed. (The time may vary depending on conditions.)

Select OFF for high-speed sampling.

Errors are displayed even when the display is set to OFF.

5.6 Number of Measurement Display Digits Setting

Setting the number of measurement display digits

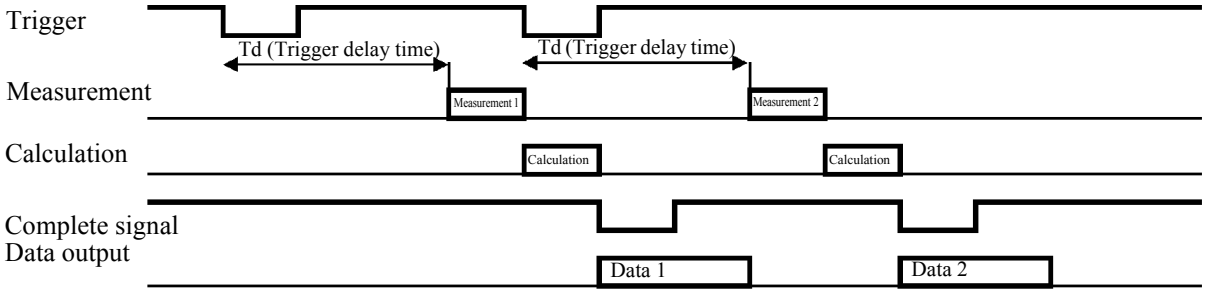
Select the number of measurement display digits by 'DIGIT' in **1 MEAS** in **MENU**.

Press **ENTER** to apply the setting.

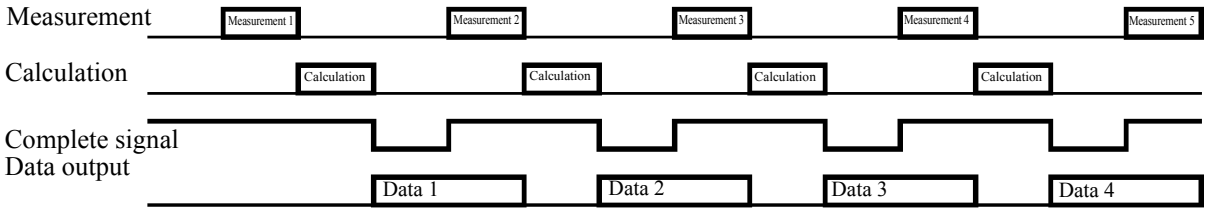
This setting is valid only for the display unit and does not affect the remote output.

5.7 Sampling Operation

Figure 5-2 shows the measurement sequence of the 7355.
A trigger event triggers a single measurement.
At the end of each measurement, a complete signal is output and data output begins.
The measurement data is output on the display or via the interface.
The complete signal output is supported by OPT+01 and OPT+03.



(a) Sampling HOLD/Trigger (MAN, BUS, EXT)



(b) Sampling RUN (IMMED)

Figure 5-2 Measurement Sequence

5.7.1 Variable Integration Function

Besides the **RATE** key on the front panel, the integration time and the sampling interval can be set as desired by using the menu.
When the integration time or sampling interval is set from the menu, the FMS indicator on the display turns off. The number of display digits varies depending on the integration time.

The integration time can be set not only in the unit of seconds but also in the unit of PLC that is synchronized with the power supply frequency.

The following table shows the setting range and resolution of the integration time (IT).

Table 5-3 Integration (IT) Setting Range and Resolution

Setting range	Resolution
200 µs to 2 s	100 µs
0.02 PLC to 100 PLC	0.01 PLC

Set the integration time according to the following procedure.

Enter the integration time by 'IT PLC' in **1 MEAS** in **MENU**.

Change the digit with ◁ and ▷ and the decimal point or subunit with △ and ▽.

Press **ENTER** to apply the setting.

If the set value is less than or equal to the resolution, the value is rounded to the nearest one digit below the resolution.

The following table shows the setting range of the sampling interval.

Table 5-4 Sampling Interval (SI) Setting Range

Setting range
4 ms to 3600 s

The sampling interval (SI) can be set arbitrarily, but must be set to a value greater than the minimum measurement time shown in Table 5-5.

When it is set below the minimum measurement time, the 7355 operates with the minimum measurement time.

Table 5-5 Integration Time and Minimum Measurement Time

Integration time	Minimum measurement time
200 µs ≤ IT ≤ 2 s	IT + 3.7 ms

Set the sampling interval according to the following procedure.

Enter the sampling interval by 'SI' in **2 TRIG** in **MENU**.

Change the digit with ◁ and ▷ and the decimal point or subunit with △ and ▽.

Press **ENTER** to apply the setting.

If the set value is less than or equal to the resolution, the value is rounded to the nearest one digit below the resolution.

5.7.2 Sampling Count Setting

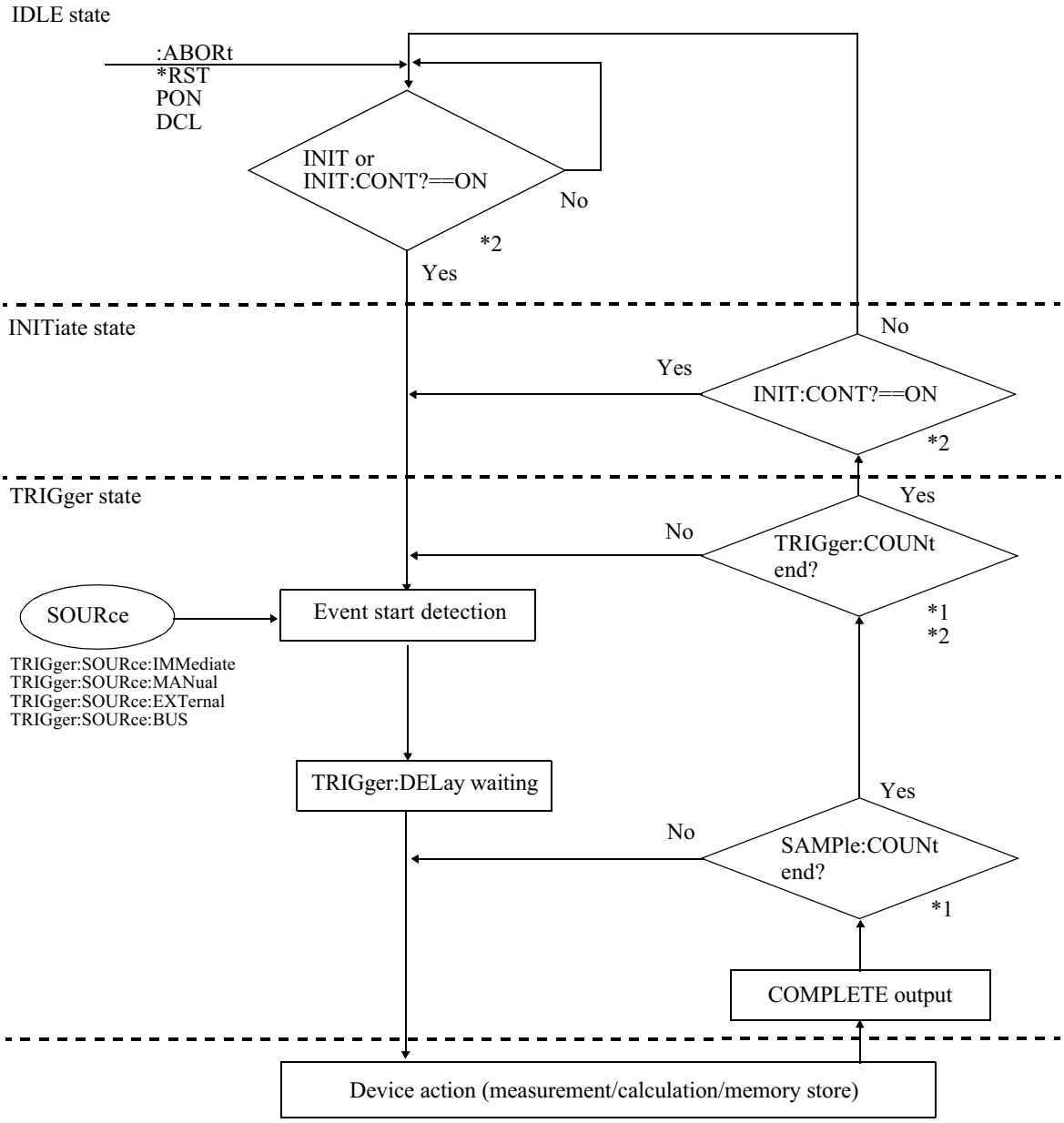
Normally sampling is performed once each time a trigger is detected, but sampling can also be performed multiple times.

Enter the number of sampling by 'N.Smpl' in **2 TRIG** in **MENU**.

Sampling count: 1 to 20000 samples/trigger

5.8 Trigger Function

The trigger system of the 7355 allows trigger detection, the sampling start to be delayed and multiple sampling by a single trigger.
The following figure shows the trigger model.



*1: Disabled when IMMediate is selected for the trigger source. (Always “Yes”)

*2: Disabled in the LOCAL mode. (Always “Yes”)

Figure 5-3 Trigger Model

5.8.1 Trigger System Operation

1. IDLE state

   The trigger system is idle when no processing is being performed on it, such as before the trigger sequence starts, after the trigger sequence ends, and at reset.

2. INITiate state

   The INITiate state means that the trigger system operation has started.

   The IDLE state is changed to the INITiate state by activating the trigger system or by turning on the continue setting.

   If the continue setting is turned on when the measurement is finished, it does not return to the IDLE state, but goes back to the INITiate state and waits for the next trigger.

3. TRIGger waiting state

   When a trigger event is detected from the selected trigger source, the device action starts after the trigger delay time.

   The complete signal is output, and after sampling is repeated a specified number of times, the trigger system returns to the INITiate state.

4. Device action

   The measurements and calculations are performed.

5.8.2 Trigger Source Selection

Select the trigger source by 'SRC' in **2 TRIG** in **MENU**.

Trigger source

IMMED	Performs continuous measurement at the sampling interval of the 7355.
MAN	Detects an event when the TRIG key is pressed or when a trigger signal is input to the external trigger signal terminal.
EXT	Detects an event when the trigger signal is input from the external trigger connector. (OPT+01, OPT+03)
BUS	Detects an event when the *TRG command or the GET command of GPIB is received from the controller.

The sampling HOLD switches between ON and OFF depending on the trigger source setting as follows.

IMMED	Samling HOLD: OFF
MAN	Samling HOLD: ON
EXT	Samling HOLD: ON
BUS	Samling HOLD: ON

Trigger key

If the **TRIG** key on the front panel is pressed when the 7355 is in the local state and the sampling HOLD is ON, the measurement starts. The trigger key can be used regardless of the trigger source selection.

5.8.3 :INITiate:CONTinuous Setting

":INITiate:CONTinuous" of the trigger system can be set by using the remote command.
This setting is for operational compatibility between the SCPI trigger system and the ADC commands.

5.8.4 Trigger Delay (Td)

Time from a trigger input to sampling start can be delayed by setting the trigger delay.

The measurement can start after the input becomes stable by coordinating the timing with other devices.

Table 5-6 shows the trigger delay time and setting resolution.

The trigger delay is set by 'DELAY' in **2 TRIG** in **MENU**.

Table 5-6 Trigger Delay Time, Setting Resolution, and Accuracy

Trigger delay setting	Resolution *1	Delay time accuracy	
		Ext trigger	Bus trigger
0 s to 600 ms	10 µs	20 to 30 µs + resolution	0.5 to 1.0 ms + resolution
600.1 ms to 6 s	100 µs		
6001 ms to 60 s	1 ms		
61 s to 3600 s	1 s	1 s	1 s

*1: If a value that is less than or equal to the resolution is set by using the panel or a remote command, the setting value is rounded to the nearest one digit below the resolution.

5.8.5 Trigger Count Setting

Normally the trigger system usually returns to the IDLE state when receiving a trigger, but it can also return to the trigger waiting state to receive triggers.

The trigger count can be set only by using a remote command.

Trigger count: 1 to 50000

5.9 Calculation Functions

This instrument supports the following eight calculation functions.

Item	Function
NULL calculation	Subtracts a fixed value from a measured value.
Smoothing calculation	Calculates the moving average.
Scaling calculation	Calculates $(\frac{M-B}{A}) \times C$.
dB calculation	Calculates $20 \log_{10}(\frac{M}{D})$.
dBm calculation	Calculates $10 \log_{10}(\frac{M^2}{D} \times \frac{1}{10^{-3}})$.
Comparator calculation	Performs a comparative calculation (HI/GO/LO).
MAX/MIN calculation	Calculates the measurement count, maximum, minimum and average values.
Statistical calculation	Calculates the maximum value, minimum value, average value, standard deviation, and dispersion for the data memory.

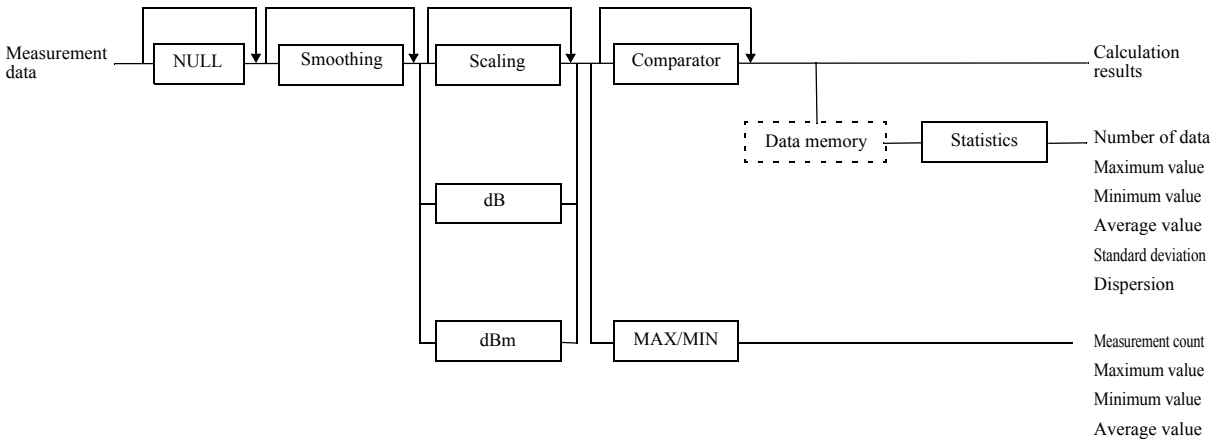


Figure 5-4 Calculation System Diagram

NOTE:

1. *The following results may be obtained from calculating the data. This is because the internal resolutions of measurement values and setting values for the calculation are higher than the resolutions of the displayed data and remote output data.*
   - *If the buzzer is set to ON in the MAX/MIN calculation, the buzzer sounds when the MAX or MIN value is renewed. However, the buzzer may sound even when the display and remote output data are not changed.*
2. *If the calculation is set to ON, the display of the calculation result is different from the ordinary measurement values and the decimal point and unit do not depend on the current measurement range.*

   <Example> DC voltage measurement, measurement range: 2000 mV, input voltage: 1 V
   Measurement value: 1000.000 mV
   Calculation result: 1.000000 V

5.9.1 NULL Calculation

1. Function

   The NULL calculation subtracts the NULL constant from the measured value and outputs the result.

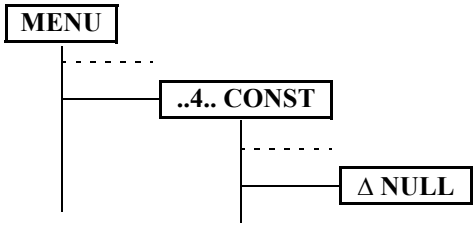
Measurement data output = Measured value - NULL constant

   Pressing the **NULL** key performs the calculation and turns on the NULL lamp on the display. When the key is re-pressed while the NULL calculation is performed, the calculation is canceled and the NULL lamp turns off.

   CAUTION: The actual measurement value is not displayed while the calculation is performed.
   Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.

2. NULL constant

   The measured value when the **NULL** key is pressed is determined as the NULL constant. The NULL calculation cannot be performed when the overload (OL) occurs. The NULL constant can be checked in **MENU** during the calculation.

   The setting range is shown below.

Setting range	Setting resolution
-9999999.E +6 to +9999999.E + 6	0.000001E - 9

   The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

3. Changing the NULL constant

   1. Display the NULL constant according to item 2 (previous description).
   2. After pressing ▽, the blinking position of the following items, which can be changed, is moved by using ◁ and ▷.
      - From the most significant digit to the least significant digit in the numeric value
      - Unit prefix (exponential part setting)
      - Decimal point

3. Change the setting value by using △ and ▽ at the blinking position.
   The unit prefix is switched according to the following order.
   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T
4. Press **ENTER** to apply the setting.

4. Canceling the NULL calculation
   The NULL calculation is canceled when any of the following conditions occur.
   - The **NULL** key is pressed while the NULL calculation is performed.
   - The *RST command is executed.

5.9.2 Smoothing Calculation

1. Function
   The smoothing function is used when the noise is superimposed on the measurement signal. Because the measured values in the specified number of times (smoothing count) are calculated by using the moving averaging method, the dispersion of the measured values becomes small.
   The smoothing calculation equation is shown below.

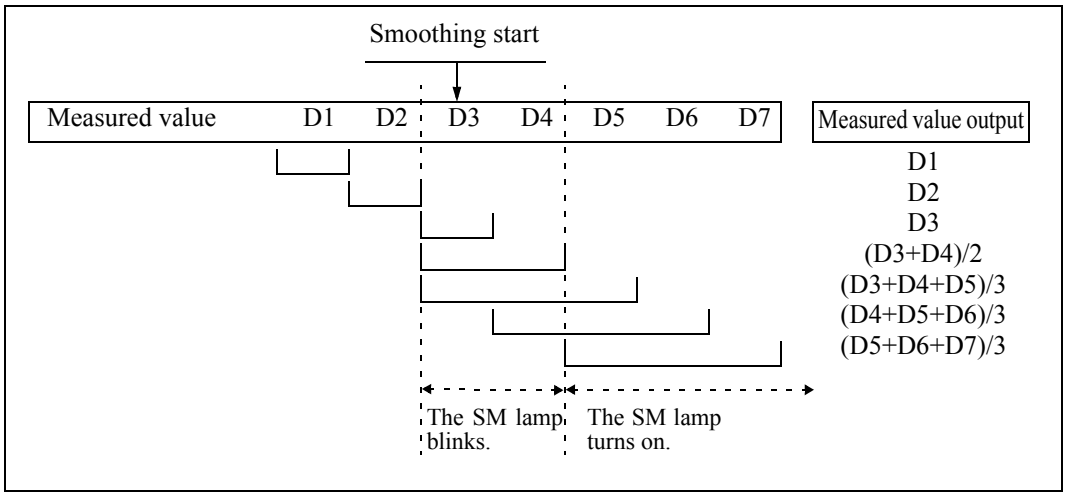
   Display = (Measured value 1 + , , , , , + Measured value N) / N

   When the **SM** key is pressed, the calculation is performed and the SM lamp on the display turns on or blinks.
   When the **SM** key is re-pressed while the smoothing calculation is performed, the calculation is canceled and the SM lamp turns off.
   The SM lamp blinks from the smoothing start to the (N - 1)th measurement and turns on after the Nth measurement.
   The (N + 1)th data is shown below.

   Display = (Measured value 2 + , , , , , + Measured value (N + 1)) / N

   When the number of measured values does not reach the number of smoothing count, the average up to that point value is displayed.

When the smoothing count, N, is set to 3, the operation is as follows.



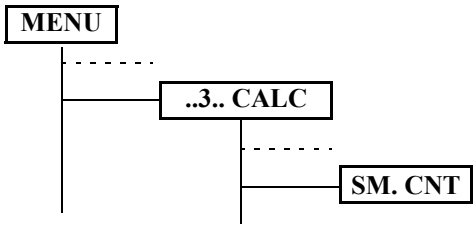
The smoothing calculation result, D (sm), at the n-th measurement, which exceeds the smoothing count, is shown below.

$$\text{Smoothing calculation result } D(sm) = \frac{1}{T}\sum_{i=n-T+1}^{n} Di$$

D (sm) :	Smoothing calculation result at the n-th measurement
Di:	Measured value (before the smoothing calculation)
T:	Smoothing count (2 to 100 can be set.)

2. Setting the smoothing count

   The smoothing count is set in **MENU**.



   The smoothing count can be set between 2 and 100.

3. Restarting the smoothing calculation

   The smoothing calculation restarts from N=1 when any of the following conditions occur.

   - The power is turned on.
   - The smoothing count is changed.
   - The NULL calculation is performed.

4. Canceling the smoothing calculation
   The smoothing calculation is canceled when any of the following conditions occur.
   - The **SM** key is pressed while the smoothing calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.
5. Smoothing calculation and overload (OL)
   If the overload occurs during the smoothing process, this measured value is ignored.
   (All measurement data except for OL data can be used in the smoothing calculation.)

5.9.3 Scaling Calculation

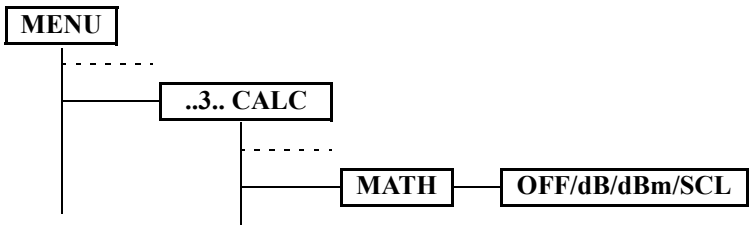
1. Function

   The scaling calculation equation is shown below.

$$\text{Display} = \frac{\text{Measured value M} - \text{Constant B}}{\text{Constant A}} \times \text{Constant C}$$

   When SCL is selected from the MATH item in **MENU**, the MATH lamp on the display turns on and the scaling calculation is performed.

   If OFF is selected from the MATH item in **MENU** while the scaling calculation is performed, the calculation is canceled and the MATH lamp turns off.

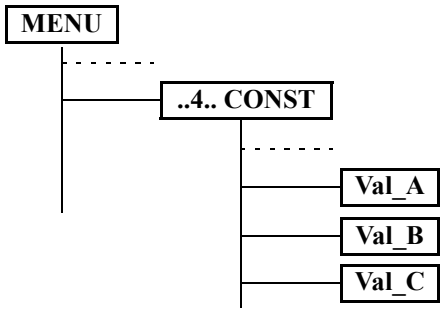


CAUTION: The actual measured value is not displayed while the scaling calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.

2. Setting scaling constants

NOTE: Press the **EXIT/LOCAL** *key if quitting the setting procedure.*

   1. Set Constants A, B, and C in **MENU**.



   2. After selecting a constant to be set, the constant can be set by pressing ▽.

3. Setting the numerical value of the constant

   The setting ranges of Constants A, B, and C are shown below.

Constant	Setting range	Minimum resolution
A	-9999999.E +6 to +9999999.E + 6	0.000001E - 9
B		0.000000E - 9
C		

   The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

   1. Press ◁ or ▷ in entering the constant setting mode, each of the following changeable items will blink.
      - From the most significant digit to the least significant digit in the numeric value
      - Unit prefix
      - Decimal point
   2. Change the numeric value and unit prefix by using △ and ▽.

      The unit prefix cycles through the following order.

      p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

      Press the **TRIG** key in this mode if setting the measured value as the scaling constant.
   3. Press **ENTER** to apply the setting.

4. SOL (scaling over)

   If the scaling calculation result exceeds ±999.9999E + 6, "±S OL" is displayed.

   In this case, the measurement range is not changed to the upper range even if the Auto-range is executed.

   (The Auto-range is determined according to the measured value acquired before the calculation is performed.)

5. Canceling the scaling calculation

   The scaling calculation is canceled when any of the following conditions occur.

   - OFF is selected from the MATH item in **MENU** while the scaling calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.
   - The dB or dBm calculation is performed.

5.9.4 dB and dBm Calculation

1. Function

   The dB calculation, in which the decibel conversion is performed, can be performed only in the voltage and current measurements.

   The dBm calculation, which is useful to calculate the power gain per 1 mV, can be performed only in the voltage measurement.

   If the zero is measured while the dB or dBm calculations is performed, the calculation error occurs and error message is displayed.

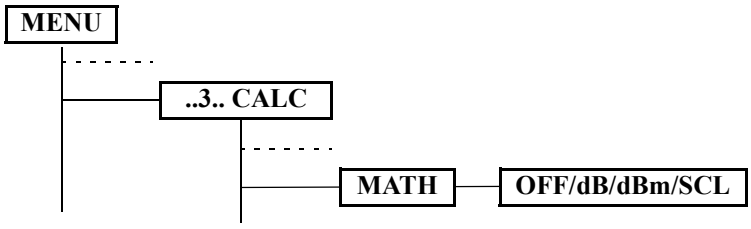
   Only one calculation can be selected from the dB, dBm, and scaling calculations at a time.

   The dB and dBm calculation equations are shown below.

$$\text{dB: Display} = 20 \log_{10} \frac{|\text{Measured value}|}{\text{Constant D}}$$

$$\text{dBm: Display} = 10 \log_{10} \frac{(\text{Measured value})^2 / \text{Constant D}}{10^{-3}}$$

   When dB or dBm is selected from the MATH item in **MENU**, the MATH lamp on the display turns on.



CAUTION: The actual measured value is not displayed while the dB or dBm calculation is performed.
Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.

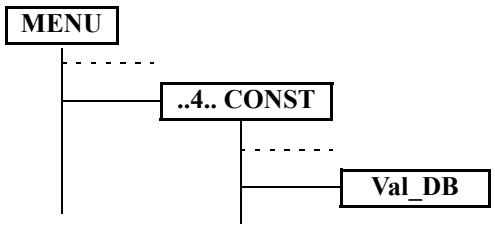
2. Setting Constant D

   The setting range of Constant D is shown below.

Setting range	Minimum resolution
0.000001E -9 to 9999999.E + 6	0.000001E - 9

The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

1. Enter the Constant D setting mode in **MENU**.



2. After entering the constant setting mode with ▽ and pressing ◁ or ▷, each of the following changeable items will blink.
   - From the most significant digit to the least significant digit in the numeric value
   - Unit prefix
   - Decimal point

3. Change the numeric value and unit prefix by using △ and ▽.

   The unit prefix cycles through the following order.

   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

   Press the **TRIG** key in this mode if setting the measured value as Constant D.

4. Press **ENTER** to apply the setting.

3. Canceling the dB and dBm calculations

   The dB and dBm calculations are canceled when any of the following conditions occur.
   - The scaling calculation is set while the dB or dBm calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.

5.9.5 Comparator Calculation

1. Function

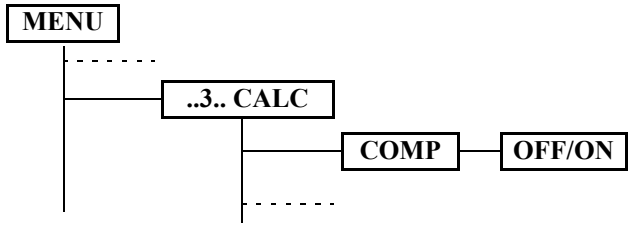
   The comparator calculation equation is shown below.

   HI = (Measured value > HI setting value)
   LO = (Measured value < LOW setting value)
   GO = (LOW setting value ≤ Measured value ≤ HI setting value)

   (Processing the special data)

   - The overload (OL) of plus (+) data is judged as HI.
   - The overload (OL) of minus (-) data is judged as LO.

   The result whether HI, GO, or LO is displayed by setting COMP in **MENU** to ON.



   The judgment result can be output to the display, buzzer, and status register.

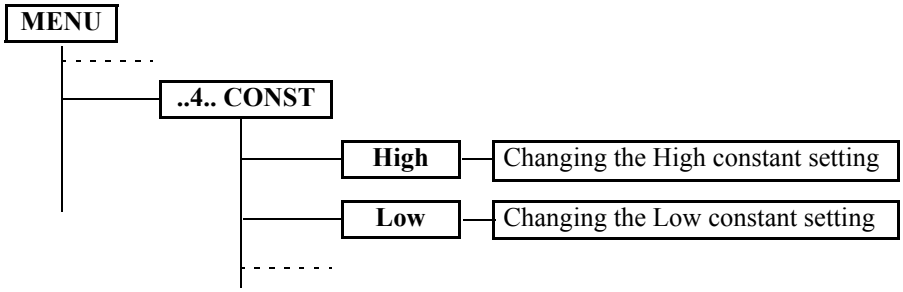
   (The buzzer sounds only when the buzzer is set to ON.)

   The comparator calculation can be applied to all measurement functions.

   However, when MAX/MIN calculation is set to ON, the comparator calculation is applied to the value before the MAX/MIN calculation.

2. Setting the judgment conditions

   The setting examples of the High constant, Low constant, and buzzer are shown below.



3. Setting the numerical value of the constant

   The setting ranges of the High and Low constants are shown below.

Constant	Setting range	Minimum resolution
High	-9999999.E +6 to +9999999.E + 6	0.000001E - 9
Low		

The exponential part is set by using a unit prefix (p, n, μ, m, k, M, G, T).

1. After entering the constant setting mode with ▽ and pressing ◁ or ▷, each of the following changeable items will blink.
   - From the most significant digit to the least significant digit in the numeric value
   - Unit prefix
   - Decimal point
2. Change the numeric value and unit prefix by using △ and ▽.

   The unit prefix cycles through the following order.

   p ↔ n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M ↔ G ↔ T

   Press the **TRIG** key in this mode if setting the measured value as the High or Low constant.
3. Press **ENTER** to apply the setting.

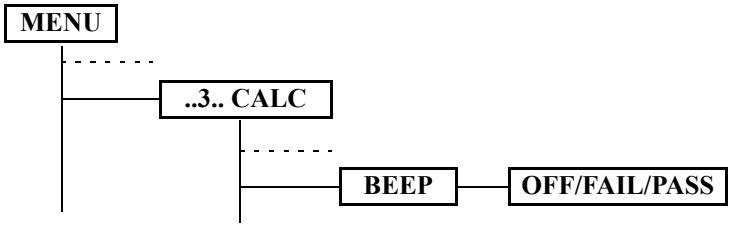
If the constants are set as “High constant < Low constant”, the calculation results may be judged as “HI and LO”. In this case, HI and LO are displayed at the same time.

4. Buzzer setting

   The buzzer output according to the PASS/FAIL judgments can be set.

   However, the buzzer output cannot be performed if •)) turns off.

   The buzzer setting is performed by BEEP in **MENU**.



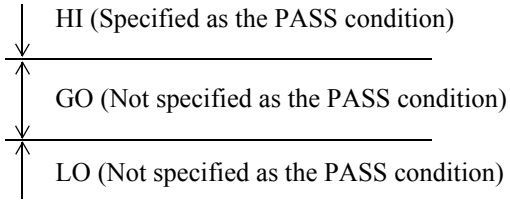
1. Press ▽ on the current setting display (BEEP) to enable the buzzer setting mode.
2. Select OFF, FAIL, or PASS by using ◁ and ▷.

5. PASS condition setting

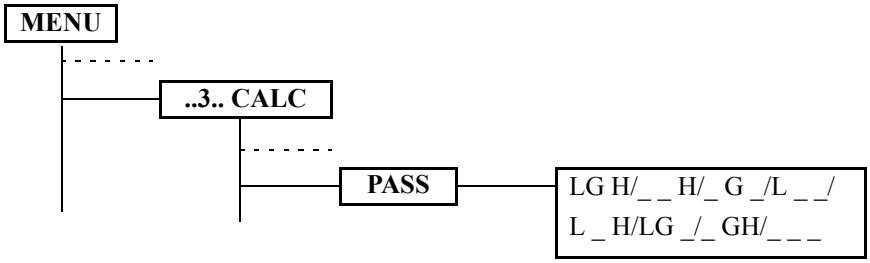
   In the PASS condition setting, each comparator calculation result (HI, GO, and LO) is specified as either PASS or not.

   For example, if HI is specified as the PASS condition and LO and GO are not specified as the PASS conditions, the comparator measurement result of HI is judged as PASS.

   The comparator calculation results of LO and GO are judged as FAIL.



   The PASS conditions are set in **MENU**.



   1. Press ▽ on the current setting display (PASS) to enable the PASS setting mode.
   2. Select the PASS condition by using ◁ and ▷.

6. Canceling the comparator calculation

   The comparator calculation is canceled when any of the following conditions occur.

   - The comparator calculation is set to OFF while the comparator calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.

7. Changing the measurement range and comparator operation

   The comparator function continues to operate even if the measurement range is changed.

   The judgment reference value remains unchanged from the previous setting value because it is unitized.

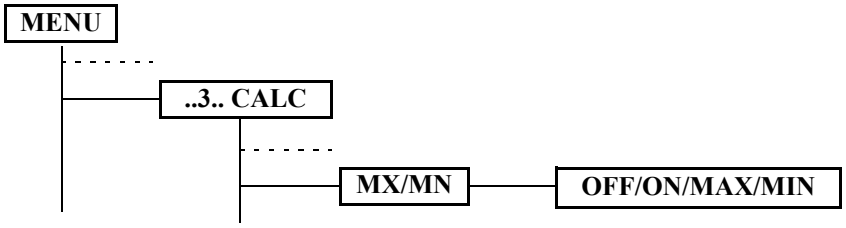
5.9.6 MAX/MIN Calculation

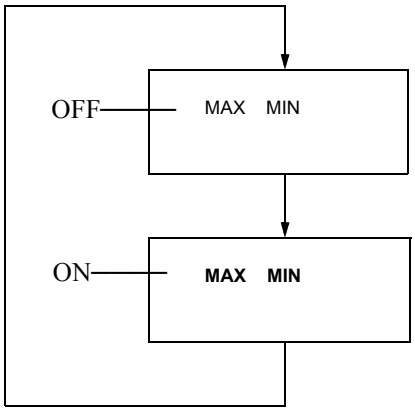
1. Function

    1. This function calculates the maximum, minimum, and average values while the MAX/MIN calculation is performed.
    2. The following calculations are performed at the same time in the MAX/MIN calculation.
        - Measurement count
        - Maximum value (MAX)
        - Minimum value (MIN)
        - Average value (AVE)
    3. All measurement data except for overload (OL) and calculation error data is used for the calculation.
    4. If the buzzer is set to ON, the buzzer sounds when the MAX or MIN value is renewed. However, although the displayed value is not changed, the buzzer may sound. This is because the resolution of measured value is higher than the resolution of the displayed value.

2. Setting the calculation

    The operation is switched as shown below by selecting MX/MN in **MENU**.





Canceling the calculation

The MAX and MIN lamps turn off.

Performing the calculation

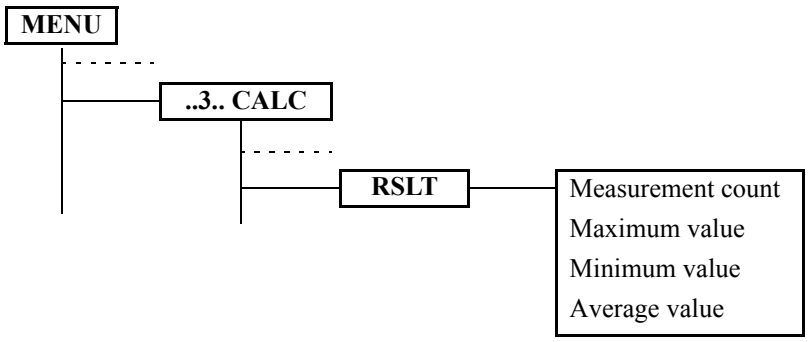
Any of the following indication (NOTE)

- The MAX and MIN lamps turn on.
- The MAX lamp blinks.
- The MIN lamp blinks.

NOTE: Refer to "4. MAX/MIN calculation value display function during the measurement" in this section.

3. How to call the MAX/MIN calculation result

    1. Select RSLT in **MENU** to see the calculation results display.



    2. Select the item to be displayed by using ◁ and ▷.

       The measurement count, maximum value, minimum value, and average value are displayed in this order.

4. MAX/MIN calculation value display function during the measurement

    1. The value during the MAX/MIN calculation can be selected and displayed from the following three items:
        - Measured value display (MAX and MIN lamps turn on.)
        - Maximum value display (MAX lamp blinks.)
        - Minimum value display (MIN lamp blinks.)

    2. How to set

       Select the item to be displayed according to "2. Setting the calculation" in this section.

       The selected value is displayed by pressing **ENTER**.

5. Canceling the MAX/MIN calculation

   The MAX/MIN calculation is canceled when any of the following conditions occur.
   - The MAX/MIN calculation is set to OFF while the MAX/MIN calculation is performed.
   - The measurement function is changed.
   - The *RST command is executed.

6. Restarting the MAX/MIN calculation

   The MAX/MIN calculation values are cleared and the MAX/MIN calculation restarts when any of the following conditions occur.
   - The power is turned on.
   - The MAX/MIN calculation is set to ON after it is set to OFF.

- Any calculation except for the comparator calculation is changed to ON or OFF.
- The NULL constant or the smoothing count is changed.

5.9.7 Statistical Calculation

1. Function

    1. This calculation obtains the maximum value, minimum value, average value, standard deviation, and dispersion of the measurement data stored in the measurement memory.

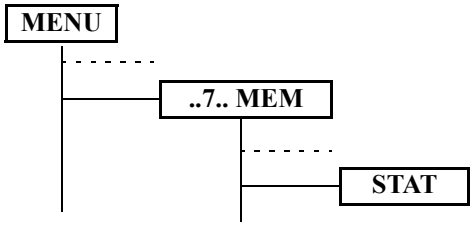
    2. In the statistical calculation, the following calculations are performed at the same time.

        - Number of samples S_{CNT} = (Number of data) - (Number of overload), in the specified range
        - Maximum value S_{MAX} = Maximum value in the target data
        - Minimum value S_{MIN} = Minimum value in the target data
        - Average value $$S_{AVE} = \frac{1}{S_{CNT}} \sum_{k=1}^{S_{CNT}} Dk$$
        - Dispersion $S_{PTP} = | S_{MAX} - S_{MIN} |$
        - Standard deviation $$S\sigma = \sqrt{\frac{1}{S_{CNT} - 1} \sum_{k=1}^{S_{CNT}} (Dk - S_{AVE})^2}$$

    3. All measurement data except for overload (OL) and calculation error data is used for the calculation.

2. Performing the calculation

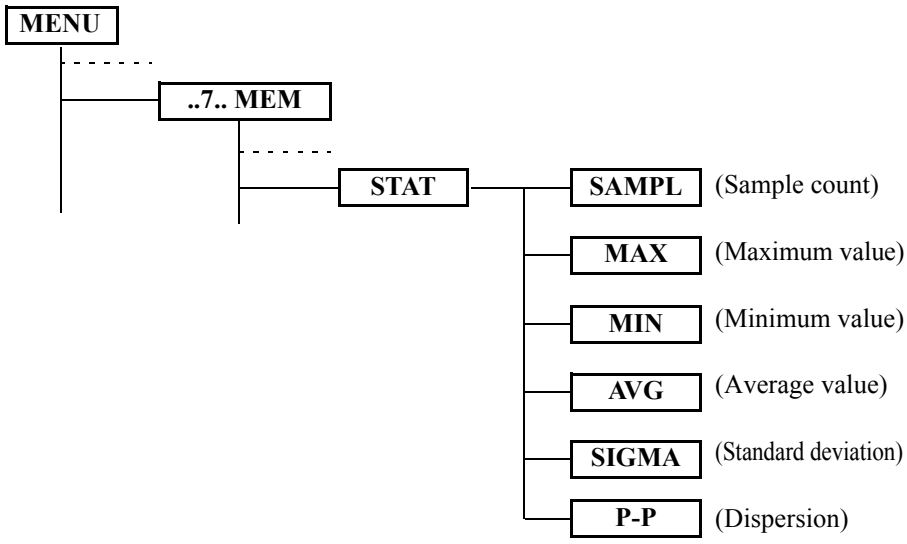
    1. Select STAT in **MENU**.



    2. Press **ENTER** to perform the calculation.
    'LAST' is displayed if no data is stored in the measurement data memory.

3. How to call the calculation result

1. Press STAT in **MENU** to see the calculation results display.



2. Select the item to be displayed by using ◁ and ▷ after the statistical calculation is performed.

5.10 Saving and Loading Measurement Condition Settings

The 7355 can store four files of measurement condition setting parameters in the non-volatile memory USER0 to USER3.

For information on setting parameters, refer to Section 8.9, “System Recovery Procedure” and Table 8-4, “Initial Values of Setting Parameters”.

5.10.1 Auto-Loading at Power ON

The 7355 starts up under the conditions saved when the power is turned off.

To always start up with fixed conditions, it can load the measurement conditions stored in USER0 at power on.

Select from the followings by ‘PON.LD’ in the category **5 PARAM** in **MENU**.

P.OFF	Starts up under the measurement conditions saved when the power is turned off.
USER0	Loads the measurement conditions in USER0.

Press **ENTER** to apply the setting.

5.10.2 Saving Setting Parameters

Select one from USER0 to USER3 by ‘SAVE’ in the category **5 PARAM** in **MENU**.

Press **ENTER** to save the setting parameters into the selected area.

5.10.3 Loading Setting Parameters

Select one from USER0 to USER3 and DFLT by ‘LOAD’ in the category **5 PARAM** in **MENU**.

Press **ENTER** to load the setting parameters from the selected area.

If DFLT is selected, the default values are loaded.

For more information on default values, refer to Section 8.9, “System Recovery Procedure”.

5.11 Storing and Recalling Measurement Data

The 7355 includes a measurement data memory to store measurement data.

5.11.1 Measurement Data Memory Store

Select ON or OFF by 'Store' in **7 MEM** in **MENU**.

Pressing **ENTER** determines the setting and turns on the 'ST' indicator.

The measurement data is stored into the measurement data memory by closing the menu and starting the measurement.

When the number of measurement results reaches 20000, the buzzer sounds and the 'ST' indicator blinks.

5.11.2 Measurement Data Memory Recall

Enter the data number to be recalled by 'RCL' in **7 MEM** in **MENU**.

Pressing **ENTER** determines the setting and turns on the 'RCL' indicator.

The data number can be increased and decreased by using ◁ and ▷ before recalling the data number.

5.11.3 Measurement Data Memory Clear

The measurement data memory is cleared when any of the following conditions occur.

- The power is turned on.
- The device clear command is executed from the GPIB interface.
- The *RST or ICL command is executed.
- The parameter initialization is performed by **6 INIT** in **MENU**.
- The memory store setting of the measurement data is switched from OFF to ON.

5.12 System Settings

5.12.1 Buzzer Setting

Select ON or OFF by 'BEEP' in **9 SYS** in **MENU**.

ON	The buzzer sounds.
OFF	The buzzer does not sound. The buzzer caused by the comparator calculation result also stops sounding.

Press **ENTER** to apply the setting.

If an error occurs, the buzzer sounds even when the buzzer is set to OFF.

5.12.2 Power Supply Frequency

When the sampling rate is set to MED, the effect of induction noise can be reduced by synchronizing the integration time with the power supply frequency.

The power supply frequency is automatically set according to the area where the 7355 is used. Use the following procedure to check or select it manually.

Select one of the following from 'LINE' in the **9 SYS** category in **MENU**.

50	The commercial power supply frequency is 50 Hz.
60	The commercial power supply frequency is 60 Hz.

5.12.3 Disabling Functions

To prevent incorrect functions from being selected because of malfunctions and to use the 7355 safely, some functions can be disabled.

Select a function to be enabled or disabled by 'F.Inhi' in **9 SYS** in **MENU**.

Press ▽ to enter in the Input layer and select ENA (Enabling) or DIS (Disabling) by using ◁ and ▷.

Press **ENTER** to apply the setting.

- Interrelationship due to disabling function setting

  Although the ENA or DIS can be set for each function, some functions are automatically disabled depending on the settings of other functions.

  If a higher function setting in the dependency relation is disabled, no lower function setting can be changed.

Table 5-7 Interrelationship due to disabling function setting

Setting function	Function depending on function setting						
	Diode	ACV (AC+DC)	ACI (AC+DC)	Continuity	ACI	FREQ	TEMP
DCV		○					
ACV		○	○		●	●	
DCI			○				
ACI			○			●	
2WΩ/LP-2WΩ	⦿			○			○
4WΩ/LP-4WΩ				○			○
FREQ							

⦿: Automatically enabled or disabled according to the setting function.
○: Disabled when any of the setting functions is set to DIS, and enabled when all related setting functions are set to ENA.
●: Disabled when any of the setting functions is set to DIS, and not enabled even when all related setting functions are set to ENA.

- Canceling the setting

  To cancel the disabling setting and enable all functions, initialize by **6 INIT** in **MENU** or reset all functions by 'F.Inhi' in **9 SYS** in **MENU**.

5.12.4 Front Panel Lock Function

This function disables all panel keys on the front panel of the 7355.

The following shows how to set the front panel lock function.

Password setting

A password is required to enable or disable the front panel keys.
A four-digit number from 0000 to 9999 can be used for the password.
Follow the steps below to set the password.

	Operation	Character display
1.	Press **MENU** and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP
3.	Select 'PWD' by using ◁ and ▷.	SYS PWD
4.	Press ▽.	PWD NEW
5.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD NEW
6.	Press **ENTER**.	PWD OLD
7.	Enter the current password according to the above procedure.	PWD OLD
8.	Press **ENTER**. If the password is correct, PASS is displayed and the password is set.	PWD PASS
	If the password is incorrect, FAIL is displayed and the procedure returns to step 3. Try it again using the correct password.	PWD FAIL
9.	Press **EXIT** to exit the menu.	

Panel lock setting

Follow the steps below to set the panel lock.

	Operation	Character display
1.	Press **MENU** and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP
3.	Select 'P.LOCK' by using ◁ and ▷.	SYS P.Lock
4.	Press ▽.	P.Lock OFF
5.	Select ON by using ◁ and ▷, and press **ENTER**.	P.Lock ON
6.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD
7.	Press **ENTER**. If the password is correct, PASS is displayed and the password is set. If the password is incorrect, FAIL is displayed and the procedure returns to step 3. Try it again using the correct password.	PWD LOCK

Panel lock release

Follow the steps below to release the panel lock.

	Operation	Character display
1.	If pressing any key while the panel lock is set, the Input layer of P.Lock is set.	PWD LOCK
2.	Select the digit in the numerical value entry section by using ◁ and ▷. Change the numerical value by using △ and ▽ to enter the password.	PWD LOCK
3.	Press **ENTER**.	
	If the password is correct, PASS is displayed and the password is set.	PWD PASS
	If the password is incorrect, FAIL is displayed and the procedure returns to step 3.	PWD FAIL

NOTE: The panel lock cannot be released if the password is forgotten. Do not forget the password. Copying the password onto paper is recommended.

5.12.5 Error Log

If any error occurs, the 7355 records the error number the 'ERR' indicator turns on.
Only the error number can be read from the front panel menu.
When queried with the remote command, the error number and error message are read back.
Up to 20 errors can be recorded.
For more information on error numbers and solutions, refer to Section 8.11, "Error Message List".

Error log reading

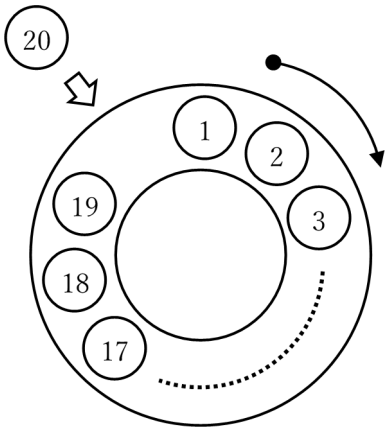
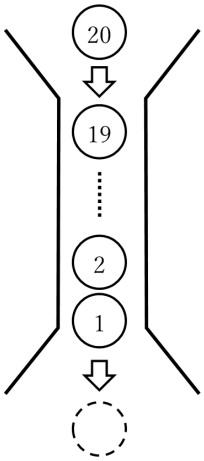
Press ▽ at 'QUEUE' in **9 SYS** in **MENU** to display the error log in order.

Error log clear

The error log is cleared when read out.

5.12.6 Event History

The errors stored in the error log described in Section 5.12.5 are also stored in the event history. The error log contents are erased by displaying or reading, however the errors stored in the event history are retained even after the power is turned off.

Event history	Error log
	
• Ring format • Latest 20 errors are stored. • 21th and later errors are overwritten to the oldest ones in order. • Errors are stored in the non-volatile memory, and are not erased by reading or turning OFF the power.	• FIFO type log operation • Up to 20 errors are stored. • 21th and later errors are not stored. • Errors are erased by reading or turning OFF the power.

5.12.6.1 Read-Out

When reading out the contents of the event history with the remote command, reset the readout first so that the errors can be read out from the oldest one.

When reading out them from the panel, temporary numbers are assigned to errors in order of oldest to newest.

	ADC command	SCPI command
Readout reset	EVR	:SYSTem:EVENt:RESet
Readout	EVH?	:SYSTem:EVENt?

6. HOW TO USE INTERFACE

The 7355 includes USB and LAN interfaces. GPIB (Option +01) and RS232 (Option +03) interfaces can be added as factory options.
However, these interfaces cannot be used at the same time. Select one of them.

Interface configuration

Interface	Standard	Option+01	Option+03
USB, CDC	✓	✓	✓
GPIB		✓	
RS232			✓
LAN	✓	✓	✓

6.1 Interface Selection

The interface can be selected and set only from the menu on the front panel.

1. The selected interface is saved in the non-volatile memory and does not change even if the power is turned off or the interface is reset.
2. A unique address for the instrument is set on the interface. When multiple digital multimeters are connected to the USB interface, an address (USB ID) is set to identify each one.
   The address is displayed by turning on the power or using the address setting in the menu.

The interface setting items and the factory defaults are shown below.

Setting item	Factory default
Interface selection (BUS)	USB
Header ON/OFF (HEADR)	ON
GPIB address (GP.Adr)	01
USB ID (USBid)	001
Baud rate (BAUD)	9600
Data length (DATA)	8bit
Parity (P-Bit)	None
Stop bit (STOP)	1bit
Echo (ECHO)	OFF
IP address (IP.Adr)	192.168.0.1
Default gateway (G.WAY)	192.168.0.0
Subnet mask (S.MASK)	255.255.255.0
Output data format (DFORM)	ASCII

Select the interface from GPIB, USB, RS232, LAN and CDC by ‘BUS’ in **8 I/F** in **MENU**.

Press **ENTER** to apply the setting.

Even when each interface is selected, only the interface equipped with the option can be used.
As standard, only USB, CDC, and LAN are available.

6.2 GPIB (Option+01)

6.2.1 Overview

Using GPIB (General Purpose Interface Bus) allows you to externally set the measurement functions and parameters on the 7355 and read out the measured data, making it simple to configure an automated measurement system.
As GPIB signals from the 7355 are electrically isolated inside the unit from the measurement signal system, connecting external devices does not affect the measured values.
The remote commands are the same as with USB.

- General specifications

  Standard: IEEE-488.2

  Code: ASCII code

  Logic level: Logical 0 “High” +2.4 V or higher
  Logical 1 “Low” +0.4 V or less

Table 6-1 Interface Functions

Code	Function
SH1	Source handshake capability
AH1	Acceptor handshake capability
T5	Basic Talker, unaddressed if MLA, Talk Only, serial poll
L4	Basic Listener, unaddressed if MTA
SR1	Service Request capability
RL1	Remote/Local switching capability
PP0	No Parallel Poll capability
DC1	Device Clear capability (The SDC and DCL commands can be used.)
DT1	Device Trigger capability (The GET command can be used.)
C0	No Controller capability
E2	Using tri-state bus drivers

6.2.2 Precautions when Using GPIB

1. Do not use connection cables to measuring instruments or bus cables to controllers that are longer than necessary. Ensure that these cables do not exceed 20 m in length. ADC CORPORATION offers the following standard bus cables.

Table 6-2 Standard Bus Cable

Length	Name
0.5 m	408JE-1P5
1 m	408JE-101
2 m	408JE-102
4 m	408JE-104

2. The bus cables have piggyback connectors. A piggyback connector has both a male connector and a female connector by itself. The male and female connectors can be stacked on top of each other. When connecting the bus cables, do not stack three or more connectors. Be sure to fix the bus cables with clamping screws.
3. Check the power requirements, grounding conditions and setting conditions of each device before turning it ON.
   Be sure to turn ON all the devices connected to the bus. The overall system operation cannot be guaranteed if any of the devices is not turned ON.
4. Connecting and disconnecting cables

   Turn OFF all the devices before connecting or disconnecting the GPIB cables. Check that the chassis of the devices on the bus are connected to each other and to the ground before connecting or disconnecting any GPIB cables.
5. ATN interrupt during message transmission

   If an ATN request interruption occurs during transfer of messages between the devices, the ATN has priority and the previous status is cleared.
6. Do not connect the controller when using the Talk-Only mode.
7. Up to 255 characters can be recognized in a single program command transmission.
   An error occurs if the program command exceeds 255 characters.
8. Retain the REN line at Low for 5 ms or longer following the transmission of a command.
9. The next command can be received before the INI (ADC), INITiate (SCPI), or *TRG command has completed execution.
   To delay the execution of the next command until the current command is complete, use the *OPC, *OPC?, or *WAI command.
   Describe the *OPC, *OPC?, or *WAI command at the end of a line in the program.

   Example “INIT;*OPC”
   “INIT;*OPC?”
   “INIT;*WAI”

6.2.3 GPIB Setting

The following setting is available when GPIB is selected as the interface.

Address setting

1. Enter any address from 0 to 30 by 'GP.Adr' in **8 I/F** in **MENU**.
2. Press **ENTER** to apply the setting.

Talk Only setting

1. Select one of the following from 'T.ONLY' in **8 I/F** in **MENU**.
   ON: Talk-Only
   OFF: Addressable
2. Press **ENTER** to apply the setting.

6.3 RS232 (Option+03)

6.3.1 Overview

Using RS232 interface allows you to externally set the measurement functions and parameters on the 7355 and read out the measured data, making it simple to configure an automated measurement system.

Because the interface is electrically isolated from the measurement signal system, measured values are not affected by external devices.

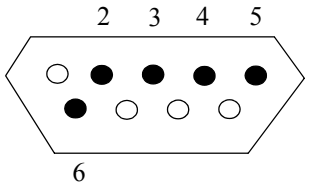
The remote commands are common to the GPIB commands.

- Specifications

Setting item			Factory default
Header of output data:	:	on, off	on
Talk-Only	:	on, off	off
Baud rate	:	19200, 9600, 4800, 2400, 1200, 600, 300	9600
Parity	:	even, odd, none	none
Number of data bits	:	8, 7	8
Number of stop bits	:	1, 2	1
Echo	:	on, off	off

For more information on how to set RS232 on the front panel, refer to Section 6.3.2, “RS232 Setting”.

The 9-pin connector (DB-9 male connector) used for the RS-232 interface is located on the rear panel.



Pin number	Input/output	Description
2	Input	Receive data (RxD)
3	Output	Transmit data (TxD)
4	Output	Data terminal ready (DTR)
5	-	Signal ground (SG)
6	Input	Data set ready (DSR)

Up to 251 characters are recognized in a single transfer of program code.

If a program code exceeds 251 characters, an error occurs.

If the DSR (data set ready) status is false when it is checked in the 7355, the transmit data (TxD) output is stopped. Data transmission restarts when DSR is true.

CAUTION: The 7355 cannot use the flow control performed by the X parameter (XON/XOFF).

6.3.2 RS232 Setting

Header of output data

1. Select the following from 'HEADR' in **8 I/F** in **MENU**.
   ON/OFF
2. Press **ENTER** to apply the setting.
   *Always OFF and unsettable when SCPI or 34401 is selected for the command language.

Talk-Only setting

1. Select the following from 'T.Only' in **8 I/F** in **MENU**.
   ON: Talk-Only
   OFF: Addressable
2. Press **ENTER** to apply the setting.

Baud rate setting

1. Select the following from 'BAUD' in **8 I/F** in **MENU**.
   9600 - 19200 - 300 - 600 - 1200 - 2400 - 4800
2. Press **ENTER** to apply the setting.

Number of data bits

1. Select the following from 'DATA' in **8 I/F** in **MENU**.
   8 bit/7 bit
2. Press **ENTER** to apply the setting.

Parity

1. Select the following from 'P-Bit' in **8 I/F** in **MENU**.
   NONE - ODD - EVEN
2. Press **ENTER** to apply the setting.

Number of stop bits

1. Select the following from 'STOP' in **8 I/F** in **MENU**.
   1 bit/2 bit
2. Press **ENTER** to apply the setting.

Echo

1. Select the following from 'ECHO' in **8 I/F** in **MENU**.
   ON/OFF
2. Press **ENTER** to apply the setting.

6.3.3 RS232 Output Data Format

The output data format of RS232 is partially different from those of other interfaces. The data output from RS232 is as follows:

1. Echo
2. Prompt
3. Measured data
4. Inquiry result (For inquiry command)

The contents and output format of each output data are shown below.

1. Echo output

   The echo output is enabled when the echo output is set to ON in the interface settings.

   The received data is output as it is, however, if <^C> (CONTROL C) and <LF> are entered, the following data is output:

   (<LF>) + (Prompt) + (Delimiter)

   NOTE: The delimiter has been set to <CR>...<LF> and it cannot be changed.

2. Prompt

   An execution result for a command received by RS232 is output by using a prompt.

   First, <LF> is output, followed by a prompt and the delimiter (<CR><LF>).

   The following two prompts are used:

Prompt	Description
=>	A command was correctly received, analyzed, and executed.
?>	An error occurred when a command was received, analyzed, and executed.

3. Measured data output (For the Only mode)

   For the Only mode, the measured data is output only if the RS232 transmission is enabled and the transmission buffer is empty after the measurement was complete.

   The delimiter (<CR><LF>) is output for each measured value.

   The measurement data output format is the same between GPIB and RS232. However, for RS232, the block delimiter has been set to <CR><LF> and it cannot be changed.

4. Output of an inquiry result executed by an inquiry command

   For the output of an inquiry result executed by an inquiry command, <LF> is output first, followed by an inquiry result and the delimiter, and finally a prompt at the end.

   <LF> + Inquiry result + Delimiter (<CR><LF>) + <LF> + Prompt (=>) + Delimiter (<CR><LF>)

6.4 USB

6.4.1 Overview

The 7355 is equipped with USB (Universal Serial Bus) conforming to USB 2.0 standard.
USB allows function settings and reading of measurement data with respect to multiple instruments connected to the bus from a PC, making it simple to configure an automated measurement system.
Also, USB also supports USB CDC, it can be used as a virtual COM port.

NOTE: All operations using a PC or hub cannot be always guaranteed

6.4.2 USB Specifications

- Standard: Complies with USB2.0 Full-Speed
- Connectors: USB type B (female)
- ID: USB ID, settable from 1 to 127
- Remote/Local: Available
- Input commands: Function setting and query with ASCII character string commands
- Output format: Measurement data and query response in ASCII character strings or binary format.
- Driver: ADC instruments USB driver
- Class: USB TMC and USB CDC

6.4.3 USB Setting

6.4.3.1 Connecting with Personal Computer

Connect the USB connector (type B) on the rear of the 7355 to that on a personal computer with a cable. Fully insert all connectors.

Use a USB hub to connect multiple instruments to a single personal computer.

6.4.3.2 Setting USB ID

The USB ID is settable when the interface is set to USB.

1. Enter the address between 1 and 127 from 'USBid' in **8 I/F** in **MENU**.
2. Press **ENTER** to apply the setting.

6.5 LAN

6.5.1 Overview

The 7355 can be remotely controlled via the LAN interface by connecting the LAN port on the rear panel to a personal computer.

Plug into a network or directly connect to a host computer to operate the 7355.

6.5.2 LAN Setting

Using the LAN interface requires the following settings. (These settings cannot be changed by remote control.)

Table 6-3 LAN Setting Items

Setting item	Function
IP address (IP.Adr)	IP address setting Default value: 192.168.0.1
Default gateway (G.WAY)	Default gateway setting Default value: 192.168.0.0
Subnet mask (S.MASK)	Subnet mask setting Default value: 255.255.255.0
MAC address (MAC.Ad) Remote command only	A unique MAC address is displayed for each unit. This address cannot be changed.

* For the setting details, verify with the network manager.

6.5.3 Communication Setting on Personal Computer

When controlling the 7355 remotely via the LAN interface, make the communication settings of the control software on the computer side as follows:

Table 6-4 Communication Settings when Using LAN

Item	Setting value
TCP/IP port number (destination)	5025
Delimiter	LF or CR + LF

6.5.4 Network Connection

When connecting the 7355 to a network, verify the necessary information with the network manager.

If an invalid address is specified for the network, both the network and the 7355 may operate unexpectedly.

NOTE: When the computer and the 7355 exist on different networks (not on the same subnet), it is necessary to set the gateway correctly.

Verify the physical connection between the 7355 and the computer according to the following procedure:

1. Set the network connection correctly without connecting cables, and turn off the 7355.
2. Connect the cables with the power off. Connect the 7355 to the computer or the network terminal such as hub by using the cables.
3. Turn on the 7355.

6.5.5 IP Connection Check

The “ping” command is used to verify connections on IP networks from computers. This command sends IP packets to network devices and receives responses from them to confirm the IP network normality.

The “ping” command is available on Windows or under UNIX. In case of Windows, enter as follows from the command prompt.

C:\>ping 192.168.0.1 (*)

(*) The argument following “ping” indicates a network device to be verified. Here, input the IP address assigned to the 7355. “192.168.0.1” is just an example. Replace this argument according to the operating environment in actual cases.

The following shows an example of “ping” command execution results when the proper connection is confirmed.

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<10ms TTL=128
Reply from 192.168.0.1: bytes=32 time=20ms TTL=128
Reply from 192.168.0.1: bytes=32 time=20ms TTL=128
Reply from 192.168.0.1: bytes=32 time=30ms TTL=128

Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\>

Here is an example of “ping” command execution results when responses cannot be received and timeout occurs due to a failure on the network.

C:\ >ping 192.168.0.1

Pinging 192.168.0.1with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

When timeout occurs, there may be a problem with the IP address settings or network connections of the 7355 or the computer.

6.5.6 Communication Method

The 7355 uses TCP/IP to communicate with a computer. The TCP/IP communication is performed according to the following procedure:

1. After turning on, the 7355 waits for the computer sending a TCP/IP connection request with the specified port number (refer to Table 6-4.)
2. The computer sends the TCP/IP connection request to the 7355.
3. The 7355 accepts the TCP/IP connection request to establish the connection.
4. After the connection establishment, the communication is performed by sending or receiving simple 7-bit ASCII codes.
5. The computer sends commands or queries to the 7355.
   The command or query messages are characters that end with CR+LF or LF.
6. The 7355 receives, interrupts and executes the commands or queries. When these queries are received correctly, it returns response messages.
   The response messages are characters with delimiters added.

Only single personal computer is connectable to the 7355.

Note for Using LAN

When sending a query command, be sure to read it out without fail.

6.6 Control Signals (Option+01, Option+03)

6.6.1 External Trigger Signal Terminal (TRIGGER IN)

The 7355 can be triggered by inputting a negative or positive logic pulse from the external trigger terminal (TRIGGER IN) on the rear panel.
Select EXTernal as the trigger source when using this terminal.
Input the TTL level or contact signal as a trigger signal.

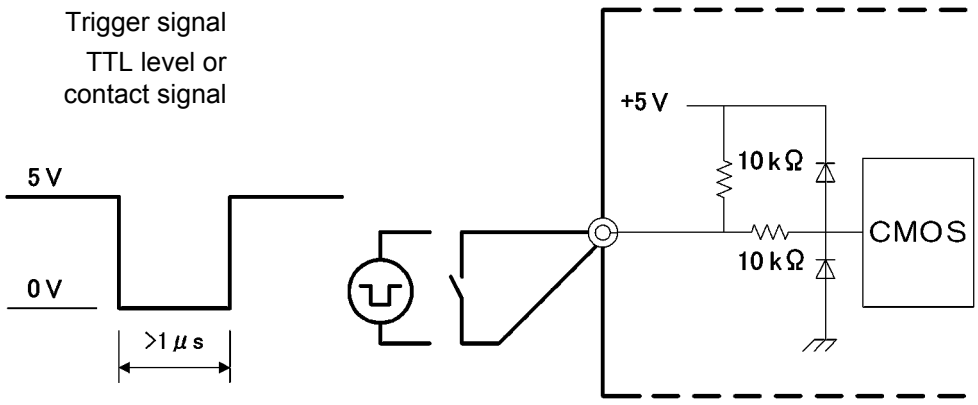


Figure 6-1 Simple Equivalent Circuit of External Trigger Signal Input Terminal

The polarity of the external trigger signal input can be switched between negative and positive logic pulses.

Select the polarity of the external trigger NEG or POS from “E.EDGE” in **9 SYS** in **MENU**.
NEG: negative logic, POS: positive logic

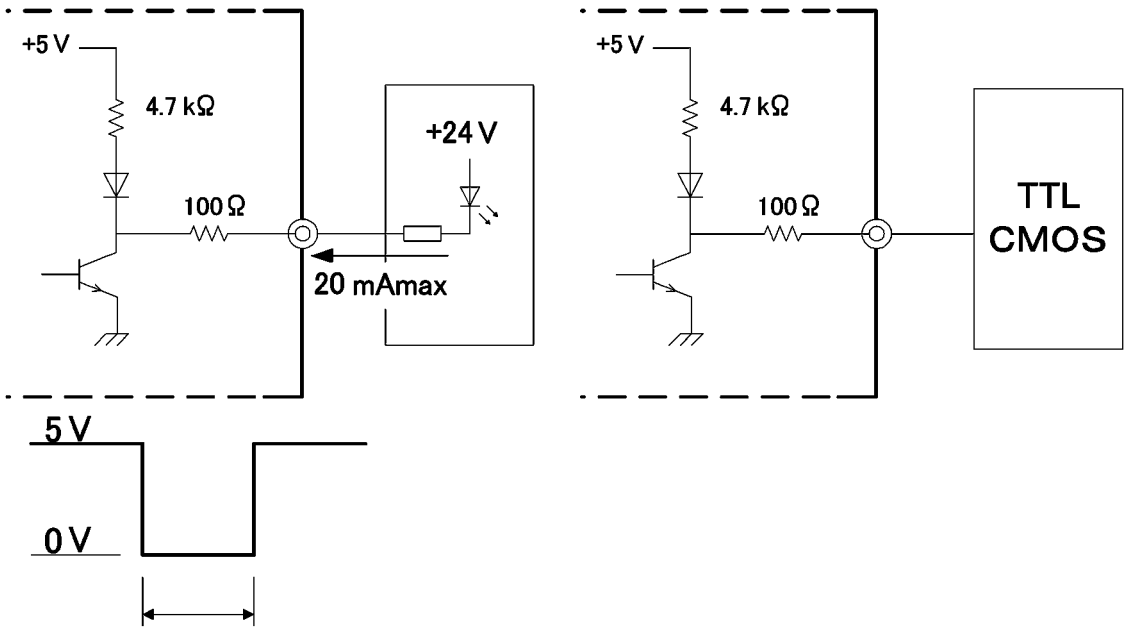
Press **ENTER** to apply the setting.

6.6.2 Complete Signal Terminal (COMPLETE OUT)

A negative logic pulse is output from the complete signal terminal (COMPLETE OUT) on the rear panel when the measurement is complete.

Pulse width: Approx. 5 µs, 100 µs or 1 ms

The complete signal can be used as the TTL level and input signals that can be directly connected to the programmable controller.



(Approx. 5 µs / 100 µs / 1 ms selectable)

Figure 6-2 Simple Equivalent Circuit of Complete Signal Output Terminal

The pulse width of the measurement complete signal output is selectable from 5 µs, 100 µs and 1 ms.

Select the pulse width of the measurement complete signal output from "C.Widt" in **9 SYS** in **MENU**.

Press **ENTER** to apply the setting.

6.7 Comparator Output (Option +03)

The comparator calculation result is output from the comparator output terminal (COMPARATOR) on the rear panel as the TTL signal and the optical semiconductor relay contact signal.

Also, the output can be changed arbitrarily by using the digital output function.
The digital output can be turned on and off from the menu, but the output setting for each terminal is available only by a remote command.

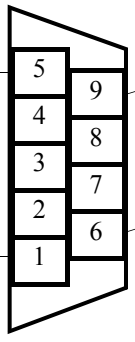
Select the digital output ON or OFF from "D.OUT" in **3 CALC** in **MENU**.
Press **ENTER** to apply the setting.

When the digital output is enabled, the comparator calculation results are not output form the output terminal.
For more information on the digital output function, refer to "Digital output" article in Section 6.8.3, "ADC Command Reference".

Specifications

Connector	Dsub 9-pin
a. Optical semiconductor relay contact	
Allowable contact voltage (for break)	30 V DC
Allowable contact current	120 mA DC
Withstand voltage between contact and GND	30 V
Contact operating time	1 ms or less
Contact output	LO: 4 pin and 8 pin HI: 3 pin and 7 pin PASS: 2 pin and 6 pin
b. TTL output	
Output level	TTL
Maximum allowable input voltage	12 V peak
Maximum allowable input current	20 mA peak

Pin name	Pin
Signal GND	5
RELAY LO	4
RELAY HI	3
RELAY PASS	2
TTL PASS	1



Pin	Pin name
9	TTL FAIL
8	RELAY LO
7	RELAY HI
6	RELAY PASS

Figure 6-3 Signal Output Pins

The calculation result signals of Hi, Go, and Lo are output to the signal output pins of HI, PASS, and LO. The output to the PASS pin depends on the PASS conditions as shown in Table 6-5 and Table 6-6.

Table 6-5 Optical Semiconductor Relay Contact and TTL Logical Output (a)

Judgment result	Contact/TTL output pin			
	RELAY HI	RELAY PASS/ TTL PASS	RELAY LO	TTL FAIL
Hi	make	Refer to Table 6-6.	break	Refer to Table 6-6.
Go	break		break	
Lo	break		make	

Table 6-6 Optical Semiconductor Relay Contact and TTL Logical Output (b)

PASS conditions				RELAY PASS TTL PASS			TTL FAIL		
				Judgment result			Judgment result		
Panel setting	Remote Command (ADC command)			Lo	Go	Hi	Lo	Go	Hi
LGH	LOP1	MIP1	HIP1	L	L	L	H	H	H
--H	LOP0	MIP0	HIP1	H	H	L	L	L	H
-G-	LOP0	MIP1	HIP0	H	L	H	L	H	L
L--	LOP1	MIP0	HIP0	L	H	H	H	L	L
L-H	LOP1	MIP0	HIP1	L	H	L	H	L	H
LG-	LOP1	MIP1	HIP0	L	L	H	H	H	L
-GH	LOP0	MIP1	HIP1	H	L	L	L	H	H
---	LOP0	MIP0	HIP0	H	H	H	L	L	L

H: break or Hi, L: make or Low

6.8 Command Reference

This section describes the command reference of the 7355.

6.8.1 Selecting Command Language

The 7355 can be operated remotely by using the following command formats.

The command selection procedure is shown below.

1. Select the **8 I/F** category from the **MENU** mode and press ▽ to enter the Select layer.
2. Select the LANG parameter by pressing ◁ and ▷, and press ▽ to enter in the Input/Run layer.
3. Select from SCPI, ADC, and 34401 by pressing ◁ and ▷, and press **ENTER** to apply the setting.

   SCPI: SCPI command format

   ADC: ADC's original command format

   34401: 34401A command format

6.8.1.1 7351-Compatible Mode

The 7351-compatible mode is the function to make the 7355 compatible with the 7351 series digital multimeters.
By setting the 7351-compatible mode, the 7355 can be controlled by application programs for the 7351 series.
In the 7351-compatible mode, the following functions are preset and will remain preset even after the initialization is performed.

NOTE:

1. *Even in the 7351-compatible mode, measurement timings and other operations may differ. Be sure to perform operation check before use.*

Function	Preset setting
Number of measurement display digits	5 1/2-digit display
Number of remote output digits	5 1/2-digit output

Select the 7351-compatible mode ON or OFF from "7351" in **9 SYS** in **MENU**.
Press **ENTER** to apply the setting.

6.8.2 Output Data Format

The output data format is a format to remotely output measurement data.
The measurement data consists of a main header part, a sub-header part, a mantissa part, and an exponent part. The number of output digits for the mantissa part can be selected between 6 1/2 digits and 5 1/2 digits. When recalling the measurement data from the measurement data memory, the IEEE 754 binary 64-bit floating point format can be selected.

Select the number of measurement data output digits between 6.5 and 5.5 from "R.DIG" in **8 I/F** in **MENU**.
Press **ENTER** to apply the setting.

Also, select the measurement data memory output between ASCII and binary format from "DFORM" in **8 I/F** in **MENU**.
Press **ENTER** to apply the setting.

When SCPI or 34401 is selected as the command language, the header output is always off.

1. Measurement data output format (1 measurement result)

   With header

   ○○○◉ S ±DD…DD E±DD △△

   1 2 3 4 5

   Without header

   ±DD…DD E±DD △△

   3 4 5

2. Output format of the measurement data memory (ASCII)

   When two or more measurement results are output from the measurement data memory, the comma "," is output as a data delimiter.

   With header

   ○○○◉ S ±DD…DD E±DD , ○○○◉ S ±DD…DD E±DD , …△△

   1 2 3 4 6 1 2 3 4 6 5

   Without header

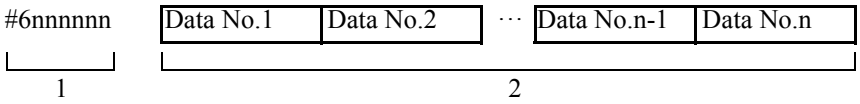
   In the above case, headers (1) and spaces (2) are not output.

1	○○○◉	Main header of 3 characters (○○○) + Sub header of 1 character (◉)
2	S	Space
3	Mantissa	Polarity + Decimal point + 6 or 7 digits
4	Exponent	E + Polarity + 2 digits
5	Block delimiter	
6	Data delimiter	

3. Output format of the measurement data memory (binary, REAL)

When the binary format is selected for measurement data memory output, the following formatted data is output.

This does not apply to the output of single data by the RN or :TRACe:RECAll? command. Also, it does not apply to the RS-232 or USBCDC interface.



1 Header

8-character ASCII data

"#": Indicates binary data

"6": Indicates the number of digits of the next data length. Always 6.

"nnnnnn": Indicates the number of bytes from data No. 1 to data No. n.
Since each data is 8 bytes, the value n × 8 is shown as a 6-digit integer in the ASCII format.

2 Measurement data

64-bit floating point data conforming to IEEE-754.

Data that is over-range or has a calculation error is expressed as NaN (Not a Number).

Main header	Measurement function
DCV	DC voltage measurement (DCV)
ACV	AC voltage measurement (ACV)
R2W	Resistance measurement (2WΩ)
R4W	Resistance measurement (4WΩ)
DCI	DC current measurement (DCI)
ACI	AC current measurement (ACI)
ADV	AC voltage (AC+DC coupling) measurement (ACV (AC+DC))
ADI	AC current (AC+DC coupling) measurement (ACI (AC+DC))
DOD	Diode measurement (Diode)
R2L	Low power resistance measurement (2WΩ)
R4L	Low power resistance measurement (4WΩ)
RCT	Continuity test (Cont)
FRQ	Frequency measurement (Freq)
TC2	Temperature measurement (TEMP-2W) °C
TC3	Temperature measurement (TEMP-3W) °C
TC4	Temperature measurement (TEMP-4W) °C
TF2	Temperature measurement (TEMP-2W) °F
TF3	Temperature measurement (TEMP-3W) °F
TF4	Temperature measurement (TEMP-4W) °F
TK2	Temperature measurement (TEMP-2W) K
TK3	Temperature measurement (TEMP-3W) K
TK4	Temperature measurement (TEMP-4W) K

Sub header	Description			Priority
O	Error information	Range over		1
O		Calculation	Scaling error	1
E			Err D (dB/dBm calculation error)	2
H	Calculation result information	Comparator result	HIGH	3
P			GO	3
L			LOW	3
M		MAX data		4
I		MIN data		4
A		AVE data		4
B		dB data		5
W		dBm data		5
S		Scaling calculation data		6
N		NULL calculation data		7
_	Other than above			8

4. Measurement data

±○○○○○○○○E±○○ (6 1/2-digit output)

±○○○○○○○E±○○ (5 1/2-digit output)

	6 1/2-digit output	5 1/2-digit output
Overload (OL)	±9.999999E+37	±9.99999E+37
Scaling calculation results OVER	±9.999999E+36	±9.99999E+36
dB/dBm calculation error	±9.999999E+35	±9.99999E+35

		6 1/2-digit output	5 1/2-digit output
DCV	200 mV	±ddd.ddddE-03	±ddd.dddE-03
	2000 mV	±dddd.dddE-03	±dddd.ddE-03
	20 V	±dd.dddddE+00	±dd.ddddE+00
	200 V	±ddd.ddddE+00	±ddd.dddE+00
	1000 V	±dddd.dddE+00	±dddd.ddE+00
ACV	200 mV	±ddd.ddddE-03	±ddd.dddE-03
	2000 mV	±dddd.dddE-03	±dddd.ddE-03
	20 V	±dd.dddddE+00	±dd.ddddE+00
	200 V	±ddd.ddddE+00	±ddd.dddE+00
	700 V	±dddd.dddE+00	±dddd.ddE+00
ACV (AC+DC)	200 mV	±ddd.ddddE-03	±ddd.dddE-03
	2000 mV	±dddd.dddE-03	±dddd.ddE-03
	20 V	±dd.dddddE+00	±dd.ddddE+00
	200 V	±ddd.ddddE+00	±ddd.dddE+00
	700 V	±dddd.dddE+00	±dddd.ddE+00
2WΩ	200 Ω	±ddd.ddddE+00	±ddd.dddE+00
	2000 Ω	±dddd.dddE+00	±dddd.ddE+00
	20 kΩ	±dd.dddddE+03	±dd.ddddE+03
	200 kΩ	±ddd.ddddE+03	±ddd.dddE+03
	2000 kΩ	±dddd.dddE+03	±dddd.ddE+03
	20 MΩ	±dd.dddddE+06	±dd.ddddE+06
	200 MΩ	±ddd.ddddE+06	±ddd.dddE+06

		6 1/2-digit output	5 1/2-digit output
4WΩ	200 Ω	±ddd.ddddE+00	±ddd.dddE+00
	2000 Ω	±dddd.dddE+00	±dddd.ddE+00
	20 kΩ	±dd.dddddE+03	±dd.ddddE+03
	200 kΩ	±ddd.ddddE+03	±ddd.dddE+03
	2000 kΩ	±dddd.dddE+03	±dddd.ddE+03
	20 MΩ	±dd.dddddE+06	±dd.ddddE+06
	200 MΩ	±ddd.ddddE+06	±ddd.dddE+06
DCI	200 mA	±ddd.ddddE-03	±ddd.dddE-03
	2000 mA	±dddd.dddE-03	±dddd.ddE-03
	10 A	±dd.dddddE+00	±dd.ddddE+00
ACI	200 mA	±ddd.ddddE-03	±ddd.dddE-03
	2000 mA	±dddd.dddE-03	±dddd.ddE-03
	10 A	±dd.dddddE+00	±dd.ddddE+00
ACI (AC+DC)	200 mA	±ddd.ddddE-03	±ddd.dddE-03
	2000 mA	±dddd.dddE-03	±dddd.ddE-03
	10 A	±dd.dddddE+00	±dd.ddddE+00
FREQ	(<10 Hz)	±d.ddddddE+00	±d.dddddE+00
	(<100 Hz)	±dd.ddddE+00	±dd.ddddE+00
	(<1000 Hz)	±ddd.ddddE+00	±ddd.dddE+00
	(<10 kHz)	±d.ddddddE+03	±d.dddddE+03
	(<100 kHz)	±dd.dddddE+03	±dd.ddddE+03
	(<1000 kHz)	±ddd.ddddE+03	±ddd.dddE+03
LP-2WΩ	200 Ω	±ddd.ddddE+00	±ddd.dddE+00
	2000 Ω	±dddd.dddE+00	±dddd.ddE+00
	20 kΩ	±dd.dddddE+03	±dd.ddddE+03
	200 kΩ	±ddd.ddddE+03	±ddd.dddE+03
	2000 kΩ	±dddd.dddE+03	±dddd.ddE+03
	20 MΩ	±dd.dddddE+06	±dd.ddddE+06
LP-4WΩ	200 Ω	±ddd.ddddE+00	±ddd.dddE+00
	2000 Ω	±dddd.dddE+00	±dddd.ddE+00
	20 kΩ	±dd.dddddE+03	±dd.ddddE+03
	200 kΩ	±ddd.ddddE+03	±ddd.dddE+03
	2000 kΩ	±dddd.dddE+03	±dddd.ddE+03
	20 MΩ	±dd.dddddE+06	±dd.ddddE+06

		6 1/2-digit output	5 1/2-digit output
DIODE	1 range	±dddd.dddE-03	±dddd.ddE-03
CONT	1 range	±dddd.dddE+00	±dddd.ddE+00
TEMP	1 range	±dddd.dddE+00	±dddd.ddE+00

The Talker format in the frequency measurement function depends on the measurement range.

6.8.3 ADC Command Reference

This section describes the ADC command reference of the 7355.

"Default" shows the status when the *RST command is executed.

CAUTION: Ensure the following when using the USB interface.
The device setting conditions are output by reading the data after executing query commands. When executing a query command, set a wait time of 20 ms after the last command.

Item		Command	Description	Default
Measurement	Function	F1	DC voltage measurement (DCV)	✓
		F2	AC voltage measurement (ACV)	
		F3	Resistance measurement (2WΩ)	
		F4	Resistance measurement (4WΩ)	
		F5	DC current measurement (DCI)	
		F6	AC current measurement (ACI)	
		F7	AC voltage (AC+DC coupling) measurement (ACV (AC+DC))	
		F8	AC current (AC+DC coupling) measurement (ACI (AC+DC))	
		F13	Diode measurement (Diode)	
		F20	Low power resistance measurement 2WΩ	
		F21	Low power resistance measurement 4WΩ	
		F22	Continuity test (Cont)	
		F42	Temperature measurement (TEMP, PT/JPT, 2W)	
		F43	Temperature measurement (TEMP, PT/JPT, 3W)	
		F44	Temperature measurement (TEMP, PT/JPT, 4W)	
		F50	Frequency measurement (Freq)	
		F?	Response: F01 to F50	
	Temperature unit	TPU0	Temperature unit °C	✓
		TPU1	Temperature unit °F	
		TPU2	Temperature unit K	
		TPU?	Response: TPU0 to TPU2	
	Platinum resistance thermometer	PT0	Platinum resistance thermometer Pt100	✓
		PT1	Platinum resistance thermometer JPt100	
		PT?	Response: PT0 or PT1	

Item		Command	Description	Default
Measurement	Invalid function	INH?n	Reads the invalid state of the function. n: 1 (DC voltage measurement) to 50 (Frequency measurement) Response 0: Valid state of the function selection 1: Invalid state of the function selection	
	Trigger	*TRG	Trigger command	
	Measurement data output request (RS-232 and LAN only)	MD?	Response: Refer to Section 6.8.2, “Output Data Format”.	
	Measurement data memory	ST0	Store OFF	✓
		ST1	Store ON	
		ST?	Response: ST0 or ST1	
		RN0[,adr]	Cancels recall execution status.	✓
		RN1[,adr]	Sets recall execution status.	
			adr: Recall data number (0 to 19999) (Omittable)	(0)
			When the recall data is read out by the talker function after setting the recall execution status, the operation is as follows. • After the output, the recall data number is incremented. • If there is no data at the specified number, the output is <EE__ +8.888888E+30>. • Data in the memory is not erased even after reading. • Cannot be used in LAN	
		RN?	Response: RN0, ddddd or RN1,ddddd	
		IRDn,m	Readout range setting n: First read number (0 to 19999) m: End read number (0 to 19999) m must be larger than n.	(0,0)
		IRO?	Data reading Response: Refer to Section 6.8.2, “Output Data Format”. (Not available for RS-232 and USB CDC)	
		IRPO?	Number of measurement results readings Response: IRPOdddd	
		IRNO?	Data range reading Response: IRNO+ddddd, +ddddd No data: IRNO+00000, -00001	
		ICL	Measurement data memory initialization	

Item		Command	Description	Default
Trigger system	Start	INI	Leaves the IDLE state.	
	Continue	INIC0	CONTINUOUS OFF	
		INIC1	CONTINUOUS ON	✓
		INIC?	Response: INIC0 or INIC1	
	Trigger abort	ABO	Forcibly moves to the IDLE state.	
	Trigger source	TRS0	IMMEDIATE	✓
		TRS1	MANUAL	
		TRS2	EXTERNAL (Optoin+01, Option+03)	
		TRS3	BUS	
		TRS?	Response: TRS0 to TRS3	
	Sampling interval	TRTn	n: 4 ms to 3600 s	(205 ms)
		TRT?	Response: TRT + d.dddddd E ± dd (The decimal point position varies depending on the setting value.)	
	Trigger delay	TRDn	n: 0 to 3600 (s)	0
		TRD?	Response: TRD + d.dddd E ± dd (The decimal point position varies depending on the setting value.)	
	Sampling count	SPNn	n: 1 to 20000 (times)	(1)
		SPN?	Response: SPNddddd	
	Trigger count	TRNn	n: 1 to 50000 (times)	(1)
		TRN?	Response: TRNNddddd	
	Complete signal width	CW0	Complete signal output width: 5 µs	
		CW1	Complete signal output width: 100 µs	✓
		CW2	Complete signal output width: 1 ms	
		CW?	Response: CW0 to CW2	
	External trigger polarity	TRP0	External trigger polarity: Negative (NEG)	✓
		TRP1	External trigger polarity: Positive (POS)	
		TRP?	Response: TRP0 or TRP1	

Item		Command	Description										Default
Measurement condition	Measurement range		Measurement range setting										
			DCV	ACV	ACV (AC+DC)	2WΩ	4WΩ	DCI	ACI	ACI (AC+DC)	LP-2WΩ	LP-4WΩ	Freq	
		R0	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	✓
		R3	200 mV	200 mV	200 mV	200 Ω	200 Ω	-	-	-	200 Ω	200 Ω	(200 mV)	
		R4	2000 mV	2000 mV	2000 mV	2000 Ω	2000 Ω	-	-	-	2000 Ω	2000 Ω	(2000 mV)	
		R5	20 V	20 V	20 V	20 kΩ	20 kΩ	-	-	-	20 kΩ	20 kΩ	(20 V)	
		R6	200 V	200 V	200 V	200 kΩ	200 kΩ	200 mA	200 mA	200 mA	200 kΩ	200 kΩ	(200 V)	
		R7	1000 V	700 V	700 V	2000 kΩ	2000 kΩ	2000 mA	2000 mA	2000 mA	2000 kΩ	2000 kΩ	(700 V)	
		R8	-	-	-	20 MΩ	20 MΩ	10 A	10 A	10 A	20 MΩ	20 MΩ	-	
		R9	-	-	-	200 MΩ	200 MΩ	-	-	-	-	-	-	
		R?	Response: R0, R3 to R9											
	Range Fix	RX	Range switching from AUTO to MANUAL											
	Sampling rate	PR0	FREE (PR0 is returned when the sampling interval or integration time is changed.)											
		PR1	FAST											
		PR2	MED											
		PR3	SLOW1											
		PR4	SLOW2 (The same as PR3 in the frequency measurement)											✓
		PR?	Response: PR0 to PR4											
	Integration time *1	ITPn	Sets the integration time in PLC units. n: 0.02 PLC to 100 PLC											(10)
		ITP?	Reply: ITP + d.dddddE ± dd (The decimal point position depends on the setting.)											
		ITSn	Sets the integration time in seconds. n: 0.0002 to 2											(0.2)
		ITS?	Response: ITS + d.dddddE ± dd (The decimal point position depends on the setting.)											
	Display digits	RE3	3 1/2-digit display											
		RE4	4 1/2-digit display											
		RE5	5 1/2-digit display											
		RE6	6 1/2-digit display											✓
		RE?	Response: RE3 to RE6											

*1: The integration time is enabled in all functions except for the frequency measurement.

Item		Command	Description	Default
Measurement condition	Auto zero	AZ0	OFF	
		AZ1	ON	✓
		AZ2	ONCE (Auto zero OFF after auto-zero execution)	
		AZ?	Response: AZ0 or AZ1	
	Continuity threshold constant	KOMn	n: 1 to 1000 (Ω), Setting resolution: 1 Ω	(10)
		KOM?	Response: KOMdddd	
Calculation	NULL calculation	NL0	OFF	✓
		NL1	ON	
		NL?	NL0 or NL1	
		KNLn	NULL constant setting n: -9999999.E + 6 to +9999999.E + 6 Setting resolution: 0.000001E-9 Note: Disabled when the NULL calculation is OFF.	(0)
		KNL?	Response: KNL±d.dddddd E±dd *2	
	Smoothing calculation	SM0	OFF	✓
		SM1	ON	
		SM?	Response: SM0 or SM1	
		TIn	Smoothing count n: 2 to 100 (times)	(10)
		TI?	Response: TIddd	
	Scaling calculation	SC0	OFF	✓
		SC1	ON	
		SC?	Response: SC0 or SC1	
		KAn	Constant A (Zero cannot be set.) n: -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(1)
		KAM	Sets the measured value to constant A.	
		KA?	Response: KA±d.dddddd E±dd *2	
		KBn	Constant B n: -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(0)
		KBM	Sets the measured value to constant B.	
		KB?	Response: KB ± d.dddddd E ± dd *2	
		KCn	Constant C n: -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(1)

*2: The decimal point position in the response is fixed.

Item		Command	Description	Default
Calculation	Scaling calculation	KCM	Sets the measured value to constant C.	
		KC?	Response: KC±d.dddddE±dd *2	
	dB/dBm calculation	DB0	OFF	✓
		DB1	dB calculation ON	
		DB2	dBm calculation ON	
		DB?	Response: DB0 to DB2	
		KDn	Constant D n: 0.000001E-9 to 9999999.E+6	(1)
		KDM	Sets the measured value to constant D.	
		KD?	Response: KD±d.dddddE±dd *2	
	MAX/MIN calculation ○○○: Header	MN0	OFF	✓
		MN1	ON	
		MN?	Response: MN0 or MN1	
		MAX?	MAX value reading Response: ○○○M ± d.dddddE ± dd *2	
		MIN?	MIN value reading Response: ○○○I ± d.dddddE ± dd *2	
		AVE?	AVE value reading Response: ○○○A ± d.dddddE ± dd *2	
		AVN?	Measurement count reading Response: AVN + d.dddddE + dd *2	
	Comparator calculation	CO0	OFF	✓
		CO1	ON	
		CO?	Response: CO0 or CO1	
		HIn	Constant HIGH n: -9999999.E + 6 to +9999999.E + 6 Setting resolution: 0.000001E-9	(0)
		HIM	Sets the measured value to constant HIGH.	
		HI?	Response: HI ± d.dddddE ± dd *2	
		LOn	Constant LOW n: -9999999.E + 6 to +9999999.E + 6 Setting resolution: 0.000001E-9	(0)
		LOM	Sets the measured value to constant LOW.	
		LO?	Response: LO ± d.dddddE ± dd *2	

*2: The decimal point position in the response is fixed.

Item		Command	Description	Default
Calculation	PASS condition range setting	LOP0	Calculation result LO is not specified as the PASS condition.	✓
		LOP1	Calculation result LO is specified as the PASS condition.	
		LOP?	Response: LOP0 or LOP1	
		MIP0	Calculation result GO is not specified as the PASS condition.	
		MIP1	Calculation result GO is specified as the PASS condition.	✓
		MIP?	Response: MIP0 or MIP1	
		HIP0	Calculation result HI is not specified as the PASS condition.	✓
		HIP1	Calculation result HI is specified as the PASS condition.	
		HIP?	Response: HIP0 or HIP1	
	Statistical calculation	SIRDn,m	Range setting and execution of statistical calculations n, m: 0 to 19999 Note: 1. An error occurs if no data exists in the setting range in the measurement data memory. 2. No parameters can be omitted.	(0,-1)
		SIRD?	Response: SIRDddddd, ddddd Default: SIRD + 00000, -00001	
		SCNT?	Sample count reading Response: SCNT + d.dddddE + dd *2	
		SMAX?	Maximum value reading Response: SMAX ± d.dddddE ± dd *2	
		SMIN?	Minimum value reading Response: SMIN ± d.dddddE ± dd *2	
		SAVE?	Average value reading Response: SAVE ± d.dddddE ± dd *2	
		SSIG?	Standard deviation value reading Response: SSIG + d.dddddE ± dd *2 However, if the sample count is 1 or less: Outputs SSIG + 9.999999E + 30.	
		SPTP?	MAX-MIN reading Response: SPTP ± d.dddddE ± dd *2	
	Digital output	DO0	Digital output disabled (comparator output enabled)	✓
		DO1	Digital output enabled (comparator output disabled)	
		DO?	Response: DO0 or DO1	

*2: The decimal point position in the response is fixed.

<table>
<tr><th colspan="2">Item</th><th>Command</th><th>Description</th><th>Default</th></tr>
<tr><td>Calculation</td><td>Digital output</td><td>DOSn</td><td>Controls arbitrarily the TTL signal or relay contact signal corresponding to 5 bits.
n (0 to 31) indicates a number that is converted from 5-bit binary to decimal.
<table><tr><td>Pin name</td><td>TTL FAIL</td><td>TTL PASS</td><td>RELAY PASS</td><td>RELAY HI</td><td>RELAY LO</td></tr><tr><td>Output signal</td><td colspan="2">TTL output</td><td colspan="3">Relay contact</td></tr><tr><td>Pin number</td><td>9</td><td>1</td><td>2-6</td><td>3-7</td><td>4-8</td></tr><tr><td>Bit</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Setting</td><td colspan="5">0: H, 1: L</td></tr></table>H: break or Hi, L: make or Low
Example: DOS0
<table><tr><td>Pin name</td><td>TTL FAIL</td><td>TTL PASS</td><td>RELAY PASS</td><td>RELAY HI</td><td>RELAY LO</td></tr><tr><td>Output</td><td>Hi</td><td>Hi</td><td>break</td><td>break</td><td>break</td></tr></table></td><td>(0)</td></tr>
<tr><td></td><td></td><td>DOS?</td><td>Response: DOS0 to DOS31</td><td></td></tr>
<tr><td>System</td><td>Buzzer</td><td>BZ0</td><td>OFF</td><td>✓</td></tr>
<tr><td></td><td></td><td>BZ1</td><td>ON</td><td></td></tr>
<tr><td></td><td></td><td>BZ?</td><td>Response: BZ0 or BZ1</td><td></td></tr>
<tr><td></td><td>Comparator calculation result buzzer *3</td><td>BP0</td><td>OFF</td><td>✓</td></tr>
<tr><td></td><td></td><td>BP1</td><td>The buzzer sounds when the comparator calculation result is FAIL.</td><td></td></tr>
<tr><td></td><td></td><td>BP2</td><td>The buzzer sounds when the comparator calculation result is PASS.</td><td></td></tr>
<tr><td></td><td></td><td>BP?</td><td>Response: BP0 to BP2</td><td></td></tr>
<tr><td></td><td>Measurement data display</td><td>DS0</td><td>OFF</td><td></td></tr>
<tr><td></td><td></td><td>DS1</td><td>ON</td><td>✓</td></tr>
<tr><td></td><td></td><td>DS?</td><td>Response: DS0 or DS1</td><td></td></tr>
<tr><td></td><td>Power supply frequency (Auto setup)</td><td>LF0</td><td>50 Hz</td><td></td></tr>
<tr><td></td><td></td><td>LF1</td><td>60 Hz</td><td></td></tr>
<tr><td></td><td></td><td>LF?</td><td>Response: LF0 or LF1</td><td></td></tr>
<tr><td></td><td>Relay switching count</td><td>RLY?</td><td>Maximum value response: ddddddddd</td><td></td></tr>
<tr><td></td><td>Initialization</td><td>*RST</td><td>Parameter initialization</td><td></td></tr>
<tr><td></td><td></td><td>Z</td><td>Former model-compatible commands</td><td></td></tr>
<tr><td></td><td></td><td>C</td><td>Device clear</td><td></td></tr>
</table>

*3: When setting to BP1 or BP2, the buzzer setting automatically turns ON (BZ1).

Item		Command	Description	Default
System	Instrument information	*IDN?	Response: ADC Corp.,nnnn,xxxxxxxxx,yyyyy ADC Corp.: Manufacturer (9 characters) nnnn: Model name (4 characters) “7355” xxxxxxxxx: Serial number (9 characters) yyyyy: Firmware revision (5 characters)	
	Option information	*OPT?	Query of option information Response: Without option “0” With option “GPIB I/F”, “RS232 I/F”, “COMP I/F”	
	MAC address	MACA?	Query of MAC address Response: BC:42:B4:nn:nn:nn	
	Measurement data header	H0	OFF	
		H1	ON	✓
		H?	Response: H0 or H1	
	Block delimiter	DL0	CRLF<EOI>	✓
		DL1	LF	
		DL2	<EOI>	
		DL?	Response: DL0 to DL2	
	Output format	DFO0	ASCII	✓
		DFO1	Binary (64-bit floating pint) (Invalid for RS-232 and USBCDC)	
		DFO?	Response: DFO0 or DFO1	
	Measurement data output digits	RDG0	5 1/2-digit output	
		RDG1	6 1/2-digit output	✓
		RDG?	Response: RDG0 or RDG1	
	SRQ	S0	ON	✓
		S1	OFF *Turns ON when the power is turned ON.	
		S?	Response: S0 or S1	
	Event history	EVR	Resets the event history readout. (to allow reading from the oldest one)	
		EVH?	Reads the event history sequentially. (After 20 readings, the history is read again from the first reading.) Response: ±ddd,"xxxxxxxxxx" Event character string (up to 80 characters) Error code • Up to 20 error events are stored. • Events after the 21st event are overwritten starting with the oldest event. • The stored contents of the event history cannot be deleted.	

Item		Command	Description	Default
System	Status	*CLS	Clears each status byte.	
		*STB?	Reads the Status Byte Register Response: ddd	
		*SREn	Sets the Service Request Enable Register (SRER). n: 0 to 255 (However, bit 6 cannot be set.)	
		*SRE?	Response: ddd	
		*ESR?	Reads the Standard Event Status Register (SESR). Response: ddd	
		*ESEn	Sets the Standard Event Status Enable Register (SESER). n: 0 to 255	
		*ESE?	Response: ddd	
		MSR?	Reads the Measurement Event Register (MER). Response: ddddd	
		MSEn	Sets the Measurement Event Enable Register (MEER). n: 0 to 65535	
		MSE?	Response: ddddd	
		QSR?	Reads the Questionable Event Register (QER). Response: ddddd	
		QSEn	Sets the Questionable Event Enable Register (QEER). n: 0 to 65535	
		QSE?	Response: ddddd	
		OSR?	Reads the Operation Event Register (OER). Response: ddddd	
		OSEn	Sets the Operation Event Enable Register (OEER). n: 0 to 65535	
		OSE?	Response: ddddd	
		*PSCn	Sets the Power On Clear Flag. Num: -32767 to +32767 When n is other than zero (0), SRER and SESER are cleared as power on. When n is set to zero (0), SRER and SESER are not cleared even at power on.	
		*PSC?	Response: 0 or 1 (When set to other than zero (0).)	

Item		Command	Description	Default
System	Operation complete	*OPC	Sets LSB in SESR after all operations are complete.	
		*OPC?	Returns 1 after all operations are complete.	
		*WAI	Waits for all operations to be completed (GPIB only)	
	Self-test	*TST?	Executes the self-test and reads the result. Response: No error “0” Error “1”	
	Setting parameter	*SAVn	Saves parameters to area n of the non-volatile memory. n: 0 to 3	
		SINI	Sets the factory-default values to areas 0 to 3 of the non-volatile memory.	
		*RCLn	Loads parameters from area n of the nonvolatile memory. n: 0 to 3	
		RINI	Loads the factory-default values as setting parameters.	
	Error log	ERR?	Reads the error contents. Response: ±ddd, "xxxxxxxx" Error character string (up to 80 characters) Error code Errors are read out one by one from the beginning of the error log number. • Up to 20 errors are stored. Errors are output in FIFO format. • If more than 20 errors occur, the last stored error is overwritten with -350, "Queue overflow". Any errors after this are not saved. • If there are no errors, +000, "No error" is returned.	
	7351-compatible mode	CMP0	7351-compatible mode OFF	✓
		CMP1	7351-compatible mode ON	
		CMP?	Response: CMP0 or CMP1	
	Calibration	CAL0	Calibration mode OFF (Writes the calibration coefficient when exiting calibration.)	✓
		CAL1	Calibration mode ON	
		CAL?	Response: CAL0 or CAL1	
		XOUT	Cancels the calibration mode (OFF). (Not write the calibration coefficient.)	
		PCn	STD value input (display count value) n: 0 to ±1999999	
		XDTn	STD value input (display value) n: Value displayed by STD	

Item		Command	Description	Default
System	Calibration	CMNT "str"	Sets the calibration information (calibration memo). The string must be enclosed in either a single or double quotation mark. The string can store up to 50 characters (half-width alphanumeric characters).	
		CMNT?	Response: CMNT "xxxx ???xxx" Up to 50 characters	

6.8.4 SCPI Command Reference

This section describes the SCPI command reference of the 7355.

The "SCPI command" shows the format in which commands are sent from an external controller to the 7355. The format consists of a command part and a parameter part. The delimiter between the command part and the parameter part is a space.

When there are multiple parameters, each parameter is separated by a comma (,). If there are three points (…) between commas, the parameters are omitted from the description.

For example, "< number 1>,.... < number 4>" requires four parameters: < number 1>, < number 2>, < number 3>, < number 4>.

Lower-case alphabetical characters in the format indicate that they can be omitted.

For example, ":CALibration:ABORt" can be abbreviated to ":CAL:ABOR".

The definitions of the symbols used in the format are as follows:

< >	Shows a parameter required for sending a command
[ ]	Shows optional data. It can be omitted.
{}	Shows that only one item need be selected from multiple items.
|	Written in curly brackets {..} and used as a delimiter for multiple items.

If the parameter is a numeric or string type, it is enclosed in angle brackets < >.
If the parameter is a choice type, it is enclosed in braces { }.

Choices are separated by (|), and if there are multiple expressions in one choice, they are separated by (,).

Example 1: {ON,1|OFF,0}
ON and 1, OFF and 0 have the same meaning, respectively, and represent a choice among four options: ON, 1, OFF, and 0.

If the input value has a range, it is represented by [Minimum|Default|Maximum] or [Minimum|Maximum].

In this manual, parameter types are described in the following format:

<int_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to an integer in the 7355.
<real_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to a real number in the 7355.
<str>	Shows alphanumeric symbols enclosed in quotation (') or double quotation (") marks.
<func>	Shows the same character string as the query response to the [:SENSe]:FUNCtion command.

Example 2: {MAXimum|MINimum|MEASurement}
If MAXimum is selected, the maximum value is set; if MINimum is selected, the minimum value is set. If MEASurement is selected, a measurement is taken and its value is set.

MEMO: The default value of the continue setting can be set to ON in the MENU (I/F-CONT).

NOTE: When using the SCPI commands via USB, note the following point.
After sending a setting command, send a query command and receive its response. Sending more than one setting command in succession may cause unintended behavior or an error to occur.
*As query command for setting commands, error check by ":SYSTem:ERRor?" and completion check by "*OPC?" are generally used.*

1. Measurement function and trigger system-related commands

Item		SCPI command	Description		Default
Measurement	Measurement function	[:SENSe]:FUNCtion "<Function>"	"<Function>"		
			"VOLTage:DC"	DC voltage measurement (DCV)	✓
			"VOLTage:AC"	AC voltage measurement (ACV)	
			"RESistance"	Resistance measurement (2WΩ)	
			"FRESistance"	Resistance measurement (4WΩ)	
			"CURRent:DC"	DC current measurement (DCI)	
			"CURRent:DC"	AC current measurement (ACI)	
			"VOLTage:ACDC"	AC voltage AC+DC coupling measurement ACV (AC+DC)	
			"CURRent:ACDC"	AC current AC+DC coupling measurement ACI (AC+DC)	
			"DIODe"	Diode measurement (Diode)	
			"RESistance:LPOWer"	Low-power resistance measurement (LP-2WΩ)	
			"FRESistance:LPOWer"	Low-power resistance measurement (LP-4WΩ)	
			"CONTinuity"	Continuity test (Cont)	
			"FREQuency"	Frequency measurement (Freq)	
			"TEMPerature"	Temperature measurement (TEMP)	
		[:SENSe]:FUNCtion?	Response: "VOLT:DC", "VOLT:AC", "RES", "FRES", "CURR:DC", "CURR:AC", "VOLT:ACDC", "CURR:ACDC", "DIOD", "RES:LPOW", "FRES:LPOW", "FREQ", "CONT" or "TEMP"		

Item		SCPI command	Description	Default
Measurement	Measurement function	:CONFigure:<Function> {<real1>|MAXimum|MINimum}, {<real2>|MAXimum|MINimum} *1	Switches to the specified measurement function. <real1>: Range setting (0 ≤ Setting value ≤ Maximum value in measurement range) (The maximum value for frequency measurement is 750 V.) Not available for continuity, diode and temperature measurements. <real2>:Resolution setting (sampling rate) For more information, see the description of "[:SENSe]:<Function> :RESolution". <reral1> and <real2> are omittable.	
		:CONFigure?	Response: "Function +d.dddddd E ±dd, +d.dddddE±dd" "+d.dddddE±dd" shows the measurement range and resolution.	
	Function prohibited	:INHibit:FUNCtion:<Function>? *1	Reads out the state of function setting prohibition. Response: 1 or 0 (1: Disabled, 0: Enabled)	
	Trigger	*TRG	Trigger command	
	Measurement operation	:MEASure?	Sets the DCV measurement function and reads the measurement data.	
		:MEASure:<Function>?, {<real1>|MAXimum|MINimum}, {<real2>|MAXimum|MINimum} *1	Performs the measurement operation and reads the measured value. <real1>: Range setting <real2>: Resolution setting (sampling rate) <reral1> and <real2> are omittable.	
		:READ?	Initializes the trigger system and reads the measurement data.	
		:FETCh?	Reads the measurement data only.	
			Response: Refer to "Output Data Format."	

*1: <Function> Specifies the function (but not enclosed in double quotes) of "measurement function."

Item		SCPI command	Description	Default
Measurement	Measurement data memory	:TRACe:STATe {ON,1|OFF,0}	Store setting	
			ON, 1: ON	
			OFF, 0: OFF	✓
		:TRACe:STATe?	Response: 1 or 0	
		:TRACe:NUMBer <int>,<int>	Reading range setting <int>: 0 to 19999	
		:TRACe:RECAll? <int>	Reads one data of the specified data number from the measurement data memory. <int>: 0 to 19999 Response: Same as the measurement data • When <int> is omitted, data of +1 to the previous data number is output. • If there is no data in the specified number, the response is <+8.888888E+30>. • Cannot be used in LAN.	(0)
		:TRACe:RECAll:NUMBer?	Query for the current specified data number Response: 0 to 19999	
		:TRACe:DATA?	Data reading Response: Refer to Section 6.8.2, "Output Data Format" (Not available for RS-232 and USB CDC)	
		:TRACe:DATA:POINts?	Reads the number of data. Response: ddddd	
		:TRACe:DATA:NUMBer?	Reads the data range Response: +ddddd, +ddddd	
		:TRACe:CLEar	Initialization	
Trigger system	Start	:INITiate	Exits the IDLE state and moves to the trigger waiting state.	
	Continue	:INITiate:CONTinuous {ON,1|OFF,0}	ON, 1: CONTINUOUS ON	
			OFF, 0: CONTINUOUS OFF	✓*2
		:INITiate:CONTinuous?	Response: 1 or 0	
	Trigger abort	:ABORt	Moves to the IDEL state forcibly.	
	Trigger source	:TRIGger:SOURce {IMMediate|MANual|EXTernal|BUS}	IMMediate	✓
			MANual	
			EXTernal (OPT+01, OPT+03)	
			BUS	
		:TRIGger:SOURce?	Response: IMM, MAN, EXT or BUS	

*2: The default value can be selected by "CONT" from [8 I/F] from [MENU]. The default value of "CONT" is OFF.

Item		SCPI command	Description	Default
Trigger system	Trigger delay	:TRIGger:DELay {<real>|MAXimum|MINimum}	<real>: 0 to 3600 (s)	(0)
		:TRIGger:DELay? {MAXimum|MINimum}	Response: +d.ddddE ± dd	
	Trigger count	:TRIGger:COUNt {<int>|MAXimum|MINimum}	<real>: 1 to 50000	(1)
		:TRIGger:COUNt? [{MAXimum|MINimum}]	Response: ddddd	
	Sampling count	:SAMPle:COUNt {<int>|MAXimum|MINimum}	<int>: 1 to 20000	(1)
		:SAMPle:COUNt? {MAXimum|MINimum}	Response: ddddd	
	Sampling interval	:TRIGger:TIMer {<real>|MAXimum|MINimum}	<real>: 0.004 to 3600 (s)	(0.205)
		:TRIGger:TIMer? [{MAXimum|MINimum}]	Response: +d.dddddE ± dd	

2. Measurement condition commands

Item		SCPI command	Description	Default
Measurement condition	Measurement range	[:SENSe]:<Function>:RANGe :AUTO {ON,1|OFF,0} *1	Auto range setting	
			ON, 1: Auto range ON	✓
			OFF, 0: Auto range OFF Not available for continuity, diode, temperature and frequency measurements.	
		[:SENSe]:<Function>:RANGe :AUTO? *1	Response: 1 or 0	
		[:SENSe]:<Function>:RANGe {<real>|MAXimum|MINimum} *1	Fixed range setting <real>: 0 ≤ Setting value ≤ Maximum value in measurement range) (The maximum value for frequency measurement is 750 V.) Not available for continuity, diode and temperature measurements.	
		[:SENSe]:<Function >:RANGe? {MAXimum|MINimum} *1	Response: +d.dddddE ±dd Returns the maximum value in the specified range.	
	Temperature measurement unit	:UNIT:TEMPerature {C|F|K}	C: °C	✓
			F: °F	
			K: K	
		:UNIT:TEMPerature?	Response: C, F or K	
	Platinum resistance thermometer measurement type	[:SENSe]:TEMPerature:TRANsducer:TYPE {RTD|TRTD|FRTD}	RTD: PT sensor 2W	✓
			TRTD: PT sensor 3W	
			FRTD: PT sensor 4W	
		[:SENSe]:TEMPerature:TRANsducer:TYPE?	Response: RTD, TRTD or FRTD	
	Platinum resistance thermometer standard	[:SENSe]:TEMPerature:TRANsducer: STANdard {PT100|JPT100}	PT100: PT100 standard	✓
			JPT100: JPT100 standard	
		[:SENSe]:TEMPerature:TRANsducer: STANdard?	Response: PT100 or JPT100	

*1: <Function> Specifies the function (but not enclosed in double quotes) of "measurement function."

Item		SCPI command	Description	Default
Measurement condition	Sampling rate	[:SENSe]:<Function>:SRATe {FAST|MED|SLOW|SSLow} *1	FAST: FAST	
			MED: MED	
			SLOW: SLOW1	
			SSLow: SLOW2	✓
			The setting apples to all functions.	
		[:SENSe]:<Function>:SRATe? *1	Response: FAST, MED, SLOW or SSL	
	Measurement resolution (sampling rate)	[:SENSe]:<Function>:RESolution {<real>|MAXimum|MINimum}	<real>: The sampling rate is set by the following inputs SLOW2: 1/10 of the minimum resolution of each measurement range SLOW1: Value of the minimum resolution for each measurement range MED: 10 times the value of the minimum resolution of each measurement range FAST: 100 times the value of the minimum resolution for each measurement range Not available for continuity, diode and temperature measurements. The setting apples to all functions.	(SLOW 2)
		[:SENSe]:<Function>:RESolution? {MAXimum|MINimum}	Response: +d.dddddE±dd	
	Measurement display digits	[:SENSe]:<Function>:DIGits {3|4|5|6|MAXimum|MINimum} *1	3: 3 1/2-digit display	
			4: 4 1/2-digit display	
			5: 5 1/2-digit display	
			6: 6 1/2-digit display	✓
			The setting apples to all functions.	
		[:SENSe]:<Function>:DIGits? {MAXimum|MINimum} *1	Response: 3, 4, 5 or 6	
	Integration time (PLC)	[:SENSe]:<Function>:NPLCycles {<real>|MAXimum|MINimum}	<real>: 0.02 to 100 PLC Not available for FREQ. The setting apples to all functions.	(10)
		[:SENSe]:<Function>:NPLCycles? {MAXimum|MINimum}	Response: +d.ddddddE ± dd	
	Integration time (s)	[:SENSe]:< Function>:APERture {<real>|MAXimum|MINimum}	<real>: 0.0002 to 2 s Not available for FREQ. The setting apples to all functions.	(0.2)
		[:SENSe]:< Function>:APERture? {MAXimum|MINimum}	Response: +d.dddddE ± dd	

*1: <Function> Specifies the function (but not enclosed in double quotes) of "measurement function."

Item		SCPI command	Description	Default
Measurement condition	Auto zero	[:SENSe]:ZERO:AUTO {ON,1|OFF,0|ONCE}	ON, 1: ON	✓
			OFF, 0: OFF	
			ONCE (Auto zero OFF after auto-zero execution)	
		[:SENSe]:ZERO:AUTO?	Response: 1 or 0	
	Continuity threshold constant	[:SENSe]:CONTinuity:THReshold {<int>|MAXimum|MINimum}	<real>: 1 to 1000 (Ω) Setting resolution: 1 Ω	(10)
		[:SENSe]:CONTinuity:THReshold? {MAXimum|MINimum}	Response: dddd	

3. Calculation function commands

Item		SCPI command	Description	Default
Calculation	NULL calculation	[:SENSe]:<Function>:REFerence :STATe {ON,1|OFF,0} *1	ON, 1: ON	
			OFF, 0: OFF	✓
		[:SENSe]:<Function>:REFerence :STATe? *1	Response: 1 or 0	
		[:SENSe]:<Function>:REFerence {<real>|MAXimum|MINimum} *1	NULL constant setting <real>: -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9 Note: Disabled when the NULL calculation is OFF.	(0)
		[:SENSe]:<Function>:REFerence? {MAXimum|MINimum}	Response: ±d.dddddddE ±dd	
	Smoothing calculation	:CALCulate:SMOothing {ON,1|OFF,0}	ON, 1: ON	
			OFF, 0: OFF	✓
		:CALCulate:SMOothing?	Response: 1 or 0	
		:CALCulate:SMOothing:POINts {<int>|MAXimum|MINimum}	Smoothing count setting <int>: 2 to 100	(10)
		:CALCulate:SMOothing:POINts? {MAXimum|MINimum}	Response: ddd	
	Scaling calculation	:CALCulate:SCALing {ON,1|OFF,0}	ON, 1: ON	
			OFF, 0: OFF	✓
		:CALCulate:SCALing?	Response: 1 or 0	
		:CALCulate:SCALing:A {<real>|MAXimum|MINimum| MEASurement}	Constant A (Zero cannot be set.) <real> -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(1)
		:CALCulate:SCALing:B {<real>|MAXimum|MINimum| MEASurement}	Constant B (Zero cannot be set.) <real> -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(0)
		:CALCulate:SCALing:C {<real>|MAXimum|MINimum| MEASurement}	Constant C (Zero cannot be set.) <real> -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(1)
		:CALCulate:SCALing:A? {MAXimum|MINimum}	Response: ±d.dddddddE ±dd	
		:CALCulate:SCALing:B? {MAXimum|MINimum}		
		:CALCulate:SCALing:C? {MAXimum|MINimum}		

*1: <Function> Specifies the function (but not enclosed in double quotes) of "measurement function."

Item		SCPI command	Description	Default
Calculation	dB/dBm calculation	:CALCulate:DB {DB|DBM|OFF}	DB: dB calculation ON Enabled for voltage or current measurement functions	
			DBM: dBm calculation ON Enabled for voltage measurement functions.	
			OFF: dB/dBm calculation OFF	✓
		:CALCulate:DB?	Response: DB, DBM or OFF	
		:CALCulate:DB:D {<real>|MAXimum|MINimum|MEASurement}	Constant D setting <real>:0.000001E-9 to 9999999.E+6	(1)
		:CALCulate:DB:D? {MAXimum|MINimum}	Response: ±d.dddddd E ±dd	
	MAX/MIN calculation	:CALCulate:AVERage {ON,1|OFF,0}	ON, 1: ON	
			OFF, 0: OFF	✓
		:CALCulate:AVERage?	Response: 1 or 0	
		:CALCulate:AVERage:MAXimum?	MAX value reading	
		:CALCulate:AVERage:MINimum?	MIN value reading	
		:CALCulate:AVERage:AVERage?	AVE value reading	
		:CALCulate:AVERage:COUNt?	Measurement count reading	
			Response: ±d.dddddd E ±dd	
	Comparator calculation	:CALCulate:LIMit {ON,1|OFF,0}	ON, 1: ON	
			OFF, 0: OFF	✓
		:CALCulate:LIMit?	Response: 1 or 0	
		:CALCulate:LIMit:UPPer {<real>|MAXimum|MINimum|MEASurement}	Constant HIGH setting <real> -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(0)
		:CALCulate:LIMit:LOWer {<real>|MAXimum|MINimum|MEASurement}	Constant LOW setting <real> -9999999.E+6 to +9999999.E+6 Setting resolution: 0.000001E-9	(0)
		:CALCulate:LIMit:UPPer? {MAXimum|MINimum}	Response: ±d.dddddd E ±dd	
		:CALCulate:LIMit:LOWer? {MAXimum|MINimum}		

Item		SCPI command	Description	Default
Calculation	PASS condition range setting	:CALCulate:LIMit:PASS:LOWer {ON,1|OFF,0}	ON, 1: Calculation result LO is specified as PASS.	
			OFF, 0: Calculation result LO is not specified as PASS.	✓
		:CALCulate:LIMit:PASS:LOWer?	Response: 1 or 0	
		:CALCulate:LIMit:PASS:MED {ON,1|OFF,0}	ON, 1: Calculation result GO is specified as PASS.	✓
			OFF, 0: Calculation result GO is not specified as PASS.	
		:CALCulate:LIMit:PASS:MED?	Response: 1 or 0	
		:CALCulate:LIMit:PASS:UPPer {ON,1|OFF,0}	ON, 1: Calculation result HI is specified as PASS.	
			OFF, 0: Calculation result HI is not specified as PASS.	✓
		:CALCulate:LIMit:PASS:UPPer?	Response: 1 or 0	
	Statistical calculation	:CALCulate:STATistics <int>,<int>	Range setting and execution of statistical calculations <int>: 0 to 19999	
		:CALCulate:STATistics:COUNt?	Sample count reading Response: ±d.dddddd±dd	
		:CALCulate:STATistics:MAXimum?	Maximum value reading Response: ±d.dddddd±dd	
		:CALCulate:STATistics:MINimum?	Minimum value reading Response: ±d.dddddd±dd	
		:CALCulate:STATistics:AVERage?	Average value reading Response: ±d.dddddd±dd	
		:CALCulate:STATistics:DEViation?	Standard deviation value reading Response: ±d.dddddd±dd	
		:CALCulate:STATistics:MAXMin?	MAX-MIN reading Response: ±d.dddddd±dd	
	Digital output disabled/enabled	:DOUTput:STATe {ON,1|OFF,0}	OFF, 0: Digital output disabled	✓
			ON, 1: Digital output enabled	
		:DOUTput:STATe?	Response: 1 or 0	
	Digital output setting	:DOUTput:BITS {0-31}	Digital output setting Refer to the item “Digital output” in Section 6.8.3, “ADC Command Reference”.	(0)
		:DOUTput:BITS? [{MAXimum|MINimum}]	Response: 0 to 31	

4. System-related commands

Item		SCPI command	Description	Default
System	Buzzer *3	:CALCulate:LIMit:BEEPer {OFF|FAIL|PASS}	OFF: OFF	✓
			FAIL: The buzzer sounds when the comparator calculation result is FAIL.	
			PASS: The buzzer sounds when the comparator calculation result is PASS.	
		:CALCulate:LIMit:BEEPer?	Response: OFF, FAIL or PASS	
		:SYSTem:BEEPer:STATe {ON,1|OFF,0}	ON, 1: Buzzer ON	
			OFF, 0: Buzzer OFF	✓
		:SYSTem:BEEPer:STATe?	Response: 1 or 0	
	Measurement data display	:DISPlay {ON,1|OFF,0}	ON, 1: ON	✓
			OFF, 0: OFF	
		:DISPlay?	Response: 1 or 0	
	Power frequency	[:SENSe]:LFRQency {50|60}	50: 50 Hz	
			60: 60 Hz	
		[:SENSe]:LFRQency?	Response: 50 or 60	
	Initialization	*RST	Instrument initialization	
	Instrument information	*IDN?	Response: ADC Corp.,nnnn,xxxxxxxxx,yyyyy ADC Corp.: Manufacturer (9 characters) nnnn: Model name (4 characters) “7355” xxxxxxxxx: Serial number (9 characters) yyyyy: Firmware revision (5 characters)	
	Option information	*OPT?	Query of option information Response: Without option “0” With option “GPIB I/F”, “RS232 I/F”, “COMP I/F”	
	MAC address	:SYSTem:MAC?	Query of MAC address Response: BC:42:B4:nn:nn:nn	
	Block delimiter	:SYSTem:GPIB:DELimiter:BLOCk {CRLF|LF|EOI}	CRLF: CRLF+EOI	✓
			LF: LF	
			EOI: EOI	
		:SYSTem:GPIB:DELimiter:BLOCk?	Response: CRLF or EOI	
	Output format	:FORMat:DATA {ASCii|REAL64}	ASCii: ASCII	✓
			REAL64: REAL 64 (Invalid for RS-232 and USBCDC)	
		:FORMat:DATA?	ASC or REAL	
	Output data digits	:FORMat:DATA:DIGits {5|6|MAXimum|MINimum}	5: 5 1/2-digit output	
			6: 6 1/2-digit output	✓
		:FORMat:DATA:DIGits? [{MAXimum|MINimum}]	Response: 5 or 6	

*3: When “:CALCulate:LIMit:BEEPer” is set to FAIL or PASS, the buzzer setting automatically turns ON.

Item		SCPI command	Description	Default
System	Status	*CLS	Clears each status byte.	
		*STB?	Reads the Status Byte Register Response: ddd	
		*SRE <int>	Sets the Service Request Enable Register. <int>: 0 to 255 (However, bit 6 cannot be set.)	
		*SRE?	Response: ddd	
		*ESR?	Reads the Standard Event Status Register. Response: ddd	
		*ESE <int>	Sets the Standard Event Status Enable Register. <int>: 0 to 255	
		*ESE?	Response: ddd	
		:STATus:MEASurement:EVENt?	Reads the Measurement Event Register. Response: ddddd	
		:STATus:MEASurement:ENABle <int>	Sets the Measurement Event Enable Register. <int>: 0 to 65535	
		:STATus:MEASurement:ENABle?	Response: ddddd	
		:STATus:QUEStionable:EVENt?	Reads the Questionable Event Register. Response: ddddd	
		:STATus:QUEStionable:ENABle <int>	Sets the Questionable Event Enable Register. <int>: 0 to 65535	
		:STATus:QUEStionable:ENABle?	Response: ddddd	
		:STATus:OPERation:EVENt?	Reads the Operation Event Register. Response: ddddd	
		:STATus:OPERation:ENABle <int>	Sets the Operation Event Enable Register. <int>: 0 to 65535	
		:STATus:OPERation:ENABle?	Response: ddddd	
		*PSC <int>	<int>: -32767 to +32767 When n is other than zero (0), SRER and SESER are cleared as power on. When n is set to zero (0), SRER and SESER are not cleared even at power on.	
		*PSC?	Response: 0 or 1 (When set to other than zero (0).)	

Item		SCPI command	Description	Default
System	Operation complete	*OPC	Sets the “Operation Complete” bit (bit 0) in the Standard Event Status register after executing the command.	
		*OPC?	Response: 1 (After all operations are complete)	
		*WAI	Waits for all operations to be completed (GPIB only)	
	Self-test	*TST?	Executes the self-test and reads the result. It takes execution time. Read out the result after executing *WAI. Response: 0 ... Pass 1 ... Fail	
	7351-compatible mode	:SYSTem:7351compatible:STATe {ON,1|OFF,0}	OFF, 0: 7351-compatible mode OFF	✓
			ON, 1: 7351-compatible mode ON	
		:SYSTem:7351compatible:STATe?	Response: 0 or 1	
	Calibratoin	:CALibration:SECure:STATe {ON,1|OFF,0}	ON, 1: Calibration mode ON	
			OFF, 0: Calibration mode OFF (Writes the calibration coefficient when exiting calibration.)	✓
		:CALibration:SECure:STATe?	Response: 1 or 0	
		:CALibration:ABORt	Cancels the calibration mode (OFF). (Not write the calibration coefficient.)	
		:CALibration <int>	STD value input (display count value) <int>: 0 to ±9999999	
		:CALibration:VALue <real>	STD value input (display value) <int>: Value displayed by STD	
		:CALibration:STRing <"str">	Sets the calibration information. <“str”>: The string must be enclosed in a double quotation mark. The string can store up to 50 characters (half-width alphanumeric characters).	
		:CALibration:STRing?	Response: “xxxx ⋯ xxx” └ Up to 50 characters	
	Setting parameter	*SAV <int>	<int>: 0 to 3 Saves the setting parameters to area [int] in the non-volatile memory.	
		*RCL <int>	<int>: 0 to 3 Loads data from area [int] in the non-volatile memory as the setting parameters.	

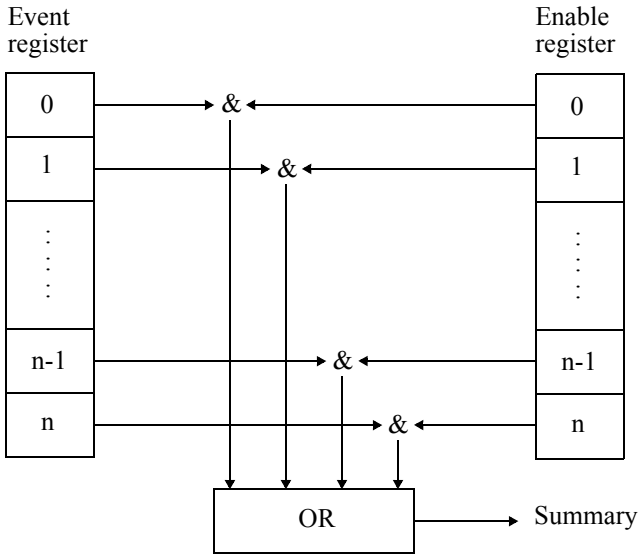
Item		SCPI command	Description	Default
System	Error log reading	:SYSTem:ERRor?	Reads the error contents.	
			Response: ±ddd, "xxxxxxxx" Error character string (up to 80 characters) Error code • Up to 20 errors are stored. Errors are output in FIFO format. • If more than 20 errors occur, the last stored error is overwritten with -350, "Queue overflow". Any errors after this are not saved. • If there are no errors, +0, "No error" is returned.	
	Event history reading	:SYSTem:EVENt?	Reads the event history contents.	
	Event history reading reset	:SYSTem:EVENt:RESet	Resets the event history reading pointer.	
	SCPI version	:SYSTem:VERSion?	Reads the SCPI version. Response: 1991.0	

6.8.5 Status Register Structure

The 7355 has a hierarchical status register structure that conforms to the IEEE standard 488.2-1987 and can send various statuses of itself to the controller. The following explains an operational model of the status structure and allocation of events.

1. Status register

   The 7355 employs a status register model as defined by the IEEE standard 488.2-1987. A status register consists of an event register and an enable register.



- Event register

  The event register latches and maintains the status for each event. (It may also hold changes.)
  Once the register is set, it remains set until it is read out by a query or cleared by *CLS.
  Data cannot be written into the event register.

- Enable register

  The enable register specifies for which bits in the Event Register a valid status summary should be generated. The logical AND operation is executed between the enable register and the event register, and the OR result is generated as a summary. The summary is written into the Status Byte Register.
  Data can be written into the enable register.

The 7355 has the following five types of status registers.

- Status Byte Register (STB)
- Standard Event Status Register (SESR)
- Measurement Event Register (MER)
- Questionable Event Register (QER)
- Operation Event Register (OER)

The status register structure in the 7355 is shown in Figure 6-4 and Figure 6-5.

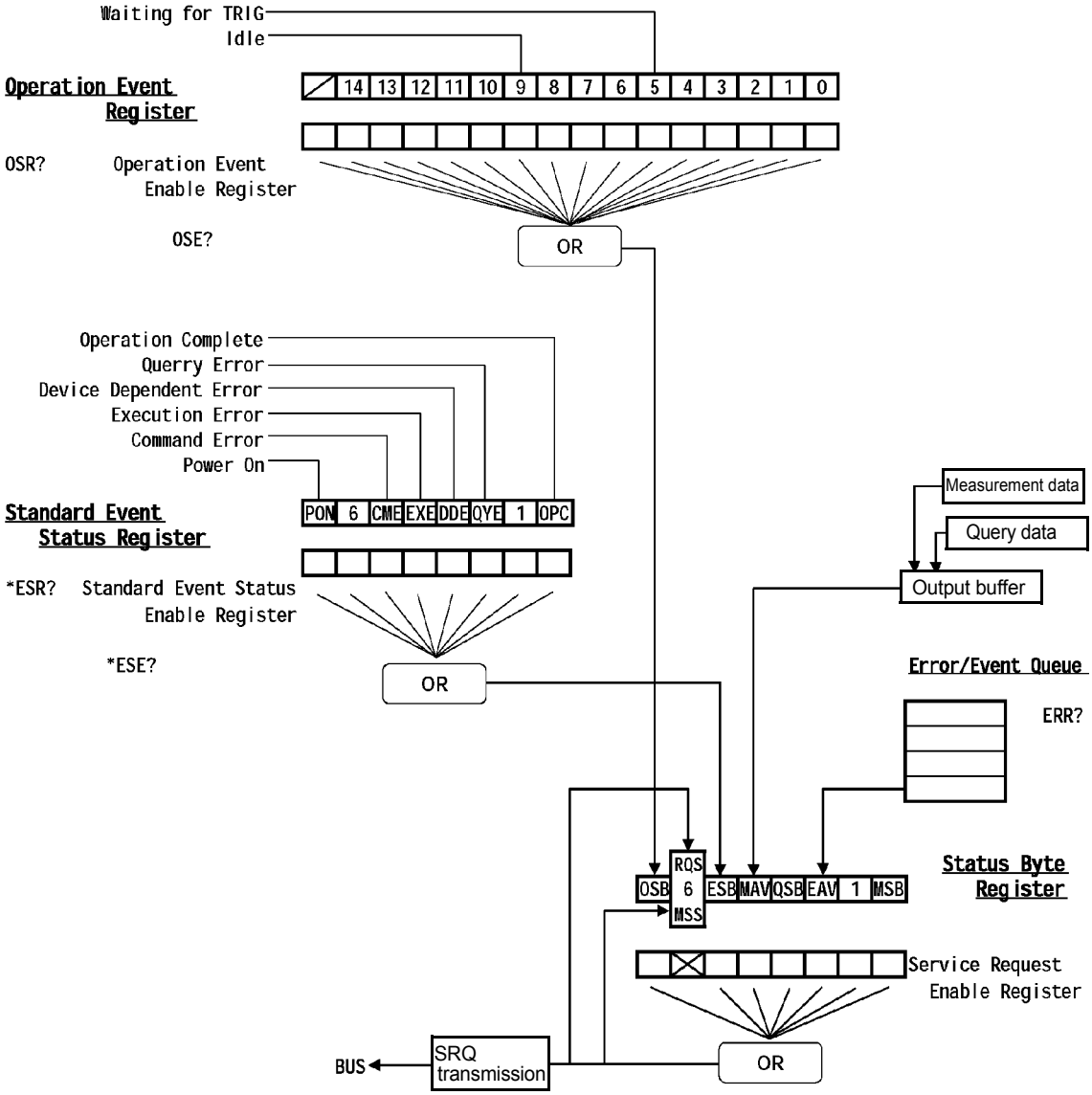


Figure 6-4 Status Register Structure (1 of 2)

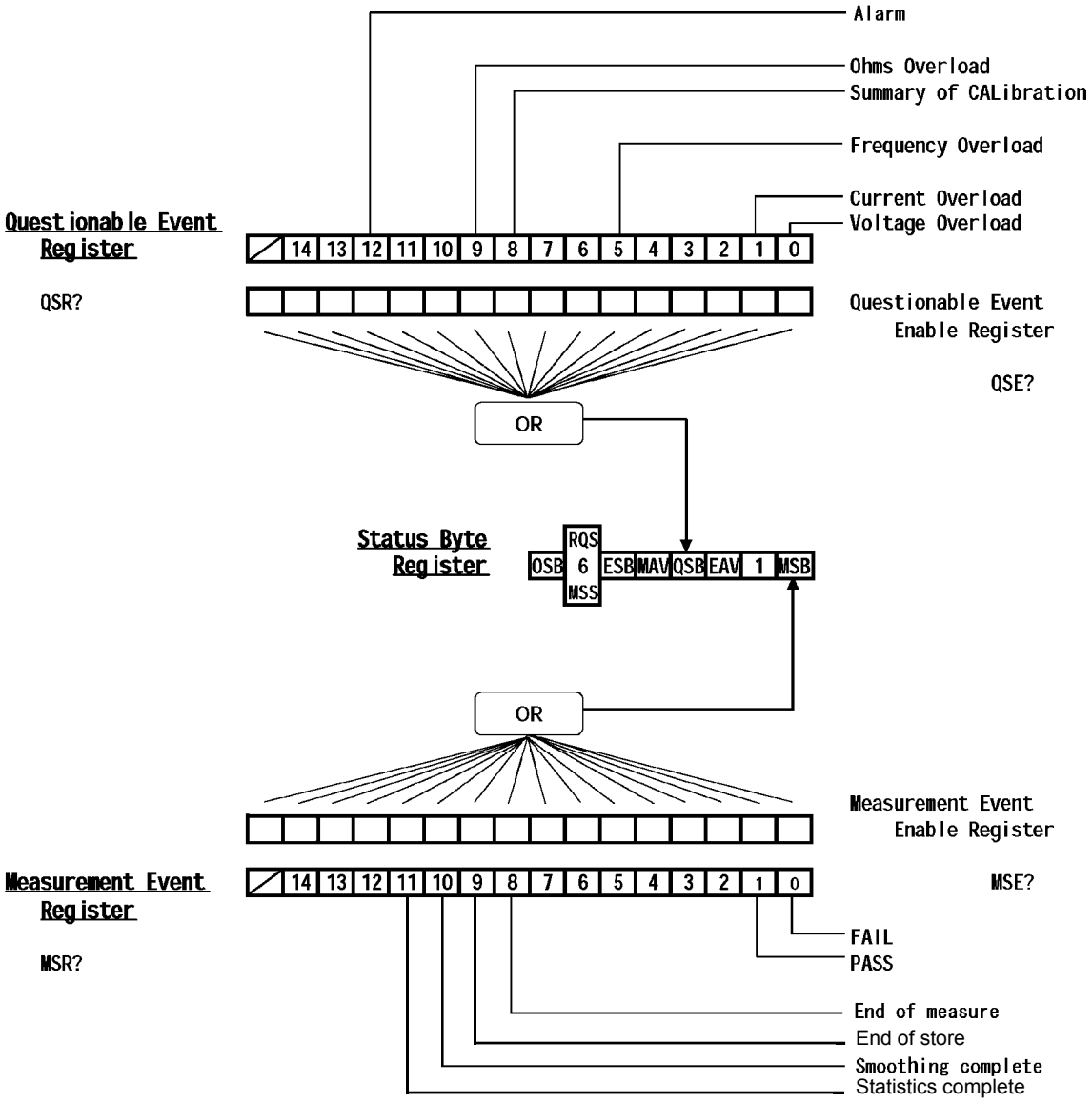


Figure 6-5 Status Register Structure (2 of 2)

2. Event enable register

   Each event register has an enable register that decides which bit is to be enabled. The enable register sets the relevant bits in decimal values.

   - Service Request Enable Register (SRER) setting: *SRE
   - Standard Event Status Enable Register (SESER) setting: *ESE
   - Measurement Event Enable Register (MEER): MSE
   - Questionable Event Enable Register (QEER): QSE
   - Operation Event Enable Register (QEER): QSE

3. Status Byte Register

   The Status Byte Register summarizes the information from the status registers. A summary of this Status Byte Register is transmitted to the controller as a service request. Consequently, the Status Byte Register is slightly different from the status register in structure. The Status Byte Register is explained in the following:
   The following figure shows the structure of the Status Byte Register.

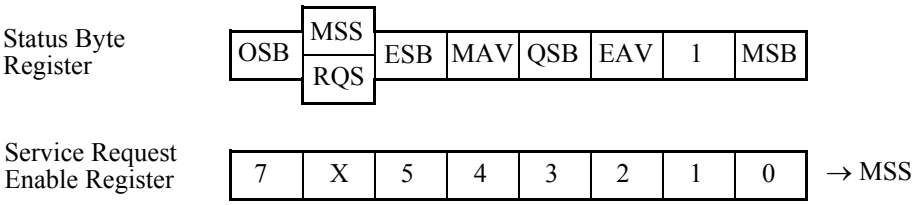


Figure 6-6 Status Byte Register Structure

The Status Byte Register conforms to the status register except the following three points.

- The summary of the Status Byte Register is written into bit 6 of the Status Byte Register.
- Bit 6 of the Enable Register is always enabled and cannot be changed.
- Bit 6 (MSS) of the Status Byte Register writes RQS (request service).

This register responds to a serial poll from the controller. When responding to the serial poll, bit 0 to bit 5, bit 7, and RQS of the Status Byte Register are read, after which RQS is reset to 0. Other bits are not cleared until their factors become 0.
The Status Byte Register, RQS, and MSS can be cleared by executing "*CLS." Accompanying this, the SRQ line also becomes false.

The following table describes each bit of the Status Byte Register

Table 6-7 Status Byte Register (STB)

bit	Name	Description
0	MSB Measurement Summary Bit	ON: Set to 1 when any event of the Measurement Event Register occurs and the bit is set to 1, if the corresponding bit of the Measurement Event Enable Register is set to 1. OFF: Set to 0 when the Measurement Event Register is cleared by being read.
1	Not in use	Always 0
2	EAV Error Available	ON: Set to 1 when the error information is stored in the Error Queue. OFF: Set to 0 when the error information is read from the Error Queue and no data exists in the Error Queue.
3	QSB Questionable Summary Bit	ON: Set to 1 when any event of the Questionable Event Register occurs and the bit is set to 1, if the corresponding bit of the Questionable Event Enable Register is set to 1. OFF: Set to 0 when the Questionable Event Register is cleared by being read.
4	MAV Message Available	ON: Set to 1 when the output data is entered in the output buffer. OFF: Set to 0 when the output buffer is read and becomes empty.
5	ESB Standard Event Status	ON: Set to 1 when any event of SESR occurs and the bit is set to 1, if the corresponding bit of SESER is set to 1. OFF: Set to 0 when the Standard Event Status Register is cleared by being read (*ESR?).
6	MSS Master Summary	ON: Set to 1 when any event of STB occurs, if the corresponding bit of SRER is set to 1.
	RQS Request Service	ON: Set to 1 when MSS is set to 1 and SRQ is generated. OFF: Set to 0 when the Status Byte Register is read by a serial poll.
7	OSB Operation Summary Bit	ON: Set to 1 when any event of the Operation Event Register occurs and the bit is set to 1, if the corresponding bit of the Operation Event Enable Register is set to 1. OFF: Set to 0 when the Operation Event Register is cleared by being read.

Common conditions in which the Status Byte Register is cleared

- Cleared when the power is turned ON.
- Cleared by *CLS except that MAV is not cleared if data exists in the output buffer.
- Not cleared even if read by *STB?

Conditions in which the Service Request Enable Register is cleared

- Cleared when the power is turned ON. (When the PSC flag is set to 1)
- Cleared when the *SRE0 command is executed.

4. Standard Event Status Register

The following tables show the allocations of the Standard Event Status Register.

Table 6-8 Standard Event Status Register (SESR)

bit	Name	Description
0	OPC Operation Complete	ON: Set to 1 when all operations are complete after receiving the *OPC command.
1	Not in use	Always 0
2	QYE Query Error	ON: Set to 1 under any of the following conditions. • When the data is read, no data exists in the output buffer and no data to be output exists. • When data in the output buffer is lost.
3	DDE Device Dependent Error	ON: Set to 1 when a device-dependent error occurs.
4	EXE Execution Error	ON: Set to 1 when a received command is not currently executable. Set to 1 when the command parameter contains an error.
5	CME Command Error	ON: Set to 1 when a received command is incorrectly spelled.
6	Not in use	Always 0
7	PON Power On	ON: Set to 1 when the power is turned ON.

Common conditions in which the Standard Event Status Register is cleared

- Cleared when the power is turned ON.
- Cleared by *CLS.
- Cleared when read by *ESR?

Conditions in which the Standard Event Status Enable Register is cleared

- Cleared when the power is turned ON. (When the PSC flag is 1)
- Cleared when the *ESE0 command is executed

5. Measurement Event Register

The following tables show the allocations of the Measurement Event Register.

Table 6-9 Measurement Event Register (MER)

bit	Name	Description
0	FL FAIL	ON: Set to 1 when the comparator calculation result satisfies the FAIL conditions.
1	PS PASS	ON: Set to 1 when the comparator calculation result satisfies the PASS conditions.
2	Not in use	Always 0
3	Not in use	Always 0
4	Not in use	Always 0
5	Not in use	Always 0
6	Not in use	Always 0
7	Not in use	Always 0
8	EOM End of measure	ON: Set to 1 when the measurement is complete.
9	EOS End of store	ON: Set to 1 when measurement data cannot be stored in the measurement memory any longer.
10	SM Smoothing complete	ON: Set to 1 when the smoothing count reaches the specified number.
11	STAT Statistics complete	ON: Set to 1 when the statistical calculation is complete.
12	Not in use	Always 0
13	Not in use	Always 0
14	Not in use	Always 0
15	Not in use	Always 0

Common conditions in which the Measurement Event Register is cleared

- Cleared when the power is turned ON.
- Cleared by *CLS.
- Cleared when the register contents are read by using the STATus :MEASurement:EVENt? command
- Cleared when the register contents are read by using the MSR? command

Conditions in which the Measurement Event Enable Register is cleared

- Cleared when the power is turned ON.
- Cleared when the STATus:MEASurement:ENABle 0 command is executed
- Cleared when the MSE0 command is executed

6. Questionable Event Register

The following tables show the allocations of the Questionable Event Register.

Table 6-10 Questionable Event Register (QER)

bit	Name	Description
0	Voltage Overload	ON: Set to 1 when OL occurs in the voltage or diode measurement.
1	Current Overload	ON: Set to 1 when OL occurs in the current measurement.
2	Not in use	Always 0
3	Not in use	Always 0
4	Not in use	Always 0
5	Frequency Overload	ON: Set to 1 when OL occurs in the frequency measurement.
6	Not in use	Always 0
7	Not in use	Always 0
8	Summary of Calibration	ON: Set 1 when the default calibration values or the calibration values acquired at the previous power ON are used due to the calibration data SUM error in the power ON check.
9	Ohms Overload	ON: Set to 1 when OL occurs in the resistance measurement.
10	Not in use	Always 0
11	Not in use	Always 0
12	Alarm	ON: Set to 1 when an alarm occurs in the measurement.
13	Not in use	Always 0
14	Not in use	Always 0
15	Not in use	Always 0

Common conditions in which the Questionable Event Register is cleared

- Cleared when the power is turned ON.
- Cleared by *CLS.
- Cleared when the register contents are read by using the :STATus:QUEStionable[:EVENt]? command
- Cleared when the register contents are read by using the QSR? command

Conditions in which the Questionable Event Enable Register is cleared

- Cleared when the power is turned ON.
- Cleared when the :STATus:QUEStionable:ENABle 0 command is executed
- Cleared when the QSE0 command is executed

7. Operation Event Register

The following tables show the allocations of the Operation Event Register.

Table 6-11 Operation Event Register (OER)

bit	Name	Description
0	Not in use	Always 0
1	Not in use	Always 0
2	Not in use	Always 0
3	Not in use	Always 0
4	Not in use	Always 0
5	Waiting for TRIG	ON: Set to 1 when entering the trigger layer.
6	Not in use	Always 0
7	Not in use	Always 0
8	Not in use	Always 0
9	Idle	ON: Set to 1 when entering the idle state.
10	Not in use	Always 0
11	Not in use	Always 0
12	Not in use	Always 0
13	Not in use	Always 0
14	Not in use	Always 0
15	Not in use	Always 0

Common conditions in which the Operation Event Register is cleared

- Cleared when the power is turned ON.
- Cleared by *CLS.
- Cleared when the register contents are read by using the :STATus:OPERation[:EVENt]? command
- Cleared when the register contents are read by using the OSR? command

Conditions in which the Operation Event Enable Register is cleared

- Cleared when the power is turned ON.
- Cleared when the :STATus:OPERation:ENABle 0 command is executed
- Cleared when the OSE0 command is executed

6.9 Sample Programs

This section describes program examples to remotely control the 7355 via USB, GPIB, RS232 or LAN. Download these programs from ADC's website.

https://www.adcmt.com/en/download/sampleprogram

Table 6-12 Sample Program List

	Description
Example 1	Captures measurement data with the trigger source set to IMMEDIATE in the Auto range of the DC voltage function.
Example 2	In the 20 kΩ range of the 2-wire resistance function, sends the measurement trigger command with the trigger source set to BUS, detects the end of measurement in the status register, and captures the data. (In the case of GPIB, the end of measurement is detected by SRQ.)
Example 2	In the 20 V range of the DC voltage function, performs 1000 measurements at the highest speed, stores the measurement data in the data memory, and captures them in ASCII format. In the case of GPIB, the example is to capture the measurement data in binary format.

- Each program offers versions for USB, USB CDC, GPIB, RS232, and LAN.
  OS: Microsoft Windows 10 Pro (32 bits, 64 bits)
  Language: Microsoft Excel Visual Basic for Applications (VBA) (32 bits, 64 bits)

  For GPIB
  GPIB adapter: National Instruments NI GPIB-USB-HS

  For USB
  Module: ADC Instrument USB Driver

  Download the ADC Instrument USB Driver from ADC's website.
  https://www.adcmt.com/en/download/usbdriver

7. SPECIFICATIONS

7.1 Specifications

Unless otherwise specified, the measurement accuracy is guaranteed for one year under the following conditions: temperature of 23±5 °C, relative humidity of 85 % or less (75 % or less in resistance measurement of 20 MΩ or more and low power resistance measurement of 2 MΩ or more).

7.1.1 DC Voltage Measurement (DCV)

5 1/2 digits

Input terminal	Range	Maximum display		Resolution		Input impedance	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
V-COM	200 mV	199.99	199.999	10 µV	1 µV	1 GΩ or higher	0.012+2	0.012+7	0.012+6	15+0.85	15+2
	2000 mV	1999.9	1999.99	100 µV	10 µV	1 GΩ or higher	0.011+2	0.011+4	0.011+2	15+0.2	15+1.5
	20 V	19.999	19.9999	1 mV	100 µV	10 MΩ ±1 %	0.015+2	0.015+5	0.015+5	20+0.25	20+1.5
	200 V	199.99	199.999	10 mV	1 mV	10 MΩ ±1 %	0.015+2	0.015+4	0.015+3	20+0.25	20+1.5
	1000 V	1099.9	1099.99	100 mV	10 mV	10 MΩ ±1 %	0.015+2	0.015+4	0.015+3	20+0.25	20+1.5

6 1/2 digits

Input terminal	Range	Maximum display		Resolution		Input impedance	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
V-COM	200 mV	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 nV	Same as 5 1/2 digits	Same as 5 1/2 digits	0.012+70	0.012+60	15+8.5	15+20
	2000 mV		1999.999		1 µV			0.011+40	0.011+20	15+2	15+15
	20 V		19.99999		10 µV			0.015+50	0.015+50	20+2.5	20+15
	200 V		199.9999		100 µV			0.015+40	0.015+30	20+2.5	20+15
	1000 V		1099.999		1 mV			0.015+40	0.015+30	20+2.5	20+15

*1: When the Auto-zero is ON
When the Auto-zero is OFF, add 2 digits (5 1/2 digits) or 20 digits (6 1/2 digits) to the digit error. “IT > 400 ms” is specified only when the Auto-zero is ON.

Additional error depending on integration time setting

Integration time	Measurement accuracy	Description
200 µs ≤ IT < 2 ms	See FAST.	4 1/2-digit display, additional error ± 10 digits
IT = 2 ms		-
2 ms < IT ≤ 15 ms		Convert the digit error to 5 1/2-digit display.
15 ms < IT < 1 PLC		Convert the digit error to 5 1/2-digit display or 6 1/2-digit display.
1 PLC ≤ IT < 100 ms	See MED.	*2
100 ms ≤ IT ≤ 200 ms	See SLOW1, 2.	*2
200 ms < IT ≤ 2 s		Additional error ± 5 digits (for 5 1/2-digit display) or ± 50 digits (for 6 1/2-digit display) *2

*2: For integer multiples of 1 PLC

Maximum allowable input voltage

V terminal – COM terminal	1000 V peak
COM terminal – chassis	500 V peak

Noise reduction ratio

RATE setting	Integration time	Effective CMRR *3		NMRR
		DC	50/60 Hz±0.08 %	50/60 Hz±0.08 %
MED/SLOW1,2	Integer multiples of 1 PLC	130 dB	120 dB	60 dB
FAST	Other	130 dB	60 dB	0 dB

*3: Unbalanced impedance of 1 kΩ

7.1.2 AC Voltage Measurement (ACV, ACV (AC+DC))

Measurement method: True RMS measurement, RMS display

Input range: 5% or more of a full scale

Crest factor: 3 : 1 at a full scale (Restricted to the maximum allowable input)

Temperature coefficient: (1/10 of measurement accuracy that includes the additional error)/˚C in each range and frequency range)

Response time: Approx. 1 s (Time until the measurement value reaches within 0.1% of the final value in the same range)

ACV 5 1/2 digits

Input terminal	Range	Maximum display		Resolution		Input impedance
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
V-COM	200 mV	199.99	199.999	10 µV	1 µV	1 MΩ ±2 %, 140 pF or less
	2000 mV	1999.9	1999.99	100 µV	10 µV	
	20 V	19.999	19.9999	1 mV	100 µV	
	200 V	199.99	199.999	10 mV	1 mV	
	700 V	749.9	749.99	100 mV	10 mV	

RATE setting	Range	Measurement accuracy*1 ±(% of reading + digits)				
		20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz
FAST	200 mV	0.38+14	0.11+12	0.1+12	0.25+15	0.7+24
	2000 mV	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	20 V	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	200 V	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	700 V	0.38+10	0.11+10	0.1+10	-	-
MED/ SLOW1,2	200 mV	0.38+140	0.11+120	0.1+100	0.25+150	0.7+240
	2000 mV	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
	20 V	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
	200 V	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
	700 V	0.38+100	0.11+100	0.1+100	-	-

ACV 6 1/2 digits

Input terminal	Range	Maximum display		Resolution		Input impedance
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
V-COM	200 mV	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 nV	Same as 5 1/2 digits
	2000 mV		1999.999		1 μV	
	20 V		19.99999		10 μV	
	200 V		199.9999		100 μV	
	700 V		749.999		1 mV	

RATE setting	Range	Measurement accuracy*1 ±(% of reading + digits)				
		20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz
MED/ SLOW1,2	200 mV	0.38+1400	0.11+1200	0.1+1000	0.25+1500	0.7+2400
	2000 mV	0.38+1400	0.11+1200	0.1+1000	0.2+1500	0.6+2400
	20 V	0.38+1400	0.11+1200	0.1+1000	0.2+1500	0.6+2400
	200 V	0.38+1400	0.11+1200	0.1+1000	0.2+1500	0.6+2400
	700 V	0.38+1000	0.11+1000	0.1+1000	-	-

*1: For sine wave input

ACV additional error depending on integration time setting

Integration time	Measurement accuracy	Description
200 μs ≤ IT < 2 ms	See FAST.	4 1/2-digit display, additional error ± 10 digits
IT = 2 ms		-
2 ms < IT ≤ 15 ms		Convert the digit error to 5 1/2-digit display.
15 ms < IT < 1 PLC		Convert the digit error to 5 1/2-digit display or 6 1/2-digit display.
1 PLC ≤ IT ≤ 2 s	See MED/ SLOW1, 2.	For integer multiples of 1 PLC

ACV (AC + DC) 5 1/2-digit display, 6 1/2-digit display

Input terminal	Range	Maximum display		Resolution		Input impedance
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
V-COM	200 mV	199.9	199.99	100 µV	10 µV	1 MΩ ±12 %, 140 pF or less
	2000 mV	1999	1999.9	1 mV	100 µV	
	20 V	19.99	19.999	10 mV	1 mV	
	200 V	199.9	199.99	100 mV	10 mV	
	700 V	749	749.9	1 V	100 mV	

RATE setting	Range	Measurement accuracy*2 ±(% of reading + digits)				
		20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz
FAST/ MED/ SLOW1,2	200 mV	0.38+14	0.11+12	0.1+12	0.25+15	0.7+24
	2000 mV	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	20 V	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	200 V	0.38+14	0.11+12	0.1+12	0.2+15	0.6+24
	700 V	0.38+10	0.11+10	0.1+10	-	-

*2: For sine wave input

ACV (AC+DC) additional error depending on integration time setting

Integration time	Reading error ± (% of reading)		Digit error (Value of measurement accuracy)	Description
	20 Hz to 45 Hz	45 Hz to 100 Hz		
200 µs ≤ IT ≤ 2 ms	3	0.15	×1	-
2 ms < IT < 1 PLC			×10 (4 1/2-digit conversion)	-
1 PLC ≤ IT < 100 ms	2	0.015	×1	For integer multiples of 1 PLC
100 ms ≤ IT ≤ 2 s	-	-		

Additional error depending on crest factor (For non-sine wave input)

1 to 2	0.05 % of range
2 to 3	0.15 % of range

Maximum allowable input voltage

V terminal – COM terminal	1000 V peak, 700 V rms, 2.2×10^7 V Hz
COM terminal – chassis	500 V peak

7.1.3 Resistance Measurement (2WΩ, 4WΩ, LP-2WΩ, LP-4WΩ)

Resistance measurement 5 1/2 digits

2WΩ/4WΩ resistance measurement

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*1, *2 ±(% of reading +digits)			Temperature coefficient*3 ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
200 Ω	199.99	199.999	10 mΩ	1 mΩ	1 mA	0.02+2	0.02+9	0.02+8	20+1	20+2
2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	1 mA	0.02+2	0.02+4	0.014+3	15+0.25	15+1.5
20 kΩ	19.999	19.9999	1 Ω	100 mΩ	100 µA	0.02+2	0.02+4	0.014+3	15+0.25	15+1.5
200 kΩ	199.99	199.999	10 Ω	1 Ω	10 µA	0.02+2	0.02+4	0.02+3	20+0.25	20+1.5
2000 kΩ	1999.9	1999.99	100 Ω	10 Ω	1 µA	0.03+2	0.03+8	0.03+6	35+2	35+5
20 MΩ	19.999	19.9999	1 kΩ	100 Ω	100 nA	0.2+2	0.2+8	0.2+6	155+2	155+5
200 MΩ	199.99	199.999	10 kΩ	1 kΩ	10 nA	1.5+2	1.5+8	1.5+6	1500+2	1500+5

2WΩ /4WΩ low-power resistance measurement

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*1, *2 ±(% of reading +digits)			Temperature coefficient*3 ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
200 Ω	199.99	199.999	10 mΩ	1 mΩ	1 mA	0.02+2	0.02+9	0.02+8	20+1	20+5
2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	100 µA	0.03+2	0.03+9	0.03+8	20+1	20+3
20 kΩ	19.999	19.9999	1 Ω	100 mΩ	10 µA	0.03+2	0.03+9	0.03+8	20+1	20+3
200 kΩ	199.99	199.999	10 Ω	1 Ω	1 µA	0.03+2	0.03+9	0.03+8	30+1	30+3
2000 kΩ	1999.9	1999.99	100 Ω	10 Ω	100 nA	0.2+2	0.2+12	0.2+10	150+2	150+5
20 MΩ	19.999	19.9999	1 kΩ	100 Ω	10 nA	1.5+5	1.5+12	1.5+10	1500+2	1500+5

*1: For 2WΩ, add the resistance of the measurement cable and the offset error of 0.3 Ω.
*2: When the Auto-zero is ON
When the Auto-zero is OFF, add 2 digits (5 1/2 digits) or 20 digits (6 1/2 digits) to the digit error. “IT > 400 ms” is specified only when the Auto-zero is ON.
*3: For 4WΩ, the temperature coefficient is applied with Auto-zero ON.

Resistance measurement 6 1/2 digits

2WΩ/4WΩ resistance measurement

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*4, *5 ±(% of reading +digits)			Temperature coefficient*6 ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
200 Ω	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 μΩ	Same as 5 1/2 digits	Same as 5 1/2 digits	0.02+90	0.02+80	20+10	20+20
2000 Ω		1999.999		1 mΩ			0.02+40	0.014+30	15+2.5	15+15
20 kΩ		19.99999		10 mΩ			0.02+40	0.014+30	15+2.5	15+15
200 kΩ		199.9999		100 mΩ			0.02+40	0.02+30	20+2.5	20+15
2000 kΩ		1999.999		1 Ω			0.03+80	0.03+60	35+20	35+50
20 MΩ		19.99999		10 Ω			0.2+80	0.2+60	155+20	155+50
200 MΩ		199.9999		100 Ω			1.5+80	1.5+60	1500+20	1500+50

2WΩ /4WΩ low-power resistance measurement

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*4, *5 ±(% of reading +digits)			Temperature coefficient*6 ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
200 Ω	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 μΩ	Same as 5 1/2 digits	Same as 5 1/2 digits	0.02+90	0.02+80	20+10	20+50
2000 Ω		1999.999		1 mΩ			0.03+90	0.03+80	20+10	20+30
20 kΩ		19.99999		10 mΩ			0.03+90	0.03+80	20+10	20+30
200 kΩ		199.9999		100 mΩ			0.03+90	0.03+80	30+10	30+30
2000 kΩ		1999.999		1 Ω			0.2+120	0.2+100	150+20	150+50
20 MΩ		19.99999		10 Ω			1.5+120	1.5+100	1500+20	1500+50

*4: For 2WΩ, add the resistance of the measurement cable and the offset error of 0.3 Ω.

*5: When the Auto-zero is ON
When the Auto-zero is OFF, add 2 digits (5 1/2 digits) or 20 digits (6 1/2 digits) to the digit error. “IT > 400 ms” is specified only when the Auto-zero is ON.

*6: For 4WΩ, the temperature coefficient is applied with Auto-zero ON.

Additional error depending on integration time setting

Integration time	Measurement accuracy	Description
200 μs ≤ IT < 2 ms	See FAST.	4 1/2-digit display, additional error ± 10 digits
IT = 2 ms		-
2 ms < IT ≤ 15 ms		Convert the digit error to 5 1/2-digit display.
15 ms < IT < 1 PLC		Convert the digit error to 5 1/2-digit display or 6 1/2-digit display.
1 PLC ≤ IT < 100 ms	See MED.	*7
100 ms ≤ IT ≤ 200 ms	See SLOW1,2.	*7
200 ms < IT ≤ 2 s		Additional error ± 5 digits (for 5 1/2-digit display) or ± 50 digits (for 6 1/2-digit display) *7

*7: For integer multiples of 1 PLC

Response time	200 MΩ: 2 s (Time until the measurement value reaches within 0.1% of the final value)
	20 MΩ: 0.5 s (Time until the measurement value reaches within 0.1% of the final value)

Open-circuit voltage:	7.5 V or less

Maximum allowable input voltage

Ω terminal – COM terminal	1000 V peak
4WΩ HI terminal – 4WΩ LO terminal	200 V peak
4WΩ HI/LO terminal – COM terminal	200 V peak
COM terminal – chassis	500 V peak
4WΩ HI/LO terminal – chassis	500 V peak

7.1.4 DC Current Measurement (DCI)

5 1/2 digits

Input terminal	Range	Maximum display		Resolution		Resistance between terminals	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
mA-COM	200 mA	199.99	199.999	10 µA	1 µA	0.5 Ω or lower	0.03+2	0.03+10	0.03+10	40+1	40+2
	2000 mA	1999.9	1999.99	100 µA	10 µA	0.5 Ω or lower	0.05+2	0.05+6	0.05+5	50+0.7	50+1.5
10A-COM	10 A	10.999	10.9999	1 mA	100 µA	0.06 Ω or lower	0.15+2	0.15+7	0.15+6	50+0.7	50+1.5

6 1/2 digits

Input terminal	Range	Maximum display		Resolution		Resistance between terminals	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
mA-COM	200 mA	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 nA	Same as 5 1/2 digits	Same as 5 1/2 digits	0.03+100	0.03+100	40+10	40+20
	2000 mA		1999.999		1 µA			0.05+60	0.05+50	50+7	50+15
10A-COM	10 A		10.99999		10 µA			0.15+70	0.15+60	50+7	50+15

*1: When the Auto-zero is ON
When the Auto-zero is OFF, add 2 digits (5½ digits) or 20 digits (6½ digits) to the digit error. “IT > 400 ms” is specified only when the Auto-zero is ON.

Additional error depending on integration time setting

Integration time	Measurement accuracy	Description
200 μs ≤ IT < 2 ms	See FAST.	4 1/2-digit display, additional error ± 10 digits
IT = 2 ms		-
2 ms < IT ≤ 15 ms		Convert the digit error to 5 1/2-digit display.
15 ms < IT < 1 PLC		Convert the digit error to 5 1/2-digit display or 6 1/2-digit display.
1 PLC ≤ IT < 100 ms	See MED.	*2
100 ms ≤ IT ≤ 200 ms	See SLOW1, 2.	*2
200 ms < IT ≤ 2 s		Additional error ± 5 digits (for 5 1/2-digit display) or ± 50 digits (for 6 1/2-digit display) *2

*2: For integer multiples of 1 PLC

	Maximum allowable input current	Input protection	Fuse replacement method
mA terminal – COM terminal	2000 mA	2A/250V fast-blow fuse which is compliant with IEC60127 sheet1	To be replaced on the rear panel
10A terminal – COM terminal	10 A	15 A/250 V fast-blow fuse with 10000 A breaking capacity	Contact us to repair

7.1.5 AC Current Measurement (ACI, ACI (AC+DC))

Measurement method: True RMS measurement, RMS display

Input range: 5% or more of a full scale

Crest factor: 3 : 1 at a full scale (Restricted to the current protection fuse rating)

Temperature coefficient: (1/10 of measurement accuracy that includes the additional error)/˚C in each range and frequency range)

Response time: Approx. 1 s (Time until the measurement value reaches within 0.1% of the final value)

ACI 5 1/2 digits

Input terminal	Range	Maximum display		Resolution		Resistance between terminals
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
mA-COM	200 mA	199.99	199.999	10 µA	1 µA	0.5 Ω or less
	2000 mA	1999.9	1999.99	100 µA	10 µA	0.5 Ω or less
10A-COM	10 A	10.999	10.9999	1 mA	100 µA	0.06 Ω or less

RATE setting	Range	Measurement accuracy*1 ±(% of reading + digits)		
		20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
FAST	200 mA	0.4+20	0.3+20	0.3+10
	2000 mA	0.5+20	0.35+10	0.35+20
	10 A	0.5+20	0.35+20	0.7+20
MED/ SLOW1,2	200 mA	0.4+200	0.3+200	0.3+100
	2000 mA	0.5+200	0.35+100	0.35+200
	10 A	0.5+200	0.35+200	0.7+200

ACI 6 1/2 digits

Input terminal	Range	Maximum display		Resolution		Resistance between terminals
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
mA-COM	200 mA	Same as 5 1/2 digits	199.9999	Same as 5 1/2 digits	100 nA	Same as 5 1/2 digits
	2000 mA		1999.999		1 μA	
10A-COM	10 A		10.99999		10 μA	

RATE setting	Range	Measurement accuracy*1 ±(% of reading + digits)		
		20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
MED/ SLOW1,2	200 mA	0.4+2000	0.3+2000	0.3+1000
	2000 mA	0.5+2000	0.35+1000	0.35+2000
	10 A	0.5+2000	0.35+2000	0.7+2000

*1: For sine wave input

ACI additional error depending on integration time setting

Integration time	Measurement accuracy	Description
200 μs ≤ IT < 2 ms	See FAST.	4 1/2-digit display, additional error ± 10 digits
IT = 2 ms		-
2 ms < IT ≤ 15 ms		Convert the digit error to 5 1/2-digit display.
15 ms < IT < 1 PLC		Convert the digit error to 5 1/2-digit display or 6 1/2-digit display.
1 PLC ≤ IT ≤ 2 s	See MED/ SLOW1, 2.	For integer multiples of 1 PLC

ACI (AC + DC) 5 1/2 digits, 6 1/2 digits

Input terminal	Range	Maximum display		Resolution		Resistance between terminals
		FAST	MED/ SLOW1,2	FAST	MED/ SLOW1,2	
mA-COM	200 mA	199.9	199.99	100 µA	10 µA	0.5 Ω or less
	2000 mA	1999	1999.9	1 mA	100 µA	0.5 Ω or less
10A-COM	10 A	10.99	10.999	10 mA	1 mA	0.06 Ω or less

RATE setting	Range	Measurement accuracy*2 ±(% of reading +digits)		
		20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
FAST/ MED/ SLOW1,2	200 mA	0.4+20	0.3+20	0.3+12
	2000 mA	0.5+20	0.35+12	0.35+20
	10 A	0.5+20	0.35+20	0.7+20

*2: For sine wave input

ACI (AC+DC) additional error depending on integration time setting

Integration time	Reading error ± (% of reading)		Digit error (Value of measurement accuracy)	Description
	20 Hz to 45 Hz	45 Hz to 100 Hz		
200 µs ≤ IT ≤ 2 ms	3	0.15	×1	-
2 ms < IT < 1 PLC			×10 (4 1/2-digit conversion)	-
1 PLC ≤ IT < 100 ms	2	0.015	×1	For integer multiples of 1 PLC
100 ms ≤ IT ≤ 2 s	-	-		

Additional error depending on crest factor (For non-sine wave input) ± (% of reading + % of range)

Range	Crest factor	
	1 to 2	2 to 3
200 mA	0+0.05	0.1+0.15
2000 mA	0+0.05	0.1+0.15
10 A	0+0.05	0.03+0.15

	Maximum allowable input current	Input protection	Fuse replacement method
mA terminal – COM terminal	2000 mA	2A/250V fast-blow fuse which is compliant with IEC60127 sheet1	To be replaced on the rear panel
10A terminal – COM terminal	10 A	15 A/250 V fast-blow fuse with 10000 A breaking capacity	Contact us to repair

7.1.6 Frequency Measurement (FREQ)

Measurement method Reciprocal

Measurement frequency range	Measurement accuracy
10 Hz to 300 kHz	0.02% of reading

A frequency over the above range is displayed but not guaranteed.

Input signal voltage range: 100 mV rms to 700 V rms and 10% of each voltage range or more (However, the input signal is restricted to the maximum allowable input voltage.)

Gate time

RATE setting	Gate time	Measurement frequency range	Measurement interval	Maximum display	
				5 1/2 digits	6 1/2 digits
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	999999	9999999
MED	100 ms	10 Hz to 300 kHz	220 ms	99999	999999
FAST	10 ms	100 Hz to 300 kHz	22 ms	9999	99999

Maximum allowable input voltage

V terminal – COM terminal	1000 V peak, 700 V rms, 2.2×10^7 V Hz
COM terminal – chassis	500 V peak

7.1.7 Diode Measurement

5 1/2 digits

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
2000 mV	1999.9	1999.99	100 µV	10 µV	1 mA	0.02+2	0.02+4	0.014+3	15+0.25	15+1.5

6 1/2 digits

Range	Maximum display		Resolution		Measurement current	Measurement accuracy*1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
2000 mV	*2	1999.999	*2	1 µV	*2	*2	0.02+40	0.014+30	15+2.5	15+15

*1: Add the offset error "(the measurement cable resistance + 0.3 Ω) × 1 mA."
*2: Same as 5 1/2 digits

The specifications other than the above are the same as those of the 2000 Ω range in 2WΩ measurement.

7.1.8 Continuity Measurement

5 1/2 digits

Range	Maximum display		Resolution		Measurement current	Measurement accuracy ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	1 mA	0.02+2	0.02+4	0.014+3	15+0.25	15+1.5

6 1/2 digits

Range	Maximum display		Resolution		Measurement current	Measurement accuracy ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
	FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto zero ON	Auto zero OFF
2000 Ω	*1	1999.999	*1	1 mΩ	*1	*1	0.02+40	0.014+30	15+2.5	15+15

*1: Same as 5 1/2 digits

Continuity judgment value: 1 to 1000 Ω

The specifications other than the above are the same as those of the 2000 Ω range in 2WΩ measurement.

7.1.9 Temperature Measurement

PT resistance thermometer	Sensor type		Pt100	JPt100
	Standard		JIS C1604-1997	JIS C1604-1981
	Measurement range		-200 to +850 °C	-200 to +649 °C
	Maximum display	5 1/2 digits	85000	64900
		6 1/2 digits	850000	649000
	Resolution	5 1/2 digits	0.01 °C	
		6 1/2 digits	0.001 °C	
	Accuracy *1		±(0.02 % of reading + 0.15 °C)	
	Temperature coefficient		±(21 ppm of reading + 0.011 °C) / °C	
	Measurement current		1 mA	
	Linearization		Digital calculation	
	Wire connection		2W, 3W or 4W	
	Allowable wire resistance		10 Ω or lower (except 2W)	
	Measurement unit		°C, °F or K	

*1 4W, RATE: SLOW2, not including the measurement probe accuracy
For 3W, add 0.1 °C to the offset error.

7.1.10 Measurement Time

RATE setting	Integration time	Measurement speed (measurement time)			
		DCV DCI 2WΩ LP-2WΩ TEMP(PT-2W)	ACV ACI Continuity Diode	ACV(AC+DC) ACI(AC+DC)	4WΩ LP-4WΩ TEMP(PT-3W) TEMP(PT-4W)
		Auto zero ON	Auto zero OFF		
FAST	2 ms	69.4 times/s (14.4 ms)	149.2 times/s (6.7 ms)	30.3 times/s (33 ms)	56.1 times/s (17.8 ms)
MED	1 PLC	19.8 times/s (50.4 ms)	40 times/s (25 ms)	20 times/s (50 ms)	18.5 times/s (53.8 ms)
SLOW1	100 ms	4.7 times/s (210.4 ms)	9.5 times/s (105 ms)	4.7 times/s (210 ms)	4.6 times/s (214 ms)
SLOW2	200 ms	2.4 times/s (410.4 ms)	4.8 times/s (205 ms)	2.4 times/s (410 ms)	2.4 times/s (414 ms)

Condition: Auto-range OFF, calculation OFF, display OFF

Maximum measurement speed and its conditions

Measurement speed	250 times/s
Target function	DCV, DCI, 2WΩ, LP-2WΩ
Integration time	200 μs
Sampling interval	4 ms
Other	*1

*1 Condition: Auto-range OFF, Auto-zero OFF

Integration time (IT) and sampling interval (SI)

Integration time (IT)	
Setting range	Resolution
200 μs to 2 s	100 μs
0.02 PLC to 100 PLC	0.01 PLC
Sampling interval (SI)	
Setting range	
4 ms to 3600 s	

Integration time and display digits

Function / Integration time (IT)	DCV/DCI 2WΩ/LP-2WΩ 4WΩ/LP-4WΩ ACV/ACI TEMP Diode/continuity	ACV(AC+DC) ACI(AC+DC)
200 μs ≤ (IT) ≤ 2 ms	4 1/2-digit display	3 1/2-digit display
2 ms < (IT) ≤ 15 ms	5 1/2-digit display	4 1/2-digit display
15 ms < (IT) ≤ 2 s	6 1/2-digit display	4 1/2-digit display

7.1.11 Calculation Functions

NULL calculation:

Display value (NULL) = Measurement value - NULL constant

Smoothing calculation:

Display value (SM) = Moving average of specified number of times

Comparator calculation:

Display (HIGH) ← HIGH setting value < Measurement value

Display (LOW) ← Measurement value < LOW setting value

Display (GO) ← LOW setting value ≤ Measurement value ≤ HIGH setting value

Scaling calculation:

Display (SCL) = (Measurement value - B)/A × C

A, B, and C are constants. (Setting values)

MAX and MIN calculation:

Display value (MAX) = Maximum measurement value after the calculation starts

Display value (MIN) = Minimum measurement value after the calculation starts

Display value (AVE) = Arithmetic mean after the calculation starts

Measurement value count = Measurement count after the calculation starts (up to 1 million)

dB and dBm calculation:

dB display = 20 log (Measurement value/D)

dBm display value = 10 log ((Measurement value)^2/D)/10^{-3}

Constant D (setting value)

Statistical calculation:

Number of samples:	Display value (SAMPLE) = Number of measurement values in the specified range of the measurement memory
Maximum value:	Display value (MAX) = Maximum measurement value in the specified range of the measurement memory
Minimum value:	Display value (MIN) = Minimum measurement value in the specified range of the measurement memory
Average value:	Display value (AVE) = Average value in the specified range of the measurement memory
Standard deviation:	Display value (SIGMA) = Standard deviation in the specified range of the measurement memory
Dispersion:	Display value (P-P) = ((Maximum measurement value) - (Minimum measurement value)) in the specified range of the measurement memory

7.2 Interface Specifications

Item	Specification
Remote control	
Remote command	Compliance with the ADC and SCPI command formats 7351-compatible mode and 34401A-comptible mode
Output format	ASCII or binary (IEEE-754 64-bit floating point)
USB (standard)	
Standard	Compliance with USB2.0 Full Speed
Connector	Type B
Class	ADC instruments USB driver, USBTMC, USBCDC
GPIB (Option +01)	
Standard	Compliance with IEEE488.2
Connector	24-pin Anphenol
Interface function	SH1, AH1, T5, L4, SR1, RL1, PP0, DC1, DT1, C0, E2
Addressing	31 kinds of talker/listener addresses can be specified from the front panel.
RS232 (Option +03)	
Standard	EIA-232
Connector	D-Sub 9 pin
Baud rate	19200, 9600, 4800, 2400, 1200, 600, 300
Parity	Even number (EVEN), odd number (ODD), or none
Number of data bits	7 bits or 8 bits
Number of stop bits	1 bit or 2 bits
Echo	ON, OFF
LAN	
Standard	IEEEE802.3
Data transmission	10BASE-T, 100BASE-TX
Connector	RJ-45

Item	Specification
Comparator output (Option+03)	
Output signal	TTL Output: PASS/FAIL Relay output: PASS/Hi/Lo (PASS/FAIL output can be set individually)
Connector	D-Sub 9 pin
a. Optical semiconductor relay contact	
Allowable contact voltage (for break)	30 V DC
Allowable contact current	120 mA DC
Withstand voltage between contact and GND	30 V
Contact operating time	Approx. 1 ms or less
b. TTL output	
Output level	TTL, selecting the positive or negative logic
Maximum allowable input voltage	12 V peak
External trigger signal (Option +01, +03)	
Connector	BNC
Signal level	TTL, falling edge or rising edge detection
Pulse width	1 µs or more
Complete signal output (Option +01, +03)	
Connector	BNC
Signal level	TTL, negative pulse
Sink current	20 mA or less
Pulse width	Approx. 5 µs, 100 µs or 1 ms

NOTE: No options can be added after purchase.

7.3 General Specifications

<table>
<tr><th>Item</th><th>Specification</th></tr>
<tr><td>Operating environment</td><td>Temperature: 0°C to +50°C
Relative humidity:
85% or less no condensation (75% or less no condensation for the resistance measurement range of 20 MΩ or higher or the low-power resistance measurement range of 2 MΩ or higher)</td></tr>
<tr><td>Storage environment</td><td>Temperature: -25°C to +70°C
Relative humidity:
85% or less, no condensation</td></tr>
<tr><td>Warm-up time</td><td>60 minutes or more</td></tr>
<tr><td>Display</td><td>Decimal 7 digits, 7-segment VA LCD</td></tr>
<tr><td>Range switching</td><td>Automatic and manual</td></tr>
<tr><td>Input method</td><td>Floating and guarded</td></tr>
<tr><td>Maximum allowable input voltage</td><td>
<table>
<tr><td></td><td></td><td></td><td></td><td>V Ω Hz</td></tr>
<tr><td></td><td></td><td></td><td>4WΩHI</td><td>200 Vpeak</td></tr>
<tr><td></td><td></td><td>4WΩLO</td><td>200 Vpeak</td><td>200 Vpeak</td></tr>
<tr><td></td><td>COM</td><td>200 Vpeak</td><td>200 Vpeak</td><td>1000 Vpeak</td></tr>
<tr><td>Chassis</td><td>500 Vpeak</td><td>500 Vpeak</td><td>500 Vpeak</td><td>1000 Vpeak</td></tr>
</table>
</td></tr>
<tr><td>Measurement method</td><td>Integration</td></tr>
<tr><td>Overload display</td><td>OL</td></tr>
<tr><td>Trigger function</td><td>External trigger signal, panel key, remote command</td></tr>
<tr><td>Memory</td><td>Measurement data memory: Up to 20,000 data
Condition setting memory: 4 (USER0 to USER3)</td></tr>
<tr><td>Power supply</td><td>AC power supply 100V/120V/220V/240V (User selectable)
<table>
<tr><td>Option No.</td><td>Standard</td><td>OPT.32</td><td>OPT.42</td><td>OPT.44</td></tr>
<tr><td>Power voldtage</td><td>100 V</td><td>120 V</td><td>220 V</td><td>240 V</td></tr>
</table>
Specify the option when ordering.
Use a power cable and a fuse that are compliant with the safety standard when changing the power supply voltage.</td></tr>
<tr><td>Power supply frequency</td><td>50 Hz/60 Hz</td></tr>
<tr><td>Power consumption</td><td>15 VA or less</td></tr>
<tr><td>Dimensions</td><td>Approx. 212 (W) x 88 (H) x 340 (D) mm</td></tr>
</table>

Item	Specification
Mass	4.5 kg or less
Safety	Compliant with IEC61010-1 Ed.3 Measurement CATII
EMC	EN61326-1 class A

8. MAINTENANCE

This chapter describes the following information which relates to the maintenance of the 7355.

8.1 Replacing Fuses

8.2 Cleaning

8.3 Calibration

8.4 Replacing Parts with Limited Life

8.5 Product Disposal and Recycling

8.6 Storage

8.7 Transport

8.8 Notes Regarding Repair, Replacement, and Periodic Calibration

8.9 System Recovery Procedure

8.10 Self-Test

8.11 Error Message List

8.1 Replacing Fuses

The 7355 includes a power fuse and a protection fuse. Replace them according to the following procedure.

CAUTION:

1. *To prevent fire and electrical shock, replace with fuses listed in Table 3-1. Never use incompatible fuses and short-circuit the fuse holder.*
2. *Whether a fuse has blown cannot be checked by observation. Measure the resistance of the fuse and judge whether the fuse has blown. (If the resistance is 15 Ω or less, the fuse has not blown.)*
3. *Do not cut any of the internal ground wiring in the 7355 or disconnect the 7355 from the ground. The 7355 may become unsafe to operate.*

8.1.1 Replacing Power Fuse

The power fuse is placed in the AC power supply connector on the rear panel.

For more information on how to check and replace the power fuse, refer to Section 3.4.2, “Changing Power Supply Voltage”.

8.1.2 Replacing Protection Fuse

The 7355 has a fuse of the current disconnect type on the rear panel for the purpose of preventing burns and fires when an overcurrent is applied to the mA current input terminal.

8.1.2.1 How to Check Protection Fuses

Measure the resistance of the protection fuse and check that the fuse is not open.

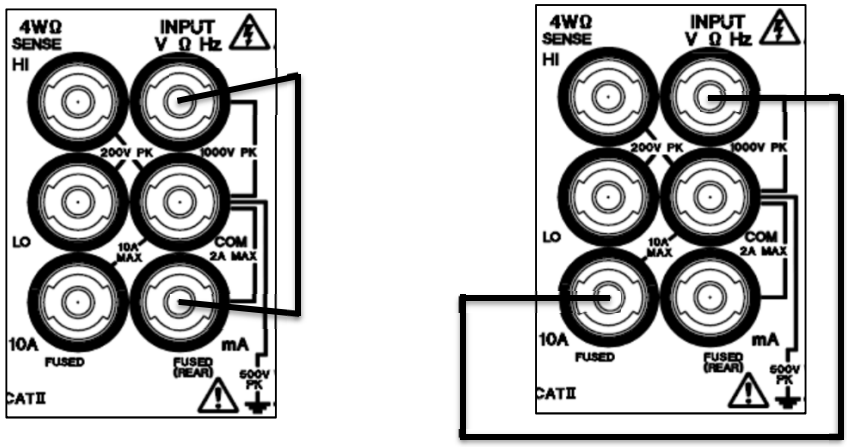
The 7355 can measure the resistance of the protection fuse by connecting the terminals as shown in Figure 8-1.

a. For the mA terminal

Connect the mA and Ω terminals.
Set the resistance measurement function (2WΩ) and measure the resistance of the protection fuse.

b. For the 10 A terminal

Connect the 10 A and Ω terminals.
Set the resistance measurement function (2WΩ) and measure the resistance of the protection fuse.



a. For the mA terminal b. For the 10 A terminal

Figure 8-1 How to Check Protection Fuses

8.1.2.2 How to Replace Protection Fuses

1. Rotate the fuse cap counter clockwise by using a flathead screwdriver and remove the fuse.
2. Replace with a fuse, which is compliant with the standard and listed in Table 3-1, and secure the fuse cap.

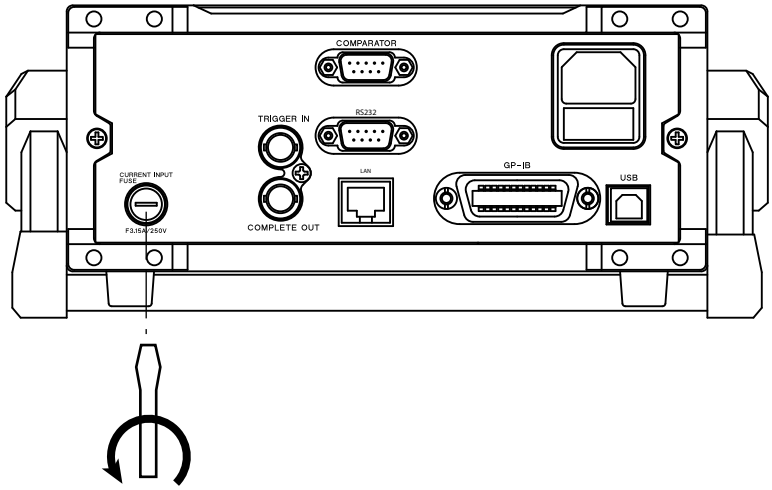


Figure 8-2 Input Terminal Protection Fuse

The protection fuse for the 10 A current input terminal is assembled inside the 7355.
Measure the resistance between the 10 A and COM terminals. If the protection fuse is open, order repair.

8.2 Cleaning

This section describes the cleaning procedure of the 7355 and warning messages.

WARNING: To prevent accidents caused by electric shocks, remove all cables from the 7355. Never attempt to remove the cover to clean the inside of the 7355.

8.2.1 Cleaning Outside

Use a soft or damp cloth to clean the outside of the 7355.

NOTE:

1. *Do not allow water inside the 7355 or lint on it.*
2. *Do not use an organic solvent such as benzene or acetone for cleaning because it degrades the plastic.*

8.2.2 Cleaning Others

Take appropriate precautions to protect the 7355 from dust.

WARNING: Periodically remove any dust which builds up on the AC power connector and plug. The tracking phenomenon, which can be caused by wet dust, may cause a fire.

8.3 Calibration

To maintain the measurement accuracy of the 7355, perform calibration at least once a year; warranty period.

8.3.1 Preparing for Calibration

1. Power supply
   Use an AC power supply in a voltage range listed in Table 3-4 and frequency of 50 Hz or 60 Hz.
2. Environment
   Perform calibration in following conditions.
   Temperature: +23±3 °C
   Humidity: 70 % or less
   An area free from dust, vibration, wind, and noise
3. Warm-up time
   Perform calibration after allowing a warm up time of 60 minutes or more.
   Sufficient warm-up time is also required for the instrument used for calibration.

8.3.2 Calibration Standards

Calibration standard	Operation range	Accuracy
DC voltage	190 mV 1900 mV - 1000 V	±18 ppm ±15 ppm
AC voltage	5 mVrms (1 kHz sine wave) 10 mVrms - 100 mVrms (1 kHz sine wave) 190 mVrms - 700 Vrms (1 kHz sine wave)	±1400 ppm ±1000 ppm ±200 ppm
Resistance	190 Ω 1.9 kΩ – 190 kΩ 1.9 MΩ 19 MΩ – 180 MΩ	±50 ppm ±20 ppm ±50 ppm ±1000 ppm
DC current	190 mA - 1.9 A 10 A	±150 ppm ±1000 ppm
AC current	5 mArms - 50 mArms (1 kHz sine wave) 100 mArms - 10 Arms (1 kHz sine wave)	±500 ppm ±1000 ppm

8.3.3 Calibration Points

1. Perform calibration for each range of each measurement function. For more information on relationships between functions and ranges, refer to Table 5-2.
2. First, calibrate the zero point in each range calibration.
3. In the DC voltage measurement and DC current measurement, calibrate the zero point and full-scale point.
4. In the 4W resistance measurement, calibrate the zero point and full-scale point, and in the 2W resistance measurement, calibrate only the zero point.
5. In the 4W low-power resistance measurement, calibrate the zero point and full-scale point, and in the 2W low-power resistance measurement, calibrate only the zero point.
6. In the AC voltage measurement and AC current measurement, calibrate the zero point, calibration point 1, calibration point 2, and calibration point 3.
7. For the ACV (AC+DC) and ACI (AC+DC) measurements, no calibration is required as those calibrations are included in the calibrations of the DC voltage and AC voltage measurements.
8. For the continuity, diode, and temperature measurements, no calibration is required as those calibrations are included in the calibrations of the DC voltage and resistance measurements.
9. For the frequency measurement, no calibration is required.

Recommended input range: +5% to -10% for each calibration point

Table 8-1 Calibratoin Points (1 of 2)

Measurement function	Range	Zero	Calibration point 1		Calibration point 2		Calibration point 3	
DCV	200 mV	Short	190 mV					
	2000 mV	Short	1900 mV					
	20 V	Short	19 V					
	200 V	Short	190 V					
	1000 V	Short	1000 V					
ACV ACV(AC+DC)	200 mV	Short	5 mV	1 kHz	19 mV	1 kHz	190 mV	1 kHz
	2000 mV	Short	50 mV	1 kHz	190 mV	1 kHz	1900 mV	1 kHz
	20 V	Short	500 mV	1 kHz	1900 mV	1 kHz	19 V	1 kHz
	200 V	Short	5000 mV	1 kHz	19 V	1 kHz	190 V	1 kHz
	700 V	Short	7 V	1 kHz	70 V	1 kHz	700 V	1 kHz

Table 8-1 Calibratoin Points (2 of 2)

Measurement function	Range	Zero	Calibration point 1	Calibration point 2	Calibration point 3
2WΩ 4WΩ	200 Ω	Short	190 Ω		
	2000 Ω	Short	1900 Ω		
	20 kΩ	Short	19 kΩ		
	200 kΩ	Short	190 kΩ		
	2000 kΩ	Short	1900 kΩ		
	20 MΩ	Short	19 MΩ		
	200 MΩ	Short	180 MΩ		
LP-2WΩ LP-4WΩ	200 Ω	Short	190 Ω		
	2000 Ω	Short	1900 Ω		
	20 kΩ	Short	19 kΩ		
	200 kΩ	Short	190 kΩ		
	2000 kΩ	Short	1900 kΩ		
	20 MΩ	Short	19 MΩ		
DCI	200 mA	Open	190 mA		
	2000 mA	Open	1900 mA		
	10 A	Open	10 A		
ACI ACI(AC+DC)	200 mA	Open	5 mA 1 kHz	19 mA 1 kHz	190 mA 1 kHz
	2000 mA	Open	50 mA 1 kHz	190 mA 1 kHz	1900 mA 1 kHz
	10 A	Open	100 mA 1 kHz	1 A 1 kHz	10 A 1 kHz

8.3.4 Calibration Procedure

Perform calibration in following procedure.

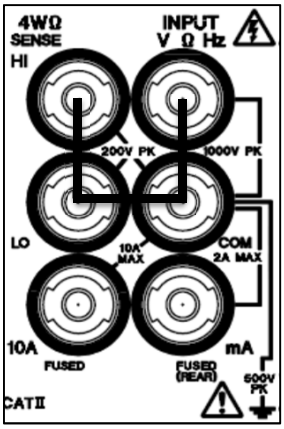
1. Set the calibration mode.
2. Short-circuit the VΩ, COM, HI and LO terminals and then open the mA and 10 A terminals.
3. Calibrate the zero point in a function and range to be calibrated.
4. Connect the calibration standard to the input terminal.
5. Calibrate the full-scale point in a function and range to be calibrated.
6. Cancel the calibration mode.

Setting the calibration mode

1. Select ON from by 'CAL' in **9 SYS** in **MENU**.
2. Press **ENTER** to apply the setting.
   The CAL indicator is turned on and indicates that the calibration mode is set.
3. Press **EXIT** to exit the menu.
4. Set a function to be calibrated.

Calibrating the zero point

5. Short-circuit between the VΩ and COM terminals and open the mA and 10 A terminals.



6. Set a range to be calibrated.
7. Pressing the **HOLD** key displays the measured value as a setting value.

8. Adjust the setting value to the calibration value by using the △, ▽, ◁, and ▷ keys.

NOTE: Press the **EXIT** *key if calibration must be cancelled due to a mistake such as entering an incorrect number.*

---

9. Press the **ENTER** key to execute the calibration.
10. To calibrate other ranges, repeat steps 6 to 9.

Calibrating the full-scale point

11. Connect the calibration standard to the input terminal.
12. Set a range to be calibrated.
    For multiple calibration points, calibrate them in the order of calibration point 1, 2, and 3.
13. Input a voltage equivalent to the calibration point.
14. Pressing the **HOLD** key displays the measurement value as a setting value.
15. Adjust the setting value to the calibration value by using the △, ▽, ◁, and ▷ keys.

NOTE: Press the **EXIT** *key if the calibration must be cancelled due to a mistake such as entering an incorrect number.*

---

16. Press the **ENTER** key to execute the calibration.
17. To calibrate other ranges, repeat steps 12 to 16.
18. To calibrate other functions, repeat steps 4 to 17.

Canceling the calibration mode

19. Select OFF from by 'CAL' in **9 SYS** in **MENU**.

NOTE: To invalidate the calibration, select EXIT. (The calibration values are not saved.)

---

20. Press **ENTER** to apply the setting.
    The CAL indicator is turned off, indicating that the calibration mode is canceled.
21. Press **EXIT** to exit the menu.

NOTE:

1. *The* **AUTO**, **RATE**, **TRIG**, **NULL** *and* **SM** *keys are disabled in the calibration mode.*
   Only **8 I/F** *and* **9 SYS** *can be selected in* **MENU**.
2. *Cancel the calibration mode before turning off the power.*
   The calibration data is stored in the internal nonvolatile memory and written in it at once when the calibration mode is canceled.

8.4 Replacing Parts with Limited Life

Parts with limited life used in the 7355 are listed in the following table.
Contact an ADC CORPORATION sales representative to replace any parts, referring to the limited lives listed in the following table.

Note that the lifetime may be shorter than that described depending on the operating environment, frequency of use, and storage environment.

MEMO: *The lifetime and recommended replacement times below are reference information and are NOT guaranteed.*

Table 8-2 Parts with Limited Life

Part name	Expected lifetime
Relays for switching functions and ranges	1,000,000 times
Panel keys	500,000 times
Fluorescent display tube	50,000 hours
USB connector	1500 times
GPIB connector	500 times
LAN connector	200 times
RS232 connector	100 times
Comparator output connector	100 times

8.5 Product Disposal and Recycling

Correctly dispose of the 7355 in accordance with local and national regulations.

Before disposal, remove the following parts from the product to prevent dispersal of substances that may adversely affect the environment, human health, or the ecosystem.

NOTE: For assistance with locating a waste disposal company, contact an ADC CORPORATION sales representative.

Name of substance or removed part	Used?	Location	Unit	Part
Capacitor containing polychlorinated biphenyls (PCBs)	No	-		
Part containing mercury	No	-		
Batteries	No	-		
Printed circuit boards	Yes	Main unit	MAIN PANEL RS232_COMP	Printed circuit boards
Toner cartridge	No	-		
Plastic containing brominated flame retardants	Yes	Main unit	BLL-021003 BLC-021004 BLB-021005	Connectors, diodes, Zener diodes, photocouplers, FET, analog ICs, logic ICs, FLASH memories, transistors
Parts containing asbestos	No	-		
Cathode-ray tubes	No	-		
Chlorofluorocarbon (CFC), hydrochlorofluorocarbon (HCFC), hydrofluorocarbon (HFC), or hydrocarbon (HC)	No	-		
Electric-discharge indicator	No	-		
LCD display of 100 cm^2 or larger	No	-		
Connecting cable	Yes	Between devices		Power cable, input and output cable
Parts containing flame-resistant ceramic fibers	No	-		
Parts containing radioactive material	No	-		

Name of substance or removed part	Used?	Location	Unit	Part
Electrolytic capacitors containing substance of concern (With height > 25 mm, dia. > 25 mm, or equivalent volume)	No	-		
Arsenic or its compounds	Yes	Main unit	Electronic components	Photocouplers, logic ICs
Nickel or its compounds	Yes	Main unit		Electronic components, mechanical components
Lead or its compounds	No	-		
PVC	Yes	Main unit		PVC components
Antimony or its compounds	Yes	Main unit		Electronic components

8.6 Storage

If not using the 7355 for a long period of time, cover it with a plastic cover or put it in a cardboard box to protect and store it at a dried place out of direct sunlight.

Storage temperature: -25 °C to +70 °C

8.7 Transport

To transport the 7355, use the original box that it came in or equivalent (5mm or more thick cardboard box).

Packing procedure

1. Wrap the 7355 with shock absorbent material and place it inside the cardboard box.
2. Put in the accessories and put in shock absorbent material again.
3. Close the cardboard box and fix the outside with packing strings.

8.8 Notes Regarding Repair, Replacement, and Periodic Calibration

8.8.1 Before Asking for Repair

Refer to Table 8-3 for checking before ordering repair.

Table 8-3 Before Ordering Repair (1 of 2)

Problem	Check	Solution
The power does not turn on.	Check the power cable	Change the power cable.
	Check the power fuse.	Replace the power fuse.
No display after turning the POWER switch ON	Check the power supply voltage setting on the fuse holder.	Set it according to the voltage of the power supply used.
Dark display after turning the POWER switch ON	Check the power supply voltage setting on the fuse holder.	Set it according to the voltage of the power supply used.
	The LCD display has reached the expected lifetime.	Contact our sales representative.
Key malfunction	Check whether the RMT indicator is on (in remote operation).	Press the LOCAL key to return to the local state.
	The panel lock functions.	Release the panel lock.
Unstable or abnormal measured values	Check that AC power is properly supplied.	Supply stable AC power.
Unmeasurable voltage and resistance	Check that the input cable is correctly connected to the VΩ and COM terminals. (For 4W resistance measurement, check the connections to the 4WΩHI/LO terminals.)	Securely connect the input cable to the VΩ and COM terminals. (For 4W resistance measurement, securely connect it to the 4WΩHI/LO terminals.)
	Check that the input cable is not disconnected.	Change the input cable.
Unmeasurable current	The input terminal differs depending on the measurement range. Check that the input cable is correctly connected to the current input terminals. 200 mA/2000 mA ranges: mA and COM terminals 10 A range: 10 A and COM terminals	Securely connect the input cable to the correct terminals.
	Check whether the protection fuse is blown. (For more information on how to check, refer to 8.1, “Replacing Fuses”.)	Replace the fuse. For the 10 A terminal, contact our sales representative.
	Check that the input cable is not disconnected.	Change the input cable.

Table 8-3 Before Ordering Repair (2 of 2)

Problem	Check	Solution
Unstable measured values for high-resistance measurement	Induction noise is superimposed on the input cable.	Shield the DUT. Use a shielded input cable.
	Check that the power supply frequency setting is the same as the local commercial power supply frequency.	Correctly set the power supply frequency to 50 Hz or 60 Hz.
No sampling	Check whether the HOLD indicator is on (in HOLD operation).	Press the TRIG or HOLD key and check whether sampling operates. (During the remote operation, panel keys do not function. Therefore, after releasing the remote operation, press any panel key.)
	A long trigger delay time is set.	Check the trigger delay time.

8.8.2 Service Request

When sending the 7355 to an ADC CORPORATION sales representative, attach a tag that indicates the following information.

- Your company name and address
- Name of the person in charge
- Serial number (on the rear panel)
- What work to request. (Repair or periodic calibration)

8.8.3 Address and Phone Number

Contact an ADC CORPORATION sales representative.

8.9 System Recovery Procedure

Initialization

The following shows how to initialize the setting parameters.

1. Select **6 INIT** from the **MENU** mode and press ▽ to enter the selection layer.
2. The 'Param' parameter is displayed. Press ▽ to enter the Input layer.
3. Press **ENTER** while 'DFLT' is displayed.

The parameters can be initialized by selecting and executing 'DFLT' when loading the parameter setting.

For more information on how to initialize the parameters, refer to Section 5.10, "Saving and Loading Measurement Condition Settings".

Factory defaults

The following shows how to set all parameters except for the calibration value and calibration memo to the factory defaults.

1. Turn on the power while pressing the **NULL** and **UP** keys.

Quitting the menu

Press **EXIT** to quit the menu operation.

The menu is forced to close at any level.

NOTE: Changing parameters, which are not determined by pressing **ENTER**, *is canceled.*

Initial values

The list of conditions, under which the setting parameters are initialized, and initial values are shown in Table 8-4.

Table 8-4 Initial Values of Setting Parameters (1 of 3)

Parameter	Factory default	MENU → INIT → DFLT	Power ON → POFF	Parameter loading		*RST	*CLS	DCL
				DFLT	USER			
Function	DCV	DCV	Saved value	DCV	USER0-3	DCV		
Range	Highest	Highest	Saved value	Highest	USER0-3	Highest		
Auto range	ON	ON	Saved value	ON	USER0-3	ON		
Number of measurement display digits	Maximum	Maximum	Saved value	Maximum	USER0-3	Maximum		
Auto zero	ON	ON	Saved value	ON	USER0-3	ON		
Temperature unit	°C	°C	Saved value	°C	USER0-3	°C		
Platinum resistance thermometer temperature sensor	PT100	PT100	Saved value	PT100	USER0-3	PT100		
Measurement data memory	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Memory read range	0,0	0,0	Saved value	0,0	USER0-3	0,0		
NULL calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
NULL constant	0	0	Saved value	0	USER0-3	0		
COMP calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
HIGH value	0	0	Saved value	0	USER0-3	0		
LOW value	0	0	Saved value	0	USER0-3	0		
MAX/MIN calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
MAX/MIN display	NORMAL	NORMAL	Saved value	NORMAL	USER0-3	NORMAL		
Smoothing calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Smoothing count	10	10	Saved value	10	USER0-3	10		
Scaling calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Constant A	1	1	Saved value	1	USER0-3	1		
Constant B	0	0	Saved value	0	USER0-3	0		
Constant C	1	1	Saved value	1	USER0-3	1		
dB calculation	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Constant D	1	1	Saved value	1	USER0-3	1		
Digital output	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Sampling	RUN	RUN	Saved value	RUN	USER0-3	RUN		
Sampling rate	SLOW2	SLOW2	Saved value	SLOW2	USER0-3	SLOW2		
Sampling interval	205 ms	205 ms	Saved value	205 ms	USER0-3	205 ms		
Sampling count	1	1	Saved value	1	USER0-3	1		

Blank: No Action

Saved value: Last setting value when the power is turned off.

Table 8-4 Initial Values of Setting Parameters (2 of 3)

Parameter	Factory default	MENU → INIT → DFLT	Power ON → POFF	Parameter loading		*RST	*CLS	DCL
				DFLT	USER			
Trigger count	1	1	Saved value	1	USER0-3	1		
Complete signal width	100 µs	100 µs	Saved value	100 µs	USER0-3	100 µs		
External trigger polarity	Negative logic	Negative logic	Saved value	Negative logic	USER0-3	Negative logic		
Integration time	200 ms	200 ms	Saved value	200 ms	USER0-3	200 ms		
Trigger delay	0	0	Saved value	0	USER0-3	0		
Continuity threshold constant	10 Ω	10 Ω	Saved value	10 Ω	USER0-3	10 Ω		
Trigger source	IMMediate	IMMediate	Saved value	IMMediate	USER0-3	IMMediate		
PASS/FAIL buzzer	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Buzzer	OFF	OFF	Saved value	OFF	USER0-3	OFF		
Interface	USB	USB	Saved value					
GPIB address	1	1	Saved value					
USB ID	1	1	Saved value					
IP address	192.168.0.1	192.168.0.1	Saved value					
Default gateway	192.168.0.0	192.168.0.0	Saved value					
Subnet mask	255.255.255.0	255.255.255.0	Saved value					
Baud rate	9600	9600	Saved value					
Data length	8	8	Saved value					
Parity	None	None	Saved value					
Stop bit	1	1	Saved value					
Echo	OFF	OFF	Saved value					
Talk only	OFF	OFF	Saved value					
Header output	ON	ON	Saved value					
Panel lock	OFF		Saved value					
Disabling function	ENA	ENA	Saved value					
Power frequency	Auto	Auto	Saved value					
PSC flag	1	1	Saved value					
Program language	ADC	ADC	Saved value					
Continue setting	OFF	OFF	Saved value			*1		
Password	0000		Saved value					

Blank: No Action

Saved value: Last setting value when the power is turned off.

*1: By selecting the initial value for continue setting

Table 8-4 Initial Values of Setting Parameters (3 of 3)

Parameter	Factory default	MENU → INIT → DFLT	Power ON → POFF	Parameter loading		*RST	*CLS	DCL
				DFLT	USER			
GPIB input/output buffer	Empty	Empty	Empty	Empty	Empty			Empty
Block delimiter	CR/ LF+EOI	CR/ LF+EOI	CR/ LF+EOI	CR/LF+EOI		CR/LF+EOI		
SRQ	ON	ON	ON	ON				
Number of measurement data output digits	6 1/2	6 1/2	Saved value	6 1/2	USER0-3	6 1/2		
Display ON/OFF	ON	ON	ON	ON	ON	ON		
Calibration mode	OFF	OFF	OFF			OFF		
Status register QEER	0	0	0					
Status register OEER	0	0	0					
Status register SESER	0	0	*2					
Status register MEER	0	0	0					
Status register SRER	0	0	*2					
Status register QER	0	0	0				0	
Status register SESR	0	0	0				0	
Status register STB	0	0	0				0	
Status register OER	0	0	0				0	
Status register MER	0	0	0				0	
Error log	Empty	Empty	Empty				Empty	
7351-compatible mode	OFF	OFF	Saved value	Saved value	Saved value	Saved value		
Calibration mode	OFF	OFF	OFF	OFF	OFF	OFF		
Calibration memo	“”							

Blank: No Action

Saved value: Last setting value when the power is turned off.

*2: By setting the *PSCn command

8.10 Self-Test

8.10.1 Self-Test

The self-test can be performed from the panel operations or by executing the remote command. Perform the self-test by selecting 'TEST' from **9 SYS** in **MENU** or executing the *TST? remote command. The test items are shown in Table 8-5.

Table 8-5 Self-Test Items

Item	Performed by panel or remote coperation	Performed at power on
ROM check sum	-	✓
RAM read/write check	-	✓
Panel display check	-	✓
Alarm status check	-	✓
Calibration data check	✓	✓
Parameter check	✓	✓
Analog to logic communication check	✓	✓
Analog section AD operation check	✓	✓
Analog section range operation check	✓	✓
Analog section reference voltage check	✓	✓
Analog section reference current check	✓	✓

8.10.2 Display and Key Tests

Turn on all segments of the display and check whether they light.
Pressing panel keys display their names.
Test according to the following procedure.

	Operations	Display
1.	Press **MENU**, and select **9 SYS** by using ◁ and ▷.	MENU SYS
2.	Press ▽.	SYS BEEP
3.	Select 'P/KEY' by using ▷.	SYS P/KEY

Operations	Display
4. Press ▽.	P/KEY ENTER
5. Press **ENTER**. Pressing **ENTER** again changes the display.	Full lighting KEY Push
6. Press a key to be checked.	KEY Key pressed
7. To quit the key test, press **SHIFT**. The display changes from "SHIFT" to "DONE".	KEY SHIFT KEY DONE
8. Press **EXIT** to exit the menu.	Measurement state

8.10.3 Model and Option Display

The model name and the option number can be displayed from the menu.

Select "MACH" from **9 SYS** in **MENU**.
Press **ENTER** to display the model name and the option number.

8.10.4 Serial Number Display

The serial number can be displayed from the menu.

Select "SER No." from **9 SYS** in **MENU**.
Press **ENTER** to display the serial number

8.11 Error Message List

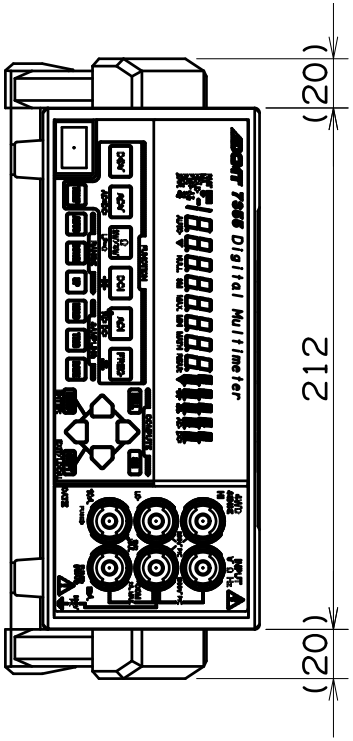
This section describes the error numbers and how to deal with errors displayed on the 7355.

No.	Error display	Error factor [ ]: Response to the ERR? command	Solution
1	Err1	RAM read/write error [RAM read/write failed]	Repair
2	Err2	Display RAM test error or communication error [Front panel does not respond]	Repair
3	Err3	Calibration data check SUM error [Calibration memory lost]	Repair
4	Err4	ROM check SUM error [ROM Chk SUM]	Repair
5	Err6	Parameter check SUM error [Save/recall memory lost]	Repair
6	Err16	FLASH memory writing error [FLASH memory error]	Repair
7	Err101	The function cannot be set. [Inhibited function]	Solution 1
8	Err140	RS232 communication error [RS232 Communication Error]	Solution 10
9	Err150	USB communication error [Illegal packet received]	Solution 2
10	Err200	AD operation IR1/IR2 ratio test error [AD Ratio 1-2]	Repair
11	Err201	AD operation IR2/IR3 ratio test error [AD Ratio 2-3]	Repair
12	Err202	AD operation IR3/IR4 ratio test error [AD Ratio 3-4]	Repair
13	Err203	AD operation IR4/IR5 ratio test error [AD Ratio 4-5]	Repair
14	Err205	Analog operation ACx1 zero measurement test error [ACx1 Zero Check]	Repair
15	Err206	Analog operation Lx1 zero measurement test error [Lx1 Zero Check]	Repair
16	Err207	Analog operation Lx10 zero measurement test error [Lx10 Zero Check]	Repair
17	Err208	Analog operation Lx100 zero measurement test error [Lx10 Zero Check]	Repair
18	Err209	Analog operation DCx10 zero measurement test error [DCx10 Zero Check]	Repair

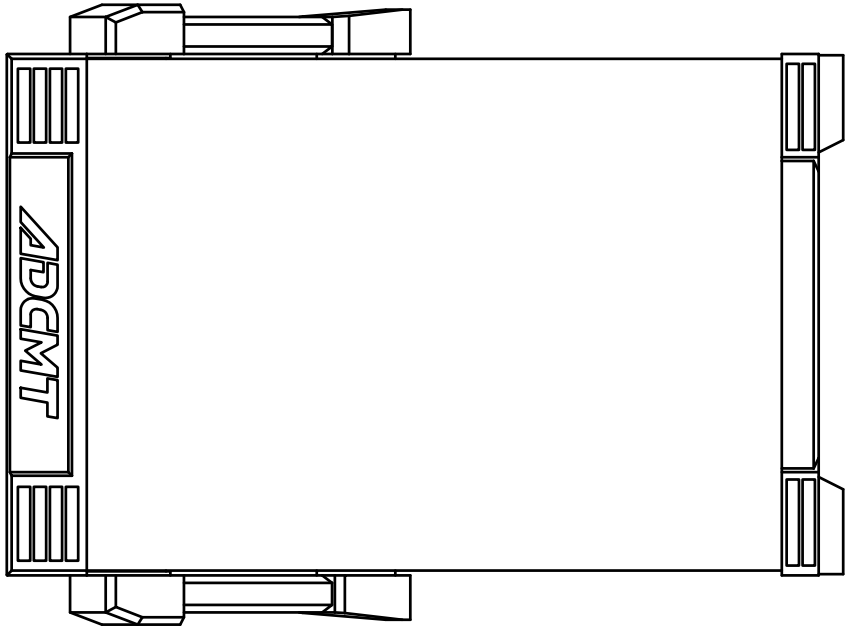
No.	Error display	Error factor [ ]: Response to the ERR? command	Solution
19	Err210	Analog section communication error [Communication error (Analog)]	Repair
20	Err215	Analog operation ATT 1/100 measurement test error [ATT 1/100 Check]	Repair
21	Err216	Analog operation ATT 1/500 measurement test error [ATT 1/500 Check]	Repair
22	Err500	Calibration data error [CAL data]	Solution 3
23	Err501	ZERO calibration is not performed. [before ZERO]	Solution 4
24	Err-102	Command syntax error [Cmd Syntax]	Solution 5
25	Err-113	The command cannot be used. [Undefined header]	Solution 5
26	Err-121	Unrecognized character as the numerical value [Invalid character in number]	Solution 5
27	Err-141	Invalid character string parameter is entered. [Invalid character data]	Solution 5
28	Err-151	Error as the character string data (Quotation mark error) [Invalid string data]	Solution 5
29	Err-200	Execution error (Command currently not executable) [Execution error]	Solution 6
30	Err-213	The INITiate command is ignored because the measurement is performing. [Init ignored]	Solution 6
31	Err-214	The READ? or MEAS? command is received when any trigger source except for IMM is set. [Trigger deadlock]	Solution 6
32	Err-222	Out-of-range parameter setting [Data out of range]	Solution 6
33	Err-313	Calibration data lost [Calibration memory lost]	Solution 7
34	Err-314	Parameters saved by “*SAV” command are lost. [Save/recall memory lost]	Solution 8
35	Err-315	Saved parameters are lost. [Configuration memory lost]	Solution 6
36	Err -330	Self-test error [Self-test failed]	Repair
37	Err-350	Error queue overflows. [Queue overflow]	Solution 9

Solution 1	Set the correct function.
Solution 2	Check the USB cable connection.
Solution 3	Calibrate according to the correct procedure.
Solution 4	Perform the ZERO cancellation.
Solution 5	Send the correct command.
Solution 6	Check the settings.
Solution 7	Re-calibrate.
Solution 8	Re-set.
Solution 9	Read out the error queue.
Solution 10	Check the RS232 cable connection

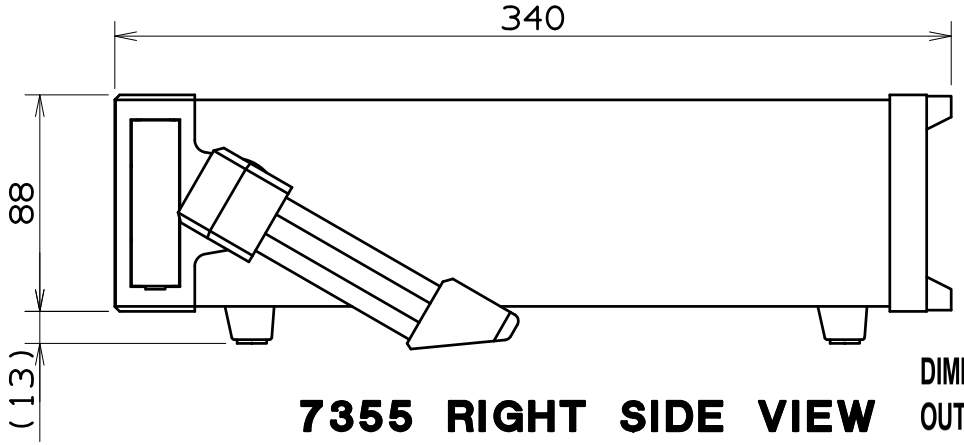
If the error cannot be fixed by the described solution, contact our sales representative.



7355 FRONT VIEW



7355 TOP VIEW



7355 RIGHT SIDE VIEW

DIMENSIONAL OUTLINE DRAWING

Unit: mm

NOTE

This drawing shows external dimensions of this instrument.

The difference in products and options used can cause a change in the appearance.

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